Lattice-based Obfuscation from NTRU and Equivocal LWE

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Abstract. Indistinguishability obfuscation (iO) turns a program unintelligible without altering its functionality and is a powerful cryptographic primitive that captures the power of most known primitives. Recent breakthroughs have successfully constructed iO from well-founded computational assumptions, yet these constructions are unfortunately insecure against quantum adversaries. In the search of post-quantum secure iO, a line of research investigates constructions from fully homomorphic encryption (FHE) and tailored decryption hint release mechanisms. Proposals in this line mainly differ in their designs of decryption hints, yet all known attempts either cannot be proven from a self-contained computational assumption, or are based on novel lattice assumptions which are subsequently cryptanalysed.

In this work, we propose a new plausibly post-quantum secure construction of iO by designing a new mechanism for releasing decryption hints. Unlike prior attempts, our decryption hints follow a public Gaussian distribution subject to decryption correctness constraints and are therefore in a sense as random as they could be. To generate such hints efficiently, we develop a general-purpose tool called primal lattice trapdoors, which allow sampling trapdoored matrices whose Learning with Errors (LWE) secret can be equivocated. We prove the security of our primal lattice trapdoors construction from the NTRU assumption. The security of the iO construction is then argued, along with other standard lattice assumptions, via a new Equivocal LWE assumption, for which we provide evidence for plausibility and identify potential targets for further cryptanalysis.

Change Log

- 2025-08-21: Added an admissibility check of $\mathbf{M}_0 = \mathbf{M}_1 \bmod q$ in the Equivocal LWE assumption (see Fig. 4) which was previously implicit but omitted. Without this check, there would be a trivial attack on the assumption (see Sections 1.4 and 5.1). The modification does not affect the XiO security proof. Expanded cryptanslysis Case L3 on the Equivocal LWE assumption (Appendix C). Fixed minor typos.
- 2025-06-15: Full version of the CRYPTO'25 publication, containing proofs in the appendix.

1 Introduction

The goal of indistinguishability obfuscation (iO) is to make programs unintelligible without altering their functionality. More specifically, it requires that, for any two functionally equivalent circuits $\Gamma_0 \equiv \Gamma_1$, their obfuscations must be computationally indistinguishable, $\mathsf{iO}(\Gamma_0) \approx_c \mathsf{iO}(\Gamma_1)$. The first candidate general-purpose obfuscator was proposed in the seminal work of Garg et al. [GGH+13]. Subsequent works have shown that iO not only implies almost all known cryptographic primitives [SW14], but even further enabling new ones [GGSW13, GGHR14, BZ14, BPR15], establishing iO as being virtually crypto-complete.

Because of its importance, the community has devoted a lot of effort on constructing general-purpose obfuscation from progressively more conservative computational assumptions [BV15, AJ15, Lin16, LV16, Lin17, JLMS19, AJL+19], leading to the recent work of Jain, Lin, and Sahai [JLS21, JLS22] who presented the first construction of iO from well-founded computational assumptions. However, even their scheme relies on a mix of different sources of hardness and, perhaps more worryingly, is insecure against quantum algorithms. Other lines of research study obfuscation with conjectured post-quantum security, for example, constructing schemes over candidate multilinear maps [BR14, BGK+14, GMM+16, CVW18], leveraging fully-homomorphic encryption (FHE) schemes from lattices [WW21, GP21, DQV+21, BDGM22], building upon noisy linear functional encryption [Agr19] and affine determinant programs [BIJ+20], among others.

Prior Lattice-based iO Attempts. Zooming into lattice-based iO constructions, more specifically we have seen the following attempts in the past:

- Schemes under the paradigm of an obfuscator evaluating some FHE-ciphertext and releasing carefully crafted decryption hint, sometimes also known as the BDGM framework [BDGM20]. To date, all attempts in this line either have only heuristic security [BDGM20], are proven under novel assumptions [GP21, WW21, DQV$^+$21] that are subsequently cryptanalysed [HJL21, JLLS23], or adopt an assumption that is highly tailored to the scheme [BDGM22] which offer limited insight to the core computational hardness.
- Schemes using noisy linear functional encryption, the latter having been constructed under a novel assumption over NTRU lattices [Agr19], with subsequent attack and fix on the scheme but not the assumption [AP20].

It is worth highlighting that, although known attacks often do not break the iO schemes themselves, almost all attempts at formulating concrete computational assumptions for these schemes have been invalidated (with the only exception of [BDGM22] which tailored the assumption to their specific scheme). This is especially unsatisfactory, since the hardness of lattice problems is among the most promising sources of post-quantum security, yet the above situation underpins our lack of understanding on lattice-based obfuscation.

This Work. Building upon prior efforts, we revisit the possibility of lattice-based iO under the BDGM framework, with the goal of eventually basing security on self-contained and reasonable computational assumptions. Below, we recall the backbone of the BDGM framework in Section 1.1, and highlight in Section 1.2 our idea of releasing a decryption hint in form of an "equivocated LWE secret" – a hint that is in a sense as randomised as possible subject to correctness constraints. In particular, our strategy involves freshly sampling and injecting a masking error into an otherwise insecure decryption hint, whose randomness is independent of the rest of the obfuscated program. This sets us subtly but fundamentally apart from all prior attempts under the BDGM framework, where (the masking of) the decryption hint were deterministically computed from randomness that were spent elsewhere (e.g. LWE errors in other LWE samples) [GP21, WW21, DQV$^+$21], hence were information-theoretically fully determined conditioned on the rest of the obfuscated program and opened up the doors for attacks [HJL21, JLLS23].

The major novelties of this work lie in the new tools we introduce for instantiating our hint-releasing strategy, summarised in Section 1.3: A new notion of *equivocal distribution* of matrices which admits super-polynomially many LWE solutions, a new primitive called *primal lattice trapdoors* allowing to sample from these LWE solutions efficiently, and a new computational assumption called *Equivocal LWE assumption*, capturing the conjectured hardness of the LWE problem over equivocated matrices in presence of an equivocated LWE secret. We provide cryptanalytic evidence of the hardness of the Equivocal LWE assumption, summarised in Section 1.4.

Concurrent Work. The concurrent and independent work of Hsieh, Jain and Lin [HJL25] also constructs iO under the BDGM framework by designing novel mechanism for releasing random decryption hints. Their construction is proven secure under a new Circular Security with Random Opening (CRO) assumption along with other standard assumptions.

1.1 The BDGM Framework

The work of Brakerski, Döttling, Garg and Malavolta (henceforth BDGM) [BDGM20] proposed a framework for constructing "exponentially-efficient" iO or XiO [LPST16], which has the same functionality as an iO but an obfuscated circuit is only required to be of size sublinear in the truth table of the obfuscated circuit. This suffices for constructing iO because of a known [BV15, AJ15, LPST16] generic transformation from XiO to iO.

The BDGM construction allows obfuscating Boolean circuits $\Gamma : [h] \times [k] \to \{0, 1\}$, whose truth table, of size $N = hk$, is more conveniently represented as an h-by-k array $\boldsymbol{\Gamma} = (\gamma_{i,j})_{i,j}$, where $\gamma_{i,j} = \Gamma(i, j)$. It calls for two ingredients:

– A (levelled) fully homomorphic encryption scheme FHE whose decryption function (for any fixed ciphertext) consists of evaluating a linear function over (an encoding of) the secret key followed by a rounding operation. The purpose of FHE is to encrypt the description of the circuit being obfuscated.
– A linearly homomorphic encryption scheme LHE for which succinct decryption hints can be generated. That is, for a ciphertext ctxt encrypting a large message msg, it is possible to release a much smaller size hint specific to ctxt such that msg can be recovered from $(\mathsf{ctxt}, \mathsf{hint})$, hopefully without compromising the security of other ciphertexts. The purpose of LHE is to encrypt the secret key $\mathsf{sk}_{\mathsf{FHE}}$ of FHE, so that the linear part of the FHE decryption can be homomorphically evaluated over the LHE encryption of $\mathsf{sk}_{\mathsf{FHE}}$.

Taking some liberty in the interpretation, the obfuscator works as follows:

(i) Generate an FHE secret key $\mathsf{sk}_{\mathsf{FHE}}$ and encrypt the circuit Γ as $\mathsf{ctxt}_{\mathsf{FHE}} = \mathsf{FHE.Enc}(\mathsf{sk}_{\mathsf{FHE}}, \Gamma)$.
(ii) Generate an LHE encryption of $\mathsf{sk}_{\mathsf{FHE}}$ as $\mathsf{ctxt}_{\mathsf{LHE}} = \mathsf{LHE.Enc}(\mathsf{pk}_{\mathsf{LHE}}, \mathsf{sk}_{\mathsf{FHE}})$.
(iii) Homomorphically evaluate a universal circuit on $\mathsf{ctxt}_{\mathsf{FHE}}$ to get $\mathsf{ctxt}'_{\mathsf{FHE}}$ encrypting $\boldsymbol{\Gamma}$. Let $\mathbf{L}$ represent the linear part of $\mathsf{FHE.Dec}(\cdot, \mathsf{ctxt}'_{\mathsf{FHE}})$.
(iv) Homomorphically evaluate the linear function $\mathbf{L}$ on $\mathsf{ctxt}_{\mathsf{LHE}}$ to obtain a $\mathsf{ctxt}'_{\mathsf{LHE}}$ encrypting $\boldsymbol{\Gamma}$.
(v) Release decryption hint hint such that $\boldsymbol{\Gamma}$ can be recovered from $\mathsf{ctxt}'_{\mathsf{LHE}}$.
(vi) The obfuscated circuit consists of $(\mathsf{ctxt}_{\mathsf{FHE}}, \mathsf{ctxt}_{\mathsf{LHE}}, \mathsf{hint})$.

With the above description, the design of the obfuscated circuit evaluator is apparent: Derive (the pertinent part of) $\mathsf{ctxt}'_{\mathsf{LHE}}$ as in the obfuscator and use hint to recover (the pertinent of) the truth table $\boldsymbol{\Gamma}$.

The "exponential-efficiency" of the construction crucially relies on the succinctness of the decryption hint hint, so that the overall size of $(\mathsf{ctxt}_{\mathsf{FHE}}, \mathsf{ctxt}_{\mathsf{LHE}}, \mathsf{hint})$ is sublinear in the size of the truth table $\boldsymbol{\Gamma}$. For security, such hint should ideally only allow the recovery of $\boldsymbol{\Gamma}$ from $\mathsf{ctxt}'_{\mathsf{LHE}}$ but not much more. For example, it should not reveal information about $\mathsf{sk}_{\mathsf{FHE}}$ which allows distinguishing $\mathsf{ctxt}_{\mathsf{FHE}}$.

The difference in the constructions [BDGM20, WW21, GP21, DQV+21, BDGM22] following the BDGM framework lies in, among others, the way that the decryption hint hint is generated. While the method of [BDGM20] is heuristic, i.e. without a security reduction from any simpler assumption, [WW21, GP21, DQV+21, BDGM22] proposed innovative methods of masking the decryption hint with security supported by various novel assumptions, unfortunately with the majority being later cryptanalysed in [HJL21, JLLS23].

1.2 New Method for Decryption Hint Generation

We consider a new method of generating the decryption hint based on a mechanism which, in a sense, allows our decryption hint to be as random as possible subject to decryption correctness.

We consider the instantiation of LHE by a packed variant of the dual-Regev encryption scheme [GPV08], also considered in [BDGM22]. Adjusting notation slightly, a ciphertext encrypting $\mathsf{sk}_{\mathsf{FHE}}$ takes the form

$$\mathsf{ctxt}_{\mathsf{LHE}} = (\mathbf{B}, \mathbf{C}) \text{ where } \mathbf{C} = \mathbf{R}\mathbf{B} + \mathbf{E} + \mathsf{Encode}(\mathsf{sk}_{\mathsf{FHE}}) \bmod q$$

where $\mathbf{R}$ is some uniformly random encryption randomness, $\mathbf{E}$ is some short noise and Encode is the encoding function of some error correcting code. In other words, the ciphertext is $\mathsf{Encode}(\mathsf{sk}_{\mathsf{FHE}})$ masked with a learning with errors (LWE) sample $\mathbf{R}\mathbf{B} + \mathbf{E} \bmod q$ with respect to $\mathbf{B}$. For such a scheme, a natural decryption hint for the evaluated ciphertext

$$\mathsf{ctxt}'_{\mathsf{LHE}} = (\mathbf{B}, \mathbf{L}\mathbf{C}) \text{ where } \mathbf{L}\mathbf{C} = \mathbf{L}\mathbf{R}\mathbf{B} + \mathbf{L}\mathbf{E} + \mathsf{Encode}(\boldsymbol{\Gamma}) \bmod q$$

is a matrix $\tilde{\mathbf{R}}$ satisfying

$$\mathbf{L}\mathbf{C} = \tilde{\mathbf{R}}\mathbf{B} + \tilde{\mathbf{E}} + \mathsf{Encode}(\boldsymbol{\Gamma}) \approx \tilde{\mathbf{R}}\mathbf{B} + \mathsf{Encode}(\boldsymbol{\Gamma}) \bmod q. \tag{1}$$

In typical instantiations of the dual-Regev encryption scheme, the public key matrix $\mathbf{B}$ is sampled uniformly at random, which implies that $\tilde{\mathbf{R}} = \mathbf{L}\mathbf{R} \bmod q$ is the only solution satisfying Eq. (1) with overwhelming probability. Releasing such an $\tilde{\mathbf{R}}$ is clearly disastrous. Indeed, due to the efficiency requirements of the obfuscator, $\mathbf{L}$ must be a tall matrix, and thus $\mathbf{R}$ may be efficiently recovered from $\mathbf{LR}$ (assuming that $\mathbf{L}$ has linearly independent columns) which then allows recovering $\mathsf{sk}_{\mathsf{FHE}}$.

Our idea is to instead instantiate $\mathbf{B}$ such that its primal lattice

$$\Lambda_q(\mathbf{B}) = \left\{\mathbf{x} : \exists \mathbf{r},\ \mathbf{x}^{\mathrm{T}} = \mathbf{r}^{\mathrm{T}}\mathbf{B} \bmod q\right\} \tag{2}$$

has a dense (i.e. containing exceptionally short vectors) sublattice. For such $\mathbf{B}$ and other appropriate parameters, Eq. (1) may admit exponentially many solutions $\tilde{\mathbf{R}}$. Thus, a natural choice of a decryption hint is a random $\tilde{\mathbf{R}}$ (following some natural distribution) satisfying Eq. (1).

Equipped with this idea, this work aims to answer the following questions:

(i) How to instantiate $\mathbf{B}$ so that Eq. (1) admits many solutions $\tilde{\mathbf{R}}$?
(ii) How to efficiently sample a random decryption hint $\tilde{\mathbf{R}}$ satisfying Eq. (1)?
(iii) Which assumptions imply the security of this decryption hint mechanism?

1.3 Our Contributions

We devise solutions to all three questions posed above, with which we arrive at a new lattice-based general-purpose obfuscation construction under the BDGM framework [BDGM20]. Below we summarise our answers to the three questions, which constitute the main novelties of this work.

Equivocal Distributions. We introduce a notion called s-equivocal distributions, denoted $\mathcal{E}$, which are distributions of matrices $\mathbf{B}$ admitting the following:

(1a) Dense Sublattice. With overwhelming probability over the choice of $\mathbf{B} \leftarrow\$ \mathcal{E}$, the lattice $\Lambda_q(\mathbf{B})$ admits a dense sublattice. Formally, we capture this by requiring that, for any vector $\mathbf{c}$, the discrete Gaussian distribution over the lattice $\Lambda_q(\mathbf{B})$ with parameter s and centre $\mathbf{c}$ has super-logarithmic min-entropy in the security parameter λ. As a consequence, an LWE sample $\mathbf{c}$ admits many solutions.

(1b) Pseudorandom with Leakage. Equivocal matrices $\mathbf{B} \leftarrow\$ \mathcal{E}$ are computationally indistinguishable from uniform even when given leakages of the form

$$x_i \cdot \tilde{\mathbf{r}}_i^{\mathsf{T}} \bmod q,$$

where x_i are uniformly random and $\tilde{\mathbf{r}}_i^{\mathsf{T}}$ are such that $\tilde{\mathbf{r}}_i^{\mathsf{T}}\mathbf{B}$ are samples from the discrete Gaussian distribution over $\Lambda_q(\mathbf{B})$ with parameter s and some known centres $\mathbf{e}_i$ of norm $\|\mathbf{e}_i\| \le s$. In particular, the joint distribution of $(\mathbf{B}, (x_i \cdot \tilde{\mathbf{r}}^{\mathsf{T}})_i)$ is indistinguishable from sampling $\mathbf{B}$, $(\mathbf{x}_i)_i$ and a matrix $\hat{\mathbf{R}}$ (independent of i) uniformly at random and outputting $(\mathbf{B}, (\mathbf{x}_i^{\mathsf{T}} \cdot \hat{\mathbf{R}})_i)$.

Naturally, the idea is to have $\mathbf{B}$ sampled from an equivocal distribution $\mathcal{E}$ in the obfuscator construction. The Dense Sublattice property guarantees that Eq. (1) admits super-polynomially many solutions $\tilde{\mathbf{R}}$, while the Pseudorandom with Leakage ensures that matrices $\mathbf{B}$ sampled from equivocal distributions are suitable for use as public keys, say in the packed dual-Regev encryption scheme.

Primal Lattice Trapdoors. Although $\mathbf{B}$ being sampled from an equivocal distribution guarantees the existence of super-polynomially many possible decryption hints $\tilde{\mathbf{R}}$, we need a way to sample them efficiently. For this, we introduce a notion called prime lattice trapdoors. While usual lattice trapdoor algorithms (e.g. [GPV08, MP12]) generate a matrix $\mathbf{B}$ together with a trapdoor td which allows to find short vectors in the kernel lattice

$$\Lambda_q^{\perp}(\mathbf{B}) = \{\mathbf{x} : \mathbf{B}\mathbf{x} = \mathbf{0} \bmod q\},$$

primal trapdoor algorithms generate an equivocal matrix $\mathbf{B}$ together with a trapdoor td which allows sampling short vectors from the primal lattice $\Lambda_q(\mathbf{B})$ (Eq. (2)) efficiently.

In more detail, a primal trapdoor scheme is a tuple of efficient algorithms ($\mathsf{pTrapGen}, \mathsf{Equivocate}$) with the following properties:

(1) The distribution $\mathcal{E} = \{\mathbf{B} \mid (\mathbf{B}, \mathsf{td}) \leftarrow \mathsf{pTrapGen}(1^\lambda)\}$ is s-equivocal.
(2) For the overwhelming majority of $(\mathbf{B}, \mathsf{td}) \leftarrow \mathsf{pTrapGen}(1^\lambda)$, any LWE secret $\mathbf{r}$, and any offset vector $\mathbf{c}$ at most distance s to $\mathbf{r}^\mathsf{T}\mathbf{B} \bmod q$, the following distributions are statistically close:

$$\{\tilde{\mathbf{r}}\mathbf{B} \bmod q \mid \tilde{\mathbf{r}} \leftarrow \mathsf{Equivocate}(\mathsf{td}, \mathbf{r}, \mathbf{c}, s)\} \approx_s \mathcal{D}_{\Lambda_q(\mathbf{B}),s,\mathbf{c}} \bmod q.$$

For $\mathbf{B}$ sampled by $\mathsf{pTrapGen}$, a random solution $\tilde{\mathbf{R}}$ of Eq. (1) can be sampled efficiently by running $\tilde{\mathbf{r}}_i \leftarrow \mathsf{Equivocate}(\mathsf{td}, \mathbf{l}_i^\mathsf{T}\mathbf{R}, \mathbf{l}_i^\mathsf{T}\mathbf{C} - \mathsf{Encode}(\mathbf{\Gamma})_i^\mathsf{T}, s)$ for each row $\mathbf{l}_i^\mathsf{T}\mathbf{R}$ of $\mathbf{LR}$ and each row $\mathbf{l}_i^\mathsf{T}\mathbf{C} - \mathsf{Encode}(\boldsymbol{\Gamma})_i^\mathsf{T}$ of $\mathbf{LC} - \mathsf{Encode}(\boldsymbol{\Gamma})$, and letting $\tilde{\mathbf{R}}$ to have the i-th row given by $\tilde{\mathbf{r}}_i^\mathsf{T}$, assuming that $\mathbf{L}$ is a low-norm matrix. By the second property of primal trapdoors, for any fixed $(\mathbf{B}, \mathbf{LC}, \boldsymbol{\Gamma})$, the distribution of the i-th row of $\tilde{\mathbf{R}}\mathbf{B} \bmod q$ induced by $\tilde{\mathbf{R}}$ is statistically close to the *public* discrete Gaussian distribution over $\Lambda_q(\mathbf{B})$ with parameter s and centred at $\mathbf{C}^\mathsf{T}\mathbf{l}_i - \mathsf{Encode}(\mathbf{\Gamma})_i$. Equivalently, the induced error term is

$$\mathbf{LC} - \tilde{\mathbf{R}}\mathbf{B} - \mathsf{Encode}(\boldsymbol{\Gamma}) \bmod q$$

which is statistically close to Gaussian over $\Lambda_q(\mathbf{B}) + \mathbf{E}^\mathsf{T}\mathbf{L}^\mathsf{T}$ with parameter s. In other words, from the decryption hint $\tilde{\mathbf{R}}$ the adversary would learn a noisy version of $\mathbf{LE}$ where the noise are independently random short vectors of $\Lambda_q(\mathbf{B})$. We construct a primal lattice trapdoor scheme and prove that it satisfies both of the above two properties. In particular, to show that $\mathbf{B}$ is pseudorandom with leakage, we rely on the NTRU assumption. We overview the construction and proofs in Section 2.2, and provide the formal details in Section 4.

Equivocal LWE Assumption. Similar to prior works, we are unable to prove the security of our obfuscator solely under standard assumptions. In addition to standard assumptions such as Learning with Errors (LWE) [Reg05, LPR10] and NTRU [SS11], we rely on a new assumption which we call Equivocal LWE.

We consider the following experiment parametrised by a distribution $\mathcal{E}$, a symmetric-key encryption scheme[3] SKE, and an encoding algorithm Encode.

1) An adversary $\mathcal{A}$ declares the following of its choice: a pair of messages $(\mathsf{msg}_0, \mathsf{msg}_1)$, a padding message $\mathsf{msg}_{\mathsf{pad}}$, and a circuit Φ.
2) Generate an SKE secret key sk. Encrypt $\mathsf{msg}_{\mathsf{pad}}$ as a padding term $\mathbf{P} \leftarrow \mathsf{SKE.Enc}(\mathsf{sk}, \mathsf{msg}_{\mathsf{pad}})$.
3) Sample an equivocal matrix $\mathbf{B} \leftarrow\$ \mathcal{E}$ and an LWE sample $\mathbf{RB} + \mathbf{E} \bmod q$. Mask $\mathsf{Encode}(\mathsf{sk})$ with the latter to yield $\mathbf{C} = \mathbf{RB} + \mathbf{E} + \mathsf{Encode}(\mathsf{sk}) \bmod q$.
4) For both $i \in \{0, 1\}$, do the following:
   (i) Encrypt msg_i as $\mathsf{ctxt}_i \leftarrow \mathsf{SKE.Enc}(\mathsf{sk}, \mathsf{msg}_i)$.
   (ii) Run the circuit[4] Φ on ctxt_i to produce a pair of matrices $(\hat{\mathbf{H}}_i, \mathbf{M}_i)$.
   (iii) Compute $\mathbf{L}_i \coloneqq \mathbf{G}^{-1}(\hat{\mathbf{H}}_i + \mathbf{1}^\mathsf{T} \otimes \mathbf{P})$, where $\mathbf{G}^{-1}(\cdot)$ is the binary decomposition operation. Assert that $\|(\mathbf{L}_i\mathbf{C} + \mathbf{M}_i) - (\mathbf{L}_i\mathbf{RB}) \bmod q\| \leq s$.
5) Assert that $\mathbf{M}_0 = \mathbf{M}_1 \bmod q$.
6) Flip a random coin $b \leftarrow\$ \{0, 1\}$. Equivocate $\mathbf{L}_b\mathbf{R}$ to obtain $\tilde{\mathbf{R}}_b$ satisfying $\tilde{\mathbf{R}}_b\mathbf{B} \approx \mathbf{L}_b\mathbf{RB} \bmod q$ with Gaussian parameter s. Set $\mathsf{hint}_b \coloneqq \tilde{\mathbf{R}}_b$.
7) Output problem instance $(\mathbf{B}, \mathbf{C}, \mathbf{P}, \mathsf{ctxt}_b, \mathsf{hint}_b)$.

The Equivocal LWE assumption states that it is infeasible to distinguish the challenge bit b from the above output. It is intended to be instantiated with an equivocal distribution $\mathcal{E}$ and SKE with pseudorandom ciphertexts.

[3] By IND-\$-CPA we mean that ciphertexts appear pseudorandomness under the chosen plaintext attack. To handle syntactical issues, in the formal assumption definition we will rely on a tagged version of SKE, a generalisation where encryption inputs also a tag, and security requires ciphertexts under any tag to be jointly pseudorandom. We omit this detail in this overview.

[4] In our application of obfuscation, SKE is intended to be a symmetric-key FHE and Φ represents the homomorphic evaluation of a function on a ciphertext. In the context of the Equivocal LWE assumption, however, the security experiment is well-defined even if SKE is not homomorphic and Φ can be any function.

The above assumption is designed to model the BDGM framework, but abstracts away complications like the choices of SKE and Encode, which we believe are irrelevant to its conjectured hardness. If the distribution $\mathcal{E}$ were instantiated as uniform, this would imply $\tilde{\mathbf{R}} = \mathbf{L}\mathbf{R} \bmod q$ w.h.p., and by further specifying appropriate SKE and Encode, the assumption would collapse to the BDGM scheme with the naive insecure decryption hint sketched in Sections 1.1 and 1.2 (in which case the assumption would be false). The Equivocal LWE assumption only differs from this trivial case in allowing $\mathcal{E}$ to be an equivocal distribution, highlighting that the latter is the source of the assumption's hardness.

The Equivocal LWE assumption has some qualitative advantage over the assumptions used in related works for constructing obfuscators. Unlike the obfuscator construction of [BDGM20][5] it does not involve any random oracle and is independent of the circuit being obfuscated. It is also non-circular and non-interactive. Importantly, upon instantiating with an equivocal distribution $\mathcal{E}$, the Equivocal LWE assumption generates decryption hints probabilistically with independent randomness. This is in contrast to the assumptions in [GP21, DQV+21] which provide decryption hints to the adversary that are deterministically generated based on a-priori committed randomness. We further note that the assumption of [GP21] is circular and allows adaptive hint queries upon given information dependent on the hidden bit b. These features are exploited in the counterexamples of [HJL21, JLLS23].

The formal assumption definition together with further discussions are found in Section 5. In Section 1.4 below we summarise some seemingly infeasible attack strategies and potential avenues for further cryptanalysis.

1.4 Cryptanalysis Attempts

We overview some unsuccessful attempts towards invalidating the Equivocal LWE assumption which are closely tied to the design of the experiments outlined above. For more elaborated discussions we refer to Section 5.1 and Appendix C.

No-hint attacks. Of an Equivocal LWE instance $(\mathbf{B}, \mathbf{C}, \mathbf{P}, \mathsf{ctxt}_b, \mathsf{hint}_b)$, only the decryption hint $\mathsf{hint}_b = \tilde{\mathbf{R}}_b$ and the challenge ciphertext ctxt_b depend on the hidden bit b. If hint_b is withheld, then the hardness of distinguishing $(\mathbf{B}, \mathbf{C}, \mathbf{P}, \mathsf{ctxt}_b)$ is guaranteed by the pseudorandomness of $\mathbf{B} \leftarrow\!\!\$\ \mathcal{E}$, the hardness of LWE w.r.t. $\mathbf{B}$ and the IND-\$-CPA-security of SKE. In particular, this rules out attacks which attempt to distinguish between $\mathbf{L}_0$ and $\mathbf{L}_1$ (deterministic given ctxt_0 and ctxt_1, respectively) by, say, trying to tell apart their distributions, and indicates that a successful counterexample necessarily makes use of the decryption hint.

Trivial distinguishing attacks based on differing quality of hints. The admissibility check in Step (iii) ensures that in both tracks $i \in \{0,1\}$ the rows of $\mathbf{W}_i = \mathbf{L}_i\mathbf{C} + \mathbf{M}_i$ are close to the lattice $\Lambda_q(\mathbf{B})$. This prevents trivial distinguishing attacks where $\mathbf{W}_0 = \mathbf{L}_0\mathbf{C} + \mathbf{M}_0$ is close to $\Lambda_q(\mathbf{B})$ but $\mathbf{W}_1 = \mathbf{L}_1\mathbf{C} + \mathbf{M}_1$ not, which would result in $\tilde{\mathbf{R}}_0$ being a much better hint than $\tilde{\mathbf{R}}_1$, i.e. $\tilde{\mathbf{R}}_0\mathbf{B}$ much closer to $\mathbf{L}_0\mathbf{C} + \mathbf{M}_0$ than $\tilde{\mathbf{R}}_0\mathbf{B}$ is to $\mathbf{L}_1\mathbf{C} + \mathbf{M}_1$. Then, the check in Step 5) makes sure that the offsets from $\Lambda_q(\mathbf{B})$ to $\mathbf{L}_0\mathbf{C}$ and $\mathbf{L}_1\mathbf{C}$ are both roughly $\mathbf{M}_0 = \mathbf{M}_1 \bmod q$, which are potentially known to the adversary. This prevents another trivial attack of testing whether $\tilde{\mathbf{R}}_b\mathbf{B} - \mathbf{L}_b\mathbf{C} \bmod q$ is close to $\mathbf{M}_0$ or $\mathbf{M}_1$.

"Zero-ising" attacks. A devastating class of attacks against many iO constructions is fuzzily known as "zero-ising attacks" [CHL+15, MSZ16, CVW18, HJL21]. We explore attack strategies against Equivocal LWE inspired by these "zero-ising attacks".

Recall that $(\tilde{\mathbf{R}}_b, \mathbf{L}_b, \mathbf{B}, \mathbf{C})$, where $\mathbf{L}_b$ is derived from $(\mathbf{P}, \mathsf{ctxt}_b)$, satisfies:

$$\begin{pmatrix} \mathbf{C} \\ \tilde{\mathbf{R}}_b\mathbf{B} \end{pmatrix} = \begin{pmatrix} \mathbf{I} \\ \mathbf{L}_b \end{pmatrix} \mathbf{R}\mathbf{B} + \begin{pmatrix} \mathbf{E} \\ \tilde{\mathbf{E}}_b \end{pmatrix} + \begin{pmatrix} \mathsf{Encode}(\mathsf{sk}) \\ \mathbf{0} \end{pmatrix} \bmod q \tag{3}$$

[5] [BDGM20] did not provide an assumption from which their construction can be based on, hence we can only compare directly with their construction.

where $\mathbf{E}, \tilde{\mathbf{E}}_b$ are short error terms.

Since (the marginal distribution of) $\mathbf{B}$ is pseudorandom by definition of equivocal distributions, under the hardness of standard SIS, it is unlikely to distinguish $\mathbf{B}$ from random by finding short vectors $\mathbf{u}$ with $\mathbf{B}\mathbf{u} = \mathbf{0} \bmod q$. Note that here we use a heuristic that the hint hint_b does not makes solving SIS w.r.t. $\mathbf{B}$ easier. On the other hand, if $\mathbf{u}$ is not short then heuristically $\begin{pmatrix} \mathbf{E} \\ \tilde{\mathbf{E}}_b \end{pmatrix} \mathbf{u} \bmod q$ would destroy all utility of the above relation. Overall, "zero-ising Eq. (3) from the right" seems difficult to execute.

The trickier case is "zero-ising from the left", that is, attempting to find (not necessarily short) vectors $(\mathbf{x}, \mathbf{t})$ such that

$$(\mathbf{t}^{\mathsf{T}}, -\mathbf{x}^{\mathsf{T}}) \begin{pmatrix} \mathbf{I} \\ \mathbf{L}_b \end{pmatrix} = \mathbf{0}^{\mathsf{T}} \bmod q \tag{4}$$

and left-multiplying $(\mathbf{t}^{\mathsf{T}}, -\mathbf{x}^{\mathsf{T}})$ to Eq. (3) to obtain a leakage vector $\mathbf{z}$ satisfying

$$\mathbf{z}^{\mathsf{T}} = (\mathbf{t}^{\mathsf{T}}, -\mathbf{x}^{\mathsf{T}}) \begin{pmatrix} \mathbf{C} \\ \tilde{\mathbf{R}}_b \mathbf{B} \end{pmatrix} = \mathbf{t}^{\mathsf{T}} \mathsf{Encode}(\mathsf{sk}) + \mathbf{t}^{\mathsf{T}} \mathbf{E} - \mathbf{x}^{\mathsf{T}} \tilde{\mathbf{E}}_b \bmod q, \tag{5}$$

with the goal of obtaining non-trivial information about various secrets, e.g. (the hidden dense sublattice of) $\Lambda_q(\mathbf{B})$, the LWE error $\mathbf{E}$ and the SKE secret key sk. Depending on the "quality" of $(\mathbf{x}, \mathbf{t})$, the distinguisher could potentially gain different information.

Below we distinguish three cases and suggest for each that either it is infeasible or does not seem to reveal useful information.

Case 1, $\mathbf{x} \neq \mathbf{0}$ short and $\mathbf{t} = \mathbf{0}$. This would be a potentially devastating attack since $\mathbf{z}^{\mathsf{T}} = \mathbf{x}^{\mathsf{T}} \tilde{\mathbf{E}}_b \bmod q$ is a short vector in $\Lambda_q(\mathbf{B})$. Fortunately, this attack amounts to solving SIS w.r.t. $\mathbf{L}_b^{\mathsf{T}}$, which we can show to be hard using the heuristic that the knowledge of hint_b does not make solving SIS w.r.t. $\mathbf{L}_b^{\mathsf{T}}$ easier. More precisely, we consider an adversary which is not given hint_b, and argue that the distribution of $\mathbf{P}$ is pseudorandom in the view of the adversary, by invoking the pseudorandomness of the equivocal matrix $\mathbf{B}$, the pseudorandomness of $\mathbf{C}$ due to LWE, and the IND-\$-CPA security of SKE. Then, we can program $\mathbf{P}$ to embed a SIS instance, such that if the adversary solves SIS w.r.t. $\mathbf{L}_b^{\mathsf{T}}$ then we can solve the given SIS instance. Furthermore, we note that the same argument can be adapted to argue about the case where the adversary solves ISIS w.r.t. $\mathbf{L}_b^{\mathsf{T}}$ and any fixed target vector.

Case 2, both $\mathbf{x}$ and $\mathbf{t}$ are short and $\mathbf{t} \neq \mathbf{0}$. Since $(\mathbf{x}, \mathbf{t})$ is short, from Eq. (4), $\mathbf{L}_b^{\mathsf{T}} \mathbf{x} = \mathbf{t}$ without modular reduction. We can therefore rewrite $(\mathbf{t}^{\mathsf{T}}, -\mathbf{x}^{\mathsf{T}}) = \mathbf{x}^{\mathsf{T}}(\mathbf{L}_b | -\mathbf{I})$. In other words, what a distinguisher can achieve by left-multiplying $(\mathbf{t}^{\mathsf{T}}, -\mathbf{x}^{\mathsf{T}})$ to Eq. (3) is no more than left-multiplying $(\mathbf{L}_b | -\mathbf{I})$ to Eq. (3), from which it learns a matrix $\mathbf{Z}_b$ equal to

$$\begin{aligned} \mathbf{Z}_b = (\mathbf{L}_b | -\mathbf{I}) \begin{pmatrix} \mathbf{C} \\ \tilde{\mathbf{R}}_b \mathbf{B} \end{pmatrix} + \mathbf{M}_b &= \mathbf{L}_b \mathbf{E} - \tilde{\mathbf{E}}_b + \mathbf{L}_b \cdot \mathsf{Encode}(\mathsf{sk}) + \mathbf{M}_b \bmod q \\ &= -\tilde{\mathbf{E}}_b + \mathbf{L}_b \cdot (\mathbf{E} + \mathsf{Encode}(\mathsf{sk})) + \mathbf{M}'_b \end{aligned} \tag{6}$$

for some $\mathbf{M}'_b$ such that $\mathbf{L}_b \cdot (\mathbf{E} + \mathsf{Encode}(\mathsf{sk})) + \mathbf{M}'_b$ is the shortest representative of $\mathbf{L}_b \cdot \mathsf{Encode}(\mathsf{sk}) + \mathbf{M}_b$ modulo q. Note that $\mathbf{L}_b \cdot \mathsf{Encode}(\mathsf{sk}) + \mathbf{M}'_b$ is guaranteed to be short relative to q by the admissibility check in Step (iii) of the experiment.

Let us examine the expression of $\mathbf{Z}_b$ in Eq. (6):

- $\mathbf{L}_b$ is a low-norm matrix known to the distinguisher.
- $\mathbf{M}'_b$ is known to the distinguisher modulo q (because $\mathbf{M}_b = \mathbf{M}'_b \bmod q$).
- $\mathbf{E}$ is Gaussian over the entire space, unknown to the distinguisher.
- $\tilde{\mathbf{E}}_b$ has rows which are Gaussian over $\Lambda_q(\mathbf{B})$, unknown to the distinguisher.

Despite crucially that $\tilde{\mathbf{E}}_b$ has rows restricted to $\Lambda_q(\mathbf{B})$ and other differences, we may view $\mathbf{L}_b \cdot (\mathbf{E} + \mathsf{Encode}(\mathsf{sk})) - \tilde{\mathbf{E}}_b$ in Eq. (6) as a so-called "integer LWE" (ILWE) sample [BDE+18] with matrix $\mathbf{L}_b$, secret $\mathbf{E} + \mathsf{Encode}(\mathsf{sk})$, and error $\tilde{\mathbf{E}}_b$. We can verify that the least-square attack in [BDE+18] against ILWE recovers $\mathbf{E} + \mathsf{Encode}(\mathsf{sk})$ given sufficiently many samples, i.e. if the number of rows of $\mathbf{L}_b$ is large enough relative to the norm difference between $\tilde{\mathbf{E}}_b$ and $\mathbf{L}_b$. Since $\|\tilde{\mathbf{E}}_b\| \geq s$ with overwhelming probability and $\mathbf{L}_b$ is a binary matrix with $h \cdot \mathsf{poly}(\lambda)$ rows[6], to avoid their attack, we must set $s \geq h \cdot \mathsf{poly}(\lambda)$ for a sufficiently large $\mathsf{poly}(\lambda)$ factor.

We note that, if the ILWE error $\tilde{\mathbf{E}}_b$ were sampled from a Gaussian distribution over the entire space, then the least-square attack in [BDE+18] is almost optimal. This means, given that we set $s \geq h \cdot \mathsf{poly}(\lambda)$, a successful attack must somehow exploit the structure of $\tilde{\mathbf{E}}_b$ that each row is Gaussian over the lattice $\Lambda_q(\mathbf{B})$. However, we caution that a too large s, being the Gaussian width of $\tilde{\mathbf{E}}_b$ whose rows are sampled from $\Lambda_q(\mathbf{B})$, may leak too much information about the "shape" of the hidden sublattice of $\Lambda_q(\mathbf{B})$, although we are not able to find strong evidence for or against this intuition.

To conclude, we are unable to exploit Eq. (6) to invalidate the Equivocal LWE assumption, although we also cannot provably rule out its usefulness. Remarkably, Eq. (6) also shares much similarity with the hint in the assumption of [DQV+21], with the crucial difference that, our leakage term Eq. (6) is masked by the fresh equivocated error $\tilde{\mathbf{E}}_b$, whereas that of [DQV+21] is masked by an error that is parallelly spent in other LWE samples. The attack of [JLLS23] against [DQV+21] recovered their masking by relying on it being reused. This may be viewed as another evidence that our hint generation, which results in new independent randomness, lands our assumption into a regime where known attack strategies do not apply. We leave Eq. (6) as a central target for further cryptanalysis, and refer to Section 5 for more comparison with [DQV+21].

Case 3, $\mathbf{x}$ *or* $\mathbf{t}$ *is long.* Clearly such $(\mathbf{x}, \mathbf{t})$ is easy to find by linear algebra. In this case, intuitively, the term $\mathbf{t}^\mathsf{T}\mathbf{E} - \mathbf{x}^\mathsf{T}\tilde{\mathbf{E}}_b \bmod q$ in Eq. (5) should destroy all utility of the leakage $\mathbf{z}$ due to modular reduction triggered by the long $\mathbf{x}$ or $\mathbf{t}$. Although, again, we are unable to formally argue this, we could reason about the lack of utility by some heuristic arguments.

As we have already showed in Case 1, solving ISIS w.r.t. $\mathbf{L}_b^\mathsf{T}$ against any fixed target without using the hint hint_b is hard assuming standard SIS. Let us again use the heuristic that the knowledge of hint_b does not make solving ISIS easier. In this circumstance, two natural ways of finding $(\mathbf{x}, \mathbf{t})$ with $\mathbf{t} = \mathbf{L}_b^\mathsf{T}\mathbf{x} \bmod q$ are 1) fix $\mathbf{x}$, then compute $\mathbf{t}$, or 2) fix $\mathbf{t}$, then solve for $\mathbf{x}$ (or a mixture of the two). In the following, we heuristically treat values not directly controllable by the adversary as uniformly random. For the first strategy, treating $\mathbf{t}$ as uniformly random, $\mathbf{t}^\mathsf{T}\mathbf{E} \bmod q$ is indistinguishable from uniform by either the leftover hash lemma or the LWE assumption with appropriate parameters. For the second strategy, treating $\mathbf{x}$ as uniformly random and writting $\tilde{\mathbf{R}}_b = \mathbf{L}\mathbf{R} + \mathbf{R}_b' \bmod q$, the Pseudorandom with Leakage property of the equivocal distribution $\mathcal{E}$ guarantees that $\mathbf{B}$ and $-\mathbf{x}^\mathsf{T}\mathbf{R}_b'$ are indistinguishable from uniformly random. Therefore, the term $\mathbf{t}^\mathsf{T}(\mathbf{E}+\mathsf{Encode}(\mathsf{sk}))$ in $\mathbf{z}^\mathsf{T} = \mathbf{t}^\mathsf{T}\mathbf{C} - \mathbf{x}^\mathsf{T}\tilde{\mathbf{R}}_b\mathbf{B} = \mathbf{t}^\mathsf{T}(\mathbf{E}+\mathsf{Encode}(\mathsf{sk})) - \mathbf{x}^\mathsf{T}\mathbf{R}_b'\mathbf{B} \bmod q$ is masked by a uniformly random vector in $\Lambda_q(\mathbf{B})$. In either case, we conclude that $\mathbf{z}$ does not seem to reveal any useful information.

2 Technical Overview

The main technical innovations of this work are (i) the construction of primal lattice trapdoors from the NTRU assumption, and (ii) techniques of instantiating the BDGM construction with equivocal matrices and proving security under the Equivocal LWE assumption. Notably, we took care to ensure that the construction can be instantiated with parameters for which the Equivocal LWE and other assumptions are plausible, and the correctness constraints are satisfied, while keeping the modulus size polynomial. We will dedicate this section to explain these technicalities.

[6] In the context of our XiO, h is the number of rows of the truth table matrix $\boldsymbol{\Gamma}$.

Our results are based on the LWE, NTRU and the new Equivocal LWE assumptions stated over a number ring $\mathcal{R}$.[7] We recall some algebraic number theory background in Section 2.1, then overview items (i) and (ii) highlighted above in Sections 2.2 and 2.3 respectively.

2.1 Cyclotomic Rings, Module Lattices, and Discrete Gaussians

Throughout this work, we will consider $\mathcal{R} = \mathbb{Z}[\zeta]$, the ring of integers of the $\mathfrak{f}$-th cyclotomic field $\mathcal{K}$ of degree $\varphi = \varphi(\mathfrak{f})$ and its quotient ring $\mathcal{R}_q \coloneqq \mathcal{R}/q\mathcal{R}$, where $q \in \mathbb{N}$. We denote by $\mathcal{R}^\times$ and $\mathcal{R}_q^\times$ the groups of units in the respective rings. Through the canonical embedding of $\mathcal{K}$ into $\mathbb{R}^\varphi$ (often represented as an isomorphic subspace of $\mathbb{C}^\varphi$), we can equip vectors over $\mathcal{R}$ with a (geometric) norm $\|\cdot\|$, and view $\mathcal{R}$-modules modules as lattices, e.g. the lattice $\Lambda_q(\mathbf{B})$ considered earlier, for $\mathbf{B} \in \mathcal{R}^{t \times k}$.

The Gaussian function $\rho_{\mathbf{s},\mathbf{c}}$ with parameter $\mathbf{s} = (s_1, \ldots, s_n) \in (\mathbb{R}^+)^{\varphi n}$ and centre $\mathbf{c} \in \mathcal{K}^n$ is defined as

$$\rho_{\mathbf{s},\mathbf{c}}(\mathbf{x}) \coloneqq \exp(-\pi \|\mathbf{x} - \mathbf{c}\|_{\mathbf{s}}^2) \quad \forall \mathbf{x} \in \mathcal{K}^n,$$

where $\|\mathbf{x} - \mathbf{c}\|_{\mathbf{s}}$ is, roughly, the norm of $\mathbf{x} - \mathbf{c}$ but with the contribution of the i-th coefficient of its canonical embedding scaled by $1/s_i^2$. For a lattice $\mathcal{L}$, the discrete Gaussian distribution over $\mathcal{L}$ with parameter $\mathbf{s}$ and centre $\mathbf{c}$, denoted by $\mathcal{D}_{\mathcal{L},\mathbf{s},\mathbf{c}}$, is then given by normalising $\rho_{\mathbf{s},\mathbf{c}}$ over all lattice points in $\mathcal{L}$. When all entries of $\mathbf{s}$ equals some s, i.e. the Gaussian distribution is spherical, we replace the subscript $\mathbf{s}$ by s, and omit it when $s = 1$. Likewise, when the centre is $\mathbf{c} = \mathbf{0}$, we omit the subscript $\mathbf{c}$. The discrete Gaussian distribution $\mathcal{D}_{\mathcal{L}+\mathbf{c},\mathbf{s}}$ over the coset $\mathcal{L} + \mathbf{c}$, with parameter $\mathbf{s}$ and centre $\mathbf{0}$, is defined similarly by normalising $\rho_{\mathcal{L}+\mathbf{c},\mathbf{s}}$ over all coset vectors in $\mathcal{L} + \mathbf{c}$. The identity $\mathcal{D}_{\mathcal{L}+\mathbf{c},\mathbf{s}} = \mathcal{D}_{\mathcal{L},\mathbf{s},-\mathbf{c}} + \mathbf{c}$ holds.

The span of a lattice $\mathcal{L}$, denoted $\mathsf{Span}(\mathcal{L})$, is the vector subspace obtained by taking all linear combinations of the basis vectors of $\mathcal{L}$ with coefficients taken from $\mathbb{R}$. In this work, somewhat unusually, we will need to consider discrete Gaussian distributions $\mathcal{D}_{\mathcal{L},\mathbf{s},\mathbf{c}}$ where the centre $\mathbf{c}$ is *outside* $\mathsf{Span}(\mathcal{L})$. Such distributions have a property that $\rho_{\mathbf{s},\mathbf{c}}(\mathbf{x})$ is strictly less than 1 for any $\mathbf{x} \in \mathsf{Span}(\mathcal{L})$. A visual example is given in Fig. 1.

2.2 Primal Lattice Trapdoors from NTRU

We propose a construction of primal lattice trapdoors ($\mathsf{pTrapGen}, \mathsf{Equivocate}$) from the NTRU assumption. We refer to Section 1.3 for an introduction to primal trapdoors and Definition 8 for a formal definition.

To recall, the (extended) NTRU assumption states that $\mathbf{b} = \mathbf{f}/d \bmod q$ where $\mathbf{f}$ and d are short Gaussians (subject to some mild constraints) is indistinguishable from a uniformly random vector $\mathbf{b}$. In this work, we rely on a weaker version which does not require the denominator d to be short.

Looking ahead, our primal trapdoor and obfuscator constructions will rely on the NTRU assumption with a rather large (but still polynomial) modulus which likely falls into the so-called "overstretched" regime of NTRU where the problem is easier than otherwise. To not break the flow, we will discuss this issue in Remark 5 in Appendix A.4.

Construction. In our construction, we let $\mathsf{pTrapGen}$ generate

$$\mathbf{B} \quad \text{uniform subject to} \quad \mathbf{d}^\mathsf{T}\mathbf{B} = \mathbf{f}^\mathsf{T} \bmod q \qquad \text{and} \qquad \mathsf{td} \coloneqq (\mathbf{B}, \mathbf{f}, \mathbf{d})$$

where $\mathbf{d} \in \mathcal{R}_q^t$ is uniform random over $\mathcal{R}_q$ and $\mathbf{f} \in \mathcal{R}^k$ is a short Gaussian vector.[8] Then, on input $(\mathsf{td}, \mathbf{r}, \mathbf{c}, s)$, $\mathsf{Equivocate}$ simply outputs

$$\tilde{\mathbf{r}} = \mathbf{r} + p \cdot \mathbf{d} \bmod q$$

[7] A number of results in this work can also be instantiated over the integers, i.e. $\mathcal{R} = \mathbb{Z}$, if we replace NTRU with its less well-known matrix variant and scale the parameters of all assumptions appropriately. However, we are unable to prove several critical results, Lemmas 13, 14 and 19 and Theorem 9, for hidden dense sublattices with module rank > 1 (cf. Remarks 6 and 8). For this reason, we will by default consider the setting where $\mathcal{R}$ is a number ring of sufficiently high degree.

[8] For technical reasons, more precisely $\mathbf{d}$ is uniformly random subject to being primitive, i.e. its entries generate $\mathcal{R}_q$, and $\mathbf{f}$ is Gaussian subject to being coprime, i.e. its entries generate $\mathcal{R}$. These are satisfied with high probability for appropriate parameters.

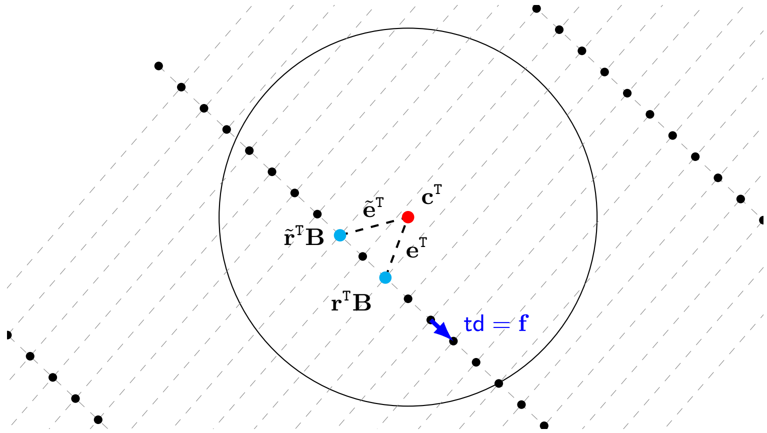

Fig. 1. Visualisation of the lattice $\Lambda_q(\mathbf{B})$ generated by an equivocal matrix $\mathbf{B}$. Black dots are lattice points of $\Lambda_q(\mathbf{B})$, in which some dimension(s) consist of short vectors. For an LWE sample $\mathbf{c}^\intercal = \mathbf{r}^\intercal\mathbf{B} + \mathbf{e}^\intercal$, its underlying LWE secret-error-tuple is not unique: Any lattice point in a ball of short radius centred at $\mathbf{c}$ (black circle) represents a possible LWE secret, e.g. we have $\mathbf{c}^\intercal = \tilde{\mathbf{r}}^\intercal\mathbf{B} + \tilde{\mathbf{e}}^\intercal$. In our primal trapdoor construction, a trapdoor td of $\mathbf{B}$ is a short vector $\mathbf{f}$ in the lattice $\Lambda_q(\mathbf{B})$ using which we can sample equivocated LWE secrets, e.g. $\tilde{\mathbf{r}}$ where $\tilde{\mathbf{r}}^\intercal\mathbf{B} = \mathbf{r}^\intercal\mathbf{B} - 2\mathbf{f}^\intercal$ as depicted. The densely packed lattice points on each hyperplane (displayed as a line) parallel to $\mathbf{f}$ constitute a coset of the lattice $\mathcal{L}(\mathbf{f}^\intercal)$ generated by $\mathbf{f}$. Note that the error $\mathbf{e}$ of $\mathbf{c}$ does not need to lie in $\mathcal{L}(\mathbf{f}^\intercal)$.

where $p \leftarrow\!\$\ \mathcal{D}_{\mathcal{R},\mathbf{s},c}$ is a short perturbation element distributed according to a Gaussian with parameter $\mathbf{s}$ and centre c. The parameter $\mathbf{s}$ is a skewed version of s with skewness defined by $\mathbf{f}$, and the centre c is defined by both $\mathbf{f}$ and the original LWE error $\mathbf{e}$ (computable given $\mathbf{B}, \mathbf{r}, \mathbf{c}$). The formal construction, together with a lemma on the sampleability of such $\mathbf{B}$, is provided in Section 4.2.

Security. The main challenge is to show that the above construction is indeed a secure primal trapdoor scheme. Recalling from Section 1.3, we want to show:

(1) The distribution of $\mathbf{B}$ sampled by $\mathsf{pTrapGen}$ is indeed equivocal, that is,
  (1a) with overwhelming probability over $\mathsf{pTrapGen}$, the lattice $\Lambda_q(\mathbf{B})$ of the sampled $\mathbf{B}$ admits a dense sublattice, i.e. for any vector $\mathbf{c} \in \mathcal{R}^k$, the distribution $\mathcal{D}_{\Lambda_q(\mathbf{B}),s,\mathbf{c}}$ has super-logarithmic min-entropy, and
  (1b) the matrix $\mathbf{B}$ is pseudorandom with leakage.
(2) The distribution of $\tilde{\mathbf{r}}^\intercal\mathbf{B} \bmod q$ induced by $\mathsf{Equivocate}$ is statistically close to the Gaussian distribution $\mathcal{D}_{\Lambda_q(\mathbf{B}),s,\mathbf{c}}$ modulo q.

Among these, we note that Properties (1a) and (2) are statistical, while (1b) is computational. For our construction, (1a) is quite intuitive, since the lattice $\Lambda_q(\mathbf{B})$ contains a lot of exceptionally short vectors by construction, i.e. all small $\mathcal{R}$-multiple of $\mathbf{f}$, and neighbouring lattice points close to each other constitute different LWE solutions to the same LWE sample. See Fig. 1 for an illustration.

To argue the above listed properties, we rely on a core lemma which states that, for appropriate s, any $\mathbf{c}$ that is an offset of some lattice point on $\Lambda_q(\mathbf{B})$ by a short vector $\mathbf{e}$, i.e. $\mathbf{c} \in \Lambda_q(\mathbf{B}) + \mathbf{e}$:

> The Gaussian distribution $\mathcal{D}_{\Lambda_q(\mathbf{B})+\mathbf{c},s}$ over the ambient lattice coset $\Lambda_q(\mathbf{B}) + \mathbf{c}$ is statistically close to the Gaussian distribution $\mathcal{D}_{\mathcal{L}(\mathbf{f}^\intercal)+\mathbf{e},s}$ over the secret dense sublattice coset $\mathcal{L}(\mathbf{f}^\intercal) + \mathbf{e}$.

Referring to Fig. 1, this is equivalent to saying that, Gaussian-sampling a lattice point over the whole lattice centred at $\mathbf{c}$ (black circle) is close to sampling only over the dense diagonal line close to $\mathbf{c}$. The intuition is that all other lattice points in $\Lambda_q(\mathbf{B})$ are simply too far away from $\mathbf{c}$ and are unlikely to be sampled.

Assuming the above core lemma for a while, we overview how to prove the desired properties (1a), (1b) and (2) of our construction. These are formalised in Appendix B.3. Below, all claims hold with overwhelming probability over the choice of $\mathbf{B}$ whenever relevant.

(1a) Dense Sublattice. Write $\Lambda = \Lambda_q(\mathbf{B})$, $\mathcal{L} = \mathcal{L}(\mathbf{f}^\mathsf{T})$ and $\mathbb{L} = \mathsf{Span}(\mathcal{L})$. First, we show that for any Gaussian parameter s and centre $\mathbf{c}$, the entropy of $\mathcal{D}_{\Lambda,s,\mathbf{c}}$ is at least that of $\mathcal{D}_{\mathcal{L},s,\mathbf{c}_\Lambda}$, where $\mathbf{c}_\Lambda$ is the shortest representative of $\mathbf{c}$ modulo $\Lambda_q(\mathbf{B})$. It thus suffices to lower-bound the entropy of $\mathcal{D}_{\mathcal{L},s,\mathbf{c}_\Lambda}$. Here, the possibility of the Gaussian centre $\mathbf{c}_\Lambda$ not being in $\mathbb{L}$ requires our attention. We show that for any centre $\mathbf{c}_\Lambda$, possibly $\notin \mathbb{L}$, the Gaussian distributions $\mathcal{D}_{\mathcal{L},s,\mathbf{c}_\Lambda}$ and $\mathcal{D}_{\mathcal{L},s,\mathsf{proj}_{\mathbb{L}}(\mathbf{c}_\Lambda)}$ are identical, where $\mathsf{proj}_{\mathbb{L}}(\mathbf{c}_\Lambda)$ is the projection of $\mathbf{c}_\Lambda$ onto $\mathbb{L}$. This allows us to consider the entropy of the latter where the centre $\mathsf{proj}_{\mathbb{L}}(\mathbf{c}_\Lambda)$ is always in $\mathbb{L}$, and which can be bounded by existing lemmas.

(1b) Pseudorandom with Leakage. To recall, we want to show that $\mathbf{B}$ is pseudorandom even when given leakages of the form $x_i \cdot \tilde{\mathbf{r}}_i \bmod q$, where $\tilde{\mathbf{r}}_i^\mathsf{T}\mathbf{B} \leftarrow\!\$\ \mathcal{D}_{\Lambda_q(\mathbf{B}),s,\mathbf{e}_i}$ for known $\mathbf{e}_i$ with $\|\mathbf{e}_i\| \leq s$. We sketch the simpler case of $t = 1$, so that $\mathbf{B} = \mathbf{b}^\mathsf{T}$ consists of only a single row and $\tilde{\mathbf{r}}_i = \tilde{r}_i \in \mathcal{R}_q$, $\mathbf{d} = d \in \mathcal{R}_q^\times$ are single elements. Using the core lemma above, we can swap $\tilde{r}_i\mathbf{b}^\mathsf{T}$ to some $w_i\mathbf{f}^\mathsf{T}$ sampled from $\mathcal{D}_{\mathcal{L}(\mathbf{f}^\mathsf{T}),s,\mathbf{e}_i}$. Moreover, we prove that such coefficient w_i is invertible over $\mathcal{R}_q$ with high probability. Then, by construction of $\mathbf{b}^\mathsf{T}$, we obtain $\tilde{r}_i = w_i \cdot d \bmod q$. Since random $\mathcal{R}_q$ multiples of $p \cdot d \bmod q$ are again random $\mathcal{R}_q$ elements, the leakage $x_i \cdot \tilde{r}_i \bmod q$ is identically distributed as x_i itself, and we successfully decouple the leakage from $\mathbf{b}^\mathsf{T}$. Now directly from the NTRU assumption, $\mathbf{b}$ is pseudorandom.[9] In the case where $\mathbf{B}$ has $t > 1$ rows, the analysis is more involved but can again be based on the NTRU assumption.

We remark that it seems difficult to construct a primal lattice trapdoor satisfying the Pseudorandom with Leakage property based instead on the LWE assumption. In brief, the LWE secret is likely to be leaked through the leakages. We refer to Remark 7 for a discussion on a natural LWE-based construction strategy and why it fails.

(2) Induced Distribution of $\tilde{\mathbf{r}}^\mathsf{T}\mathbf{B} \bmod q$. We show that the distribution of the equivocated error, i.e. $\mathbf{c}^\mathsf{T} - \tilde{\mathbf{r}}^\mathsf{T}\mathbf{B} \bmod q$, induced by the output of $\mathsf{Equivocate}$ is precisely $\mathcal{D}_{\mathcal{L}(\mathbf{f}^\mathsf{T})+\mathbf{e},s}$ in the core lemma above. This implies the desired property of $\tilde{\mathbf{r}}\mathbf{B} \bmod q \approx_s \mathbf{c}^\mathsf{T} - \mathcal{D}_{\Lambda_q(\mathbf{B})+\mathbf{c},s} \bmod q = \mathcal{D}_{\Lambda_q(\mathbf{B}),s,\mathbf{c}} \bmod q$. To recall, an equivocated LWE secret takes the form $\tilde{\mathbf{r}} = \mathbf{r} + p\mathbf{d} \bmod q$, hence $\mathbf{c}^\mathsf{T} - \tilde{\mathbf{r}}^\mathsf{T}\mathbf{B} = \mathbf{e}^\mathsf{T} - p\mathbf{f}^\mathsf{T} \bmod q$, where $p \leftarrow\!\$\ \mathcal{D}_{\mathcal{R},\mathbf{s},c}$. The claim is shown by checking that both the skewness of Gaussian parameter $\mathbf{s}$ introduced by $\mathbf{f}$ and the offset c are precisely cancelled out when multiplying p to $\mathbf{f}$, so that the quantity $\mathbf{e} - p\mathbf{f}$ is distributed precisely according to $\mathcal{D}_{\mathcal{L}(\mathbf{f}^\mathsf{T})+\mathbf{e},s}$.

Finally, we summarise the high-level proof steps for the above useful core lemma as a chain of technical lemmas collected in Appendix B.2.

Step 1. Characterising short vectors. In Lemma 14, we show that all short vectors in $\Lambda_q(\mathbf{B})$ are also contained in the hidden sub-lattice $\mathcal{L}(\mathbf{f}^\mathsf{T}) = \mathbf{f} \cdot \mathcal{R}$. Note that the difference between $\Lambda_q(\mathbf{B})$ and $\mathcal{L}(\mathbf{f}^\mathsf{T})$ is that $\Lambda_q(\mathbf{B})$ consists of all vectors which are congruent to some $\mathcal{R}$-multiple of $\mathbf{f}$ modulo q, while $\mathcal{L}(\mathbf{f}^\mathsf{T})$ only contains those which are exact $\mathcal{R}$-multiples of $\mathbf{f}$. This claim reflects the intuition that, for a vector to be contained in $\Lambda_q(\mathbf{B})$ but not $\mathcal{L}(\mathbf{f}^\mathsf{T})$, it has to be congruent to a large $\mathcal{R}$-multiple of $\mathbf{f}$ modulo q to trigger the modular reduction, and reducing a large $\mathcal{R}$-multiple of $\mathbf{f}$ (which would be long) modulo q would not make it much shorter, because the coefficients would evenly spread modulo q.

Step 2. Closeness of Gaussian masses. In Lemma 15, we show that for a suitable Gaussian parameter s, which is greater than the smoothing parameter[10] of $\mathcal{L}(\mathbf{f}^\mathsf{T})$ and yet significantly smaller than q, and a sufficiently short vector $\mathbf{e}$ not necessarily in $\mathsf{Span}(\mathcal{L}(\mathbf{f}^\mathsf{T}))$, the Gaussian masses $\rho_{s,\mathbf{e}}(\Lambda_q(\mathbf{B}))$ and $\rho_{s,\mathbf{e}}(\mathcal{L}(\mathbf{f}^\mathsf{T}))$ are negligibly close. This result is a variant of a similar classic result which requires $\mathbf{e} \in \mathsf{Span}(\mathcal{L})$, yet a proof in our case requires much more care since $\mathbf{e} \notin \mathsf{Span}(\mathcal{L})$ forbids the direct use of most of the existing lemmas.

[9] In the context of NTRU encryption, the element d needs to be short and acts as the decryption key. In our setting, however, we can afford to have uniformly random d subject to it being invertible over $\mathcal{R}_q$.

[10] The smoothing parameter [MR04] of a lattice specifies the amount of Gaussian noise needed to "smooth out" the discreteness of the lattice.

For brevity, write $\Lambda = \Lambda_q(\mathbf{B})$, $\mathcal{L} = \mathcal{L}(\mathbf{f}^{\mathrm{T}})$ and $\mathbb{L} = \mathsf{Span}(\mathcal{L})$. The core idea of the proof of Lemma 15 is to decompose $\mathbf{e} = \mathbf{e}_{\mathbb{L}} + \mathbf{e}_{\mathbb{P}}$, where $\mathbf{e}_{\mathbb{L}} = \mathsf{proj}_{\mathbb{L}}(\mathbf{e})$ and $\mathbf{e}_{\mathbb{P}}$ is orthogonal to $\mathbf{e}_{\mathbb{L}}$, so that we can use classic techniques to handle the $\mathbf{e}_{\mathbb{L}}$ part, while dealing with the $\mathbf{e}_{\mathbb{P}}$ part using the fact that it is short.

In more detail, applying this idea on $\rho_{s,\mathbf{e}}(\mathcal{L})$, we can easily express

$$\rho_{s,\mathbf{e}}(\mathcal{L}) = \rho_{s,\mathbf{e}_{\mathbb{L}}}(\mathcal{L}) \cdot \rho_s(\mathbf{e}_{\mathbb{P}}).$$

We next turn to $\rho_{s,\mathbf{e}}(\Lambda)$. First, we split the mass over the lattice Λ into a sum of masses over each coset of $\mathcal{L}$, i.e.

$$\rho_{s,\mathbf{e}}(\Lambda) = \textstyle\sum_{\mathbf{k} \in \Lambda/\mathcal{L}} \rho_{s,\mathbf{e}}(\mathcal{L} + \mathbf{k}).$$

Then, for each $\mathbf{k} \in \Lambda/\mathcal{L}$, we invoke the smoothing guarantee over $\mathcal{L}$ and get

$$\rho_{s,\mathbf{e}}(\mathcal{L} + \mathbf{k}) \approx \rho_{s,\mathbf{e}_{\mathbb{L}}}(\mathcal{L}) \cdot \rho_s(\mathbf{k}_{\mathbb{P}} - \mathbf{e}_{\mathbb{P}})$$

where $\mathbf{k}_{\mathbb{P}}$ is the component of $\mathbf{k}$ orthogonal to $\mathbb{L}$. Substituting the above into the sum, we obtain

$$\rho_{s,\mathbf{e}}(\Lambda) \approx \rho_{s,\mathbf{e}_{\mathbb{L}}}(\mathcal{L}) \cdot \rho_s(\mathbf{e}_{\mathbb{P}}) \cdot \left(1 + O\left(\textstyle\sum_{\mathbf{k} \in (\Lambda/\mathcal{L}) \setminus \{\mathbf{0}\}} \rho_s(\mathbf{k}_{\mathbb{P}} - \mathbf{e}_{\mathbb{P}})\right)\right).$$

It thus remains to show that the expression in the big O is negligible. To this end, we use the result from Step 1 to argue that any vector $\mathbf{k} \in (\Lambda/\mathcal{L}) \setminus \{\mathbf{0}\}$, hence not in $\mathcal{L}$, must be long. On the other hand, since $\mathbf{k}_{\mathbb{L}}$ is at most of the same length as the shortest vector in $\mathcal{L}$, the orthogonal component $\mathbf{k}_{\mathbb{P}}$ must be long. Then, since $\mathbf{e}_{\mathbb{P}}$ is short, $\mathbf{k}_{\mathbb{P}} - \mathbf{e}_{\mathbb{P}}$ must be long, and therefore $\rho_s(\mathbf{k}_{\mathbb{P}} - \mathbf{e}_{\mathbb{P}})$ is negligible. The claim then goes through by showing that $\rho_s(\mathbf{k}_{\mathbb{P}} - \mathbf{e}_{\mathbb{P}})$ is so small that, after summing over $\mathbf{k} \in (\Lambda/\mathcal{L}) \setminus \{\mathbf{0}\}$, it is still negligible.

Step 3. Closeness of Gaussian distributions. Lemma 16 is our core lemma, stating that for an s mildly large, $\mathcal{D}_{\Lambda_q(\mathbf{B})+\mathbf{c},s}$ and $\mathcal{D}_{\mathcal{L}(\mathbf{f}^{\mathrm{T}})+\mathbf{e},s}$ are statistically close. We first apply the identities

$$\mathcal{D}_{\Lambda_q(\mathbf{B})+\mathbf{c},s} = \mathcal{D}_{\Lambda_q(\mathbf{B})+\mathbf{e},s} = \mathcal{D}_{\Lambda_q(\mathbf{B}),s,-\mathbf{e}} + \mathbf{e} \qquad \text{and} \qquad \mathcal{D}_{\mathcal{L}(\mathbf{f}^{\mathrm{T}})+\mathbf{e},s} = \mathcal{D}_{\mathcal{L}(\mathbf{f}^{\mathrm{T}}),s,-\mathbf{e}} + \mathbf{e}.$$

This simplifies the task to showing $\mathcal{D}_{\Lambda_q(\mathbf{B}),s,-\mathbf{e}} \approx_s \mathcal{D}_{\mathcal{L}(\mathbf{f}^{\mathrm{T}}),s,-\mathbf{e}}$. By some routine calculations, the statistical distance between the two distributions is exactly

$$1 - \frac{\rho_{s,-\mathbf{e}}(\mathcal{L}(\mathbf{f}^{\mathrm{T}}))}{\rho_{s,-\mathbf{e}}(\Lambda_q(\mathbf{B}))}.$$

Using the bound derived in Step 2, we see that the above quantity is negligible.

2.3 Obfuscator Construction and Security Analysis

Equipped with the Equivocal LWE assumption, we sketch how we can prove our obfuscator secure. We instantiate SKE with the symmetric-key variant of the GSW encryption scheme [GSW13]. In the security experiment, the adversary $\mathcal{A}$ first declares two circuits Γ_0, Γ_1 which are functionally equivalent, i.e. they have the same truth table $\boldsymbol{\Gamma}$. The experiment passes $\mathsf{Obf}(\Gamma_b) = (\mathsf{ctxt}_b, \mathbf{B}, \mathbf{C}, \mathsf{hint}_b)$ for $b \in \{0,1\}$ to $\mathcal{A}$, where $\mathsf{ctxt}_b = \mathbf{V}_b$ corresponds to a symmetric-key GSW encryption of the circuit Γ_b under key sk, $\mathbf{B}$ is a random equivocal matrix sampled using $\mathsf{pTrapGen}$, $\mathbf{C}$ is a packed-dual-Regev encryption of sk under the public key $\mathbf{B}$, and $\mathsf{hint}_b = \tilde{\mathbf{R}}_b$ is an equivocated LWE secret sampled using the primal trapdoor td which serves as a decryption hint. The adversary $\mathcal{A}$ wins if it can distinguish between $b = 0$ and $b = 1$. We argue that this does not happen, except with negligible probability, with the following sequence of steps:

(i) Let D_b denote the distribution of $\mathsf{Obf}(\Gamma_b)$. Let $\mathbf{L}_b$ be the linear transform which encodes the linear part of the GSW decryption algorithm on the ciphertext derived by homomorphically evaluating a universal circuit on $\mathbf{V}_b$.[11] This transform is to be homomorphically evaluated on the dual-Regev ciphertext $\mathbf{C}$. Let

[11] Actually, after the evaluation, we further evaluate a no-operation gate which corresponds to adding an encryption of 0. These ciphertexts of 0 are the padding term $\mathbf{P}$ in the Equivocal LWE assumption.

$\mathbf{M}_b := -\lfloor q/2 \rfloor \boldsymbol{\Gamma}$ be the negative of the encoding of the truth table. Denote by D'_b a similar distribution where the equivocated LWE secret $\tilde{\mathbf{R}}_b$ is instead sampled inefficiently according to the ideal distribution $\tilde{\mathbf{R}}\mathbf{B} \bmod q \sim \mathcal{D}_{\Lambda_q(\mathbf{B}),s,\mathbf{LC}+\mathbf{M}_b} \bmod q$, without using td. By the statistical indistinguishability property of $\mathsf{Equivocate}$, i.e. property (2) in Section 2.2, one has $D_b \approx_s D'_b$.

(ii) Invoke the Equivocal LWE assumption to conclude that $D'_0 \approx_c D'_1$

The construction, as described above, requires a modulus q exponential in the depth of the circuit $\Gamma : [h] \times [k] \to \{0,1\}$ being obfuscated. This is because, to evaluate an obfuscated circuit on input (i,j), an evaluator needs to homomorphically evaluate a universal circuit on the GSW ciphertext $\mathbf{V}$, where the universal circuit is hardcoded with the tuple (i,j). That is, the universal circuit hardwired with (i,j) maps Γ to $\Gamma(i,j)$. This results in an exponential increase in the noise magnitude of the GSW ciphertext relative to the depth d of such universal circuit. To maintain correctness, the modulus q must scale accordingly with the depth of Γ. We avoid this drawback by employing a "levelled" version of GSW: we let the obfuscator additionally sample GSW keys sk_i for $i \in [d]$ and provide, as part of the obfuscation of Γ, GSW encryptions of key sk_{i+1} under key sk_i, for $i \in [d]$. This allows us to pick a polynomial-size modulus q. Notice that no "circular loop" of secret keys is present and the IND-CPA security can still be derived assuming plain LWE only. These components can be used to run the bootstrapping circuit [GSW13] after each multiplication gate of the circuit, which ensures that the noise norm remains bounded throughout the evaluation.

We refer to Section 6 for the formal obfuscator construction.

3 Preliminaries

Due to space limit we refer the readers to Appendix A for an extended preliminaries, which include: standard cryptographic notations, algebraic number theory background and treatment on geometric norms (Appendix A.1), standard definitions on module lattices (Appendix A.2) and discrete Gaussians over them (Appendix A.3), standard lattice-based computational assumptions including LWE and a discussion on why our definition of extended-NTRU is *weaker* than the standard version (Appendix A.4), cryptographic objects such as homomorphic computation and bootstrapping (Appendix A.5), and established lemmas on each of these topics.

Below we state selected definitions that are required for understanding our formal definitions and claims in the main matter. These definitions are also restated in Appendix A for ease of referencing.

The LWE Module Lattice and Gaussians. We consider the Minkowski space $\mathbb{H} \subseteq \mathbb{C}^{\mathbb{Z}_\mathfrak{f}^\times}$ defined as $\mathbb{H} := \left\{\mathbf{x} \in \mathbb{C}^{\mathbb{Z}_\mathfrak{f}^\times} : x_i = \overline{x_{\varphi-i}} \ \forall i \in \mathbb{Z}_\mathfrak{f}^\times\right\} \cong \mathbb{R}^\varphi$. An element $a = \sum_{i=0}^{\varphi-1} a_i \zeta^i \in \mathcal{R}$ (or $\mathcal{R}_q$) is represented by its coefficient embedding $(a_i)_{i \in \mathbb{Z}_\varphi} \in \mathbb{Z}^\varphi$ (or $\mathbb{Z}_q^\varphi$), or canonical embedding $\sigma(a) := (\sigma_j(a))_{j \in \mathbb{Z}_\mathfrak{f}^\times} \in \mathbb{H}$ where σ_j runs through all φ ring homomorphisms from $\mathcal{K}$ to $\mathbb{C}$ fixing elements of $\mathbb{Q}$. For a matrix $\mathbf{B} \in \mathcal{R}_q^{t \times k}$, we define the q-ary lattice

$$\Lambda_q(\mathbf{B}) := \left\{\sigma(\mathbf{x}) \in \mathbb{H}^k : \exists \mathbf{r} \in \mathcal{R}_q^t,\ \mathbf{r}^\mathsf{T}\mathbf{B} = \mathbf{x}^\mathsf{T} \bmod q\right\}.$$

Definition 1 (Distance to Lattice). *For $\beta > 0$, a lattice $\mathcal{L}$, and a matrix $\mathbf{W}$ with the i-th row given by $\mathbf{w}_i^\mathsf{T}$, we write $\mathsf{Dist}(\mathbf{W}, \mathcal{L}) \leq \beta$ if, for each row $\mathbf{w}_i^\mathsf{T}$, it holds that $\min_{\mathbf{x} \in \mathcal{L}} \|\mathbf{x}^\mathsf{T} - \mathbf{w}_i^\mathsf{T}\| \leq \beta$.*

Definition 2 (Gaussian Function). *For a Gaussian (width) parameter $\mathbf{s} \in (\mathbb{R}^+)^{\varphi m}$, $\mathbf{x} \in \mathbb{H}^m$, centre $\mathbf{c} \in \mathbb{H}^m$, define the (elliptical) Gaussian function $\rho_{\mathbf{s},\mathbf{c}}(\mathbf{x}) := \exp(-\pi\|\mathbf{x} - \mathbf{c}\|_\mathbf{s}^2)$, where $\|\mathbf{x}\|_\mathbf{s} := \left(\overline{\mathbf{x}}^\mathsf{T}\mathsf{diag}(\mathbf{s})^{-2}\mathbf{x}\right)^{1/2} \in \mathbb{R}_{\geq 0}$. In the special case where $\mathbf{s} = \mathbf{1}_{\varphi m} \cdot s$ for some $s > 0$, where $\mathbf{1}_{\varphi m}$ denotes the all-one vector, we write $\rho_{s,\mathbf{c}}(\mathbf{x}) := \exp(-\pi\|\mathbf{x} - \mathbf{c}\|^2/s^2)$. We omit the subscripts s and $\mathbf{c}$ in cases $s = 1$ and $\mathbf{c} = \mathbf{0}$ respectively.*

Definition 3 (Smoothing Parameter [MR04]). *Let $\mathbf{s} \in (\mathbb{R}^+)^{\varphi m}$, $\mathcal{L} \subseteq \mathbb{H}^m$ be a lattice, and $\epsilon \geq 0$. We say that $\mathbf{s} \geq \eta_\epsilon(\mathcal{L})$ if $\rho_{1/\mathbf{s}}(\mathcal{L}^\vee) \leq 1 + \epsilon$, where the inverse $1/\mathbf{s}$ is taken component-wise.*

Definition 4 (Decisional (extended-)NTRU Assumption [SS11]). *Let the parameters* $\mathsf{params} = (\mathcal{R}, q, m, \chi)$ *be parametrised by* λ*. The* $\mathsf{dNTRU}_{\mathsf{params}}$ *assumption states that for any PPT adversary* $\mathcal{A}$ *it holds that*

$$\left|\Pr\left[\mathcal{A}(\mathbf{b}) = 1 \;\middle|\; \begin{array}{l} \mathbf{f} \leftarrow\!\$\; \mathcal{D}_{\mathcal{R}^m,\chi};\;\; d \leftarrow\!\$\; \mathcal{R}_q^\times \\ \mathbf{b} := \mathbf{f}/d \bmod q \end{array}\right] - \Pr\left[\mathcal{A}(\mathbf{b}) = 1 \;\middle|\; \mathbf{b} \leftarrow\!\$\; \mathcal{R}_q^m\right]\right| \le \mathsf{negl}(\lambda).$$

We note that for $\mathcal{R} = \mathbb{Z}$, the NTRU problem as stated above is probably not hard, at least for polynomial-size q. To cover the case $\mathcal{R} = \mathbb{Z}$, we could consider the matrix variant of NTRU, which is less well-known but nevertheless used in constructions (e.g. [GGH+19, BIP+22, ADDG24]) and considered in cryptanalysis (e.g. [DvW21]).

For technical reasons[12], We define the notion of a tagged symmetric-key encryption (SKE), a generalisation of SKE where encryption additionally inputs a tag. By letting the tag space to be singleton, we recover the usual SKE notion.

Definition 5 (Tagged-SKE). *A tagged symmetric-key encryption (tagged-SKE) scheme for a key space* $\mathcal{S}$*, a tag space* $\mathcal{T}$*, message spaces* $\{\mathcal{M}_{\mathsf{tag}}\}_{\mathsf{tag}\in\mathcal{T}}$ *and ciphertext spaces* $\{\mathcal{C}_{\mathsf{tag}}\}_{\mathsf{tag}\in\mathcal{T}}$ *is a tuple of PPT algorithms* $(\mathsf{Enc}, \mathsf{Dec})$*:*

- $\mathsf{ctxt} \leftarrow \mathsf{Enc}(\mathsf{sk}, \mathsf{tag}, \mathsf{msg})$*: For any tag* $\mathsf{tag} \in \mathcal{T}$*, generate a ciphertext* $\mathsf{ctxt} \in \mathcal{C}_{\mathsf{tag}}$ *encrypting message* $\mathsf{msg} \in \mathcal{M}_{\mathsf{tag}}$ *under the secret key* $\mathsf{sk} \in \mathcal{S}$*.*
- $\mathsf{msg} \leftarrow \mathsf{Dec}(\mathsf{sk}, \mathsf{tag}, \mathsf{ctxt})$*: Decrypt a ciphertext* ctxt *using a secret key* sk *and a tag* tag *to obtain a message* $\mathsf{msg} \in \mathcal{M}$ *or a failure symbol* $\perp$*.*

The IND-\$-CPA security of a tagged-SKE says that, for all queried message-tag tuples under the coupling spaces, the resulting ciphertexts are jointly indistinguishable from random samples over the corresponding ciphertext spaces. We refer to Definitions 14 and 15 for formal definitions of correctness and security.

Definition 6 (Indistinguishability Obfuscation [GGH+13]). *An indistinguishability obfuscation for a circuit family* $\mathcal{C} = \{\mathcal{C}_\lambda\}_{\lambda\in\mathbb{N}}$ *is a pair of PPT algorithms* $\mathsf{iO} = (\mathsf{Obf}, \mathsf{Eval})$ *satisfying the following properties:*

- *Correctness: For all* $\lambda \in \mathbb{N}$*, all circuit* $\Gamma \in \mathcal{C}_\lambda$*, all inputs* x *it holds that* $\mathsf{Eval}(\mathsf{Obf}(\Gamma), x) = \Gamma(x)$*.*
- *Security: For any PPT adversary* $\mathcal{A}$*, any pair of circuits* $(\Gamma_0, \Gamma_1) \in \mathcal{C}_\lambda$ *such that* $|\Gamma_0| = |\Gamma_1|$ *and* $\Gamma_0(x) = \Gamma_1(x)$ *on all inputs* x*, it holds that*

$$|\mathsf{Pr}\left[\mathcal{A}(\mathsf{Obf}(\Gamma_0)) = 1\right] - \mathsf{Pr}\left[\mathcal{A}(\mathsf{Obf}(\Gamma_1)) = 1\right]| \le \mathsf{negl}(\lambda).$$

The following theorem states that, under the assumption that the LWE problem is sub-exponentially hard, it suffices to consider circuits with a polynomial-size input domain and obfuscators that output obfuscated circuits of size slightly sublinear in size of the truth table of the circuit.

Theorem 1 ([LPST16, BDGM22]). *Assuming sub-exponentially hard LWE, if there exists a shallow sub-exponentially secure indistinguishability obfuscator for* $\mathsf{P}^{\mathsf{log}}/\mathsf{poly}$ *with non-trivial efficiency, then there exists an indistinguishability obfuscator for* P/poly *with sub-exponential security.*[13]

4 Equivocal Distribution and Primal Lattice Trapdoors

We define an equivocal distribution and a primal lattice trapdoor scheme in Section 4.1. We provide a primal lattice trapdoor construction in Section 4.2, and prove in Appendices B.2 and B.3 that it satisfies the defined desiderata.

[12] More specifically, this is to take care of syntactical issues due to the use of key-switching in our XiO construction.

[13] $\mathsf{P}^{\mathsf{log}}/\mathsf{poly}$ denotes the class of circuits with description size $\ell = \mathsf{poly}(\lambda)$ and input space size $N = \mathsf{poly}(\lambda)$. Non-trivial efficiency means that the size of the obfuscated circuit is bounded by $o(N)\mathsf{poly}(\ell, \lambda)$. The theorem poses no restriction on the runtime of the obfuscator, which can be as large as $N\mathsf{poly}(\ell, \lambda)$, except that its depth should be at most $\mathsf{poly}(\log N)$.

$\mathsf{ExpPRwL}^0_{\mathcal{E},\mathcal{A},r,(\mathbf{e}_i)_{i\in[Q]}}(1^\lambda)$	$\mathsf{ExpPRwL}^1_{\mathcal{E},\mathcal{A},r,(\mathbf{e}_i)_{i\in[Q]}}(1^\lambda)$
$\mathbf{B} \leftarrow\!\!\$\; \mathcal{E}$	$\mathbf{B} \leftarrow\!\!\$\; \mathcal{R}_q^{t\times k} : \mathbf{B}\mathcal{R}_q^k = \mathcal{R}_q^t$
for $i \in [Q]$ **do**	$\hat{\mathbf{R}} \leftarrow\!\!\$\; \mathcal{R}_q^{r\times t} : \hat{\mathbf{R}}\mathcal{R}_q^t = \mathcal{R}_q^r$
$\quad\tilde{\mathbf{r}}_i^{\mathsf{T}}\mathbf{B} \leftarrow\!\!\$\; \mathcal{D}_{\Lambda_q(\mathbf{B}),s,\mathbf{e}_i} \bmod q$	**for** $i \in [Q]$ **do**
$\quad x_i \leftarrow\!\!\$\; \mathcal{R}_q$	$\quad\mathbf{x}_i \leftarrow\!\!\$\; \mathcal{R}_q^r$
$\quad\mathbf{l}_i^{\mathsf{T}} := x_i\tilde{\mathbf{r}}_i^{\mathsf{T}} \bmod q$	$\quad\mathbf{l}_i^{\mathsf{T}} := \mathbf{x}_i^{\mathsf{T}}\hat{\mathbf{R}} \bmod q$
$b' \leftarrow \mathcal{A}(\mathbf{B}, \mathbf{l}_1, \ldots, \mathbf{l}_Q)$	$b' \leftarrow \mathcal{A}(\mathbf{B}, \mathbf{l}_1, \ldots, \mathbf{l}_Q)$
return b'	**return** b'

Fig. 2. Experiments for pseudorandomness with leakage.

4.1 Definitions

In the common LWE setting, the public matrix $\mathbf{B}$ is uniformly random and, given some LWE samples $\mathbf{c}^{\mathsf{T}} = \mathbf{r}^{\mathsf{T}}\mathbf{B} + \mathbf{e}^{\mathsf{T}} \bmod q$ where $\mathbf{e}$ is some short vector, the solution tuple $(\mathbf{r}, \mathbf{e})$ is uniquely (although supposedly not efficiently) determined by $\mathbf{c}$. We define the notion of *equivocal distribution*, where for a matrix $\mathbf{B}$ sampled from such distribution, its LWE samples $\mathbf{c}^{\mathsf{T}} = \mathbf{r}^{\mathsf{T}}\mathbf{B} + \mathbf{e}^{\mathsf{T}} \bmod q$ admit super-polynomially many solution tuples $(\mathbf{r}', \mathbf{e}')$. In lattice terminology, this means $\Lambda_q(\mathbf{B})$ contains exceptionally short vectors, so that any short coset vector $\tilde{\mathbf{e}} \in \Lambda_q(\mathbf{B}) + \mathbf{c}$ is also a possible error of $\mathbf{c}$. Pictorially (e.g. see Fig. 1), the lattice $\Lambda_q(\mathbf{B})$ consists of some dimensions which are densely packed with lattice points. We further require that such $\mathbf{B}$ is pseudorandom even in presence of leakages in the form of random multiples of short vectors in $\Lambda_q(\mathbf{B})$, which assures that such $\mathbf{B}$ is compatible for use as, e.g., a public key of another encryption scheme.

Definition 7 (Equivocal Distribution). *Let $\mathcal{R}, q, t, k, s$ be parametrised by λ. A distribution $\mathcal{E} \sim \mathcal{R}_q^{t\times k}$ over primitive matrices, i.e. $\mathbf{B}\mathcal{R}_q^k = \mathcal{R}_q^t$ for all $\mathbf{B} \in \mathcal{E}$, is said to be s-equivocal, if both of the following hold:*

- *(Dense Sublattice.) With probability $1 - \mathsf{negl}(\lambda)$ over the randomness of $\mathbf{B} \leftarrow\!\!\$\; \mathcal{E}$, for any $\mathbf{c} \in \mathcal{R}^k$, it holds that*
$$H_\infty(\mathcal{D}_{\Lambda_q(\mathbf{B}),s,\mathbf{c}}) \geq \omega(\log\lambda)$$
where $H_\infty(X) := -\log\max_x \Pr[X = x]$ denotes the min-entropy.
- *(Pseudorandom with Leakage.) There exists a module rank $1 \leq r \leq t$ such that for any PPT $\mathcal{A}$, any $Q \in \mathsf{poly}(\lambda)$, and any $\mathbf{e}_1, \ldots, \mathbf{e}_Q \in \mathcal{R}^k$ with $\|\mathbf{e}_i\| \leq s$ for all $i \in [Q]$, the quantity*
$$\left|\Pr\left[\mathsf{ExpPRwL}^0_{\mathcal{E},\mathcal{A},r,(\mathbf{e}_i)_{i\in[Q]}}(1^\lambda) = 1\right] - \Pr\left[\mathsf{ExpPRwL}^1_{\mathcal{E},\mathcal{A},r,(\mathbf{e}_i)_{i\in[Q]}}(1^\lambda) = 1\right]\right|$$
$\leq \mathsf{negl}(\lambda)$, *where the experiments $\mathsf{ExpPRwL}^b_{\mathcal{E},\mathcal{A},r,(\mathbf{e}_i)_{i\in[Q]}}$ are defined in Fig. 2.*

The Pseudorandom with Leakage property models pseudorandomness of $\mathbf{B}$ and post-processing on samples $\tilde{\mathbf{r}}_i$ from the distribution $\mathcal{D}_{\Lambda_q(\mathbf{B}),s,\mathbf{e}_i}$ jointly. It guarantees that when (only) observing random multiples $x_i \cdot \tilde{\mathbf{r}}_i$'s, these are jointly indistinguishable from random samples over the row-span of a random rank-r wide matrix $\hat{\mathbf{R}}$, where r represents the rank of the hidden dense sublattice of $\Lambda_q(\mathbf{B})$. In other words, for any adversary who performs random left-multiplication on $\tilde{\mathbf{r}}_i$'s, this ensures nothing can be learnt after the fact.[14]

Under Definition 7, the common LWE setting with a unique solution $(\mathbf{r}, \mathbf{e})$ corresponds to $H_\infty(\mathcal{D}_{\Lambda_q(\mathbf{B}),s,\mathbf{c}}) \approx 0$, and w.h.p. the Pseudorandom with Leakage property does not hold, since the overwhelming probability mass of each $\tilde{\mathbf{r}}_i$ in $\mathsf{ExpPRwL}^0$ is concentrated at one point, and a collection of multiple leakages will likely not lie in the same row-span of some random wide $\hat{\mathbf{R}}$.

[14] This does not, however, guarantee that nothing can be learnt from $(x_i, \tilde{\mathbf{r}}, x_i\tilde{\mathbf{r}}_i)_i$ jointly, but doing so would require utilising non-trivial correlations between $(x_i)_i$ and $(\tilde{\mathbf{r}}_i)_i$ which seems difficult.

Remark 1 (Primitive matrices). In Definition 7 we restrict $\mathcal{E}$ to be supported by primitive matrices so that notation such as $\tilde{\mathbf{r}}_i^\mathsf{T}\mathbf{B} \leftarrow\$ \mathcal{D}_{\Lambda_q(\mathbf{B}),s,\mathbf{e}_i} \bmod q$ is without ambiguity – such an $\tilde{\mathbf{r}}_i$ is unique modulo q. We could lift this restriction at the cost of complicating notation. For example, instead of $\tilde{\mathbf{r}}_i^\mathsf{T}\mathbf{B} \leftarrow\$ \mathcal{D}_{\Lambda_q(\mathbf{B}),s,\mathbf{e}_i} \bmod q$, we would need to first sample $\mathbf{y}^\mathsf{T} \leftarrow\$ \mathcal{D}_{\Lambda_q(\mathbf{B}),s,\mathbf{e}_i} \bmod q$ and then sample $\tilde{\mathbf{r}}_i^\mathsf{T}$ uniformly at random subject to $\tilde{\mathbf{r}}_i^\mathsf{T}\mathbf{B} = \mathbf{y}^\mathsf{T} \bmod q$.

Remark 2 (Pseudorandom without leakage.). In the definition of the Pseudorandom with Leakage property in Definition 7, $Q = 0$ captures the case where no leakage is provided to the adversary. One could relax the definition to require pseudorandomness to hold only for $Q = 0$ instead of for all $Q = \mathsf{poly}(\lambda)$, if the application does not require pseudorandomness with leakage. As we will discuss in Appendix C, pseudorandomness with leakage is desirable for our obfuscator.

A primal trapdoor scheme consists of two algorithms called pTrapGen and Equivocate. The former allows to sample a matrix $\mathbf{B}$ following an equivocal distribution together with a trapdoor td, and the latter, given td and an LWE solution, samples from the super-polynomially large set of possible LWE solutions. We call the resulting new solution the *equivocated* LWE solution. A primal trapdoor scheme can be seen as the dual of the (ordinary) lattice trapdoor generation and (SIS) preimage sampling algorithms [GPV08]. We require that the equivocated LWE solution behaves like an honestly-sampled LWE solution, that is, the distribution of $\tilde{\mathbf{r}}^\mathsf{T}\mathbf{B} \bmod q$ induced by the output of Equivocate is distributed close to Gaussian over $\Lambda_q(\mathbf{B})$ with centre $\mathbf{c}$; equivalently, the induced error term is distributed close to Gaussian over the coset $\Lambda_q(\mathbf{B}) + \mathbf{c}$.

Definition 8 (Primal Trapdoor). *A primal trapdoor scheme with parameters $(\mathcal{R}, q, t, k, s)$ parametrised by λ consists of the tuple of PPT algorithms* (pTrapGen, Equivocate) *with the following syntax:*

$(\mathbf{B}, \mathsf{td}) \leftarrow \mathsf{pTrapGen}(1^t, 1^k, q)$*: On input parameters t, k, q, the primal trapdoor generation algorithm return a matrix $\mathbf{B} \in \mathcal{R}_q^{t\times k}$ along with a trapdoor* td*.*

$\tilde{\mathbf{r}} \leftarrow \mathsf{Equivocate}(\mathsf{td}, \mathbf{r}, \mathbf{c}, s)$*: On input the trapdoor* td*, vectors $\mathbf{r} \in \mathcal{R}_q^t$, $\mathbf{c} \in \mathcal{R}^k$, and a Gaussian parameter $s > 0$, return a vector $\tilde{\mathbf{r}} \in \mathcal{R}_q^t$.*

A primal trapdoor scheme should satisfy the following properties:

– *(Equivocality.)* $\mathcal{E} = \left\{\mathbf{B} \;\middle|\; (\mathbf{B}, \mathsf{td}) \leftarrow \mathsf{pTrapGen}(1^t, 1^k, q)\right\}$ *is s-equivocal.*
– *(Induced Distribution of Equivocate.) With probability $1 - \mathsf{negl}(\lambda)$ over the randomness of $(\mathbf{B}, \mathsf{td}) \leftarrow \mathsf{pTrapGen}(1^t, 1^k, q)$, for all $\mathbf{r} \in \mathcal{R}_q^t$ and $\mathbf{c} \in \mathcal{R}^k$ satisfying $\|\mathbf{c}^\mathsf{T} - \mathbf{r}^\mathsf{T}\mathbf{B} \bmod q\| \leq s$,*[15] *the following distributions (both parametrised by $(\mathbf{B}, \mathbf{r}, \mathbf{c})$) are statistically close in λ:*

$$\left\{\tilde{\mathbf{r}}\mathbf{B} \bmod q \;\middle|\; \tilde{\mathbf{r}} \leftarrow \mathsf{Equivocate}(\mathsf{td}, \mathbf{r}, \mathbf{c}, s)\right\} \approx_s \mathcal{D}_{\Lambda_q(\mathbf{B}),s,\mathbf{c}} \bmod q.$$

Remark 3. The "Induced Distribution of Equivocate" property in Definition 8 is equivalent to requiring that the equivocated LWE error satisfies

$$\left\{\tilde{\mathbf{e}} \;\middle|\; \begin{array}{l} \tilde{\mathbf{r}} \leftarrow \mathsf{Equivocate}(\mathsf{td}, \mathbf{r}, \mathbf{c}, s) \\ \tilde{\mathbf{e}}^\mathsf{T} := \mathbf{c}^\mathsf{T} - \tilde{\mathbf{r}}^\mathsf{T}\mathbf{B} \bmod q \end{array}\right\} \approx_s \mathcal{D}_{\Lambda_q(\mathbf{B})+\mathbf{c},s} \bmod q$$

for $\mathbf{r}, \mathbf{c}$ satisfying the same constraints. This is because $\mathbf{c}^\mathsf{T} - \mathcal{D}_{\Lambda_q(\mathbf{B}),s,\mathbf{c}} \bmod q = \mathcal{D}_{\Lambda_q(\mathbf{B})+\mathbf{c},s}$.

4.2 Primal Trapdoor Construction

We construct a primal lattice trapdoor scheme Π_{PTD}, given in Fig. 3. Lemma 1 below states that our constructed pTrapGen can be efficiently implemented, whose proof is given in Appendix B.1.

[15] We write $\|\mathbf{x} \bmod q\|$ for the norm of the shortest representative of $\mathbf{x} \in \mathcal{R}_q^k$ modulo q.

$(\mathbf{B}, \mathsf{td}) \leftarrow \mathsf{pTrapGen}(1^t, 1^k, q)$	$\tilde{\mathbf{r}} \leftarrow \mathsf{Equivocate}(\mathsf{td}, \mathbf{r}, \mathbf{c}, s)$
$\mathbf{d} \leftarrow\$ \mathcal{R}_q^t : \mathbf{d}^{\mathsf{T}}\mathcal{R}_q^t = \mathcal{R}_q$	$\mathbf{s} := s/\sigma(\bar{\mathbf{f}}^{\mathsf{T}}\mathbf{f})$ // component-wise inversion
$\mathbf{f} \leftarrow\$ \mathcal{D}_{\mathcal{R}^k,\chi_f} : \mathbf{f}^{\mathsf{T}}\mathcal{R}^k = \mathcal{R}$	$\mathbf{e}_{\mathbb{L}} := \mathsf{proj}_{\mathsf{Span}(\mathcal{L}(\mathbf{f}^{\mathsf{T}}))}(\mathbf{r}^{\mathsf{T}}\mathbf{B} - \mathbf{c}^{\mathsf{T}} \bmod q)$
$\mathbf{B} \leftarrow\$ \mathcal{R}_q^{t\times k} : \mathbf{d}^{\mathsf{T}}\mathbf{B} = \mathbf{f}^{\mathsf{T}} \bmod q$	// project error onto $\mathsf{Span}(\mathcal{L}(\mathbf{f}^{\mathsf{T}}))$
$\mathsf{td} := (\mathbf{B}, \mathbf{f}, \mathbf{d})$	$-c \cdot \mathbf{1}_k := \mathbf{e}_{\mathbb{L}}/\mathbf{f}$ // component-wise inversion
return $(\mathbf{B}, \mathsf{td})$	$p \leftarrow\$ \mathcal{D}_{\mathcal{R},\mathbf{s},c}$
	$\tilde{\mathbf{r}}^{\mathsf{T}} := \mathbf{r}^{\mathsf{T}} + p\mathbf{d}^{\mathsf{T}} \bmod q$
	return $\tilde{\mathbf{r}}$

Fig. 3. Construction Π_{PTD} of primal trapdoor algorithms.

Lemma 1. *Fix arbitrary $\mathbf{d} \in \mathcal{R}_q^t$ such that $\mathbf{d}^{\mathsf{T}}\mathcal{R}_q^t = \mathcal{R}_q$ and $\mathbf{f} \in \mathcal{R}_q^k$. Let $\mathcal{B} = \mathcal{B}_{\mathbf{d},\mathbf{f}}$ denote the set*

$$\mathcal{B} = \left\{\mathbf{B} \in \mathcal{R}_q^{t\times k} : \mathbf{d}^{\mathsf{T}}\mathbf{B} = \mathbf{f}^{\mathsf{T}} \bmod q\right\}.$$

Then, there exists a PPT algorithm Sample *that on input $\mathbf{d}, \mathbf{f}$ returns $\mathbf{B} \in \mathcal{B}$ sampled from the uniform distribution over $\mathcal{B}$, whenever $\mathcal{B}$ is not empty.*

Theorems 2 and 3 state that Π_{PTD} satisfies the two desired properties in Definition 8 with proofs collected in Appendices B.2 and B.3. In particular, Appendix B.2 establishes the core lemma on Gaussian distribution informally stated in Section 2.2, which is used in Appendix B.3 to establish the theorems below.

Theorem 2 (Equivocality). *Let $\mathcal{R}, q, t, k, \chi_f, s, \beta_\Lambda$ be parametrised by λ, where q is a rational prime unramified in $\mathcal{R}$ and splitting into g prime ideals. Denote $\kappa := \frac{1}{2} \cdot q^{\frac{1}{g}} \cdot \varphi^{\frac{1}{4}}/(\varphi+1)^{1+\frac{2}{\varphi}}$. Let* $\mathsf{pTrapGen}$ *be defined as in Fig. 3.*

Let $\varphi \geq \lambda$, $q > 4\beta_\Lambda \chi_f \sqrt{\varphi}$, $q^{\frac{1}{g}} \geq s\sqrt{k}$, $\beta_\Lambda \geq s(\sqrt{k\varphi}+1)$, $s \geq 2\chi_f k\varphi$ and $k \geq 2(t-1)\log q + 1$. If $t > 1$ then further let $\kappa > 2\beta_\Lambda \chi_f \sqrt{(k-1)\varphi}$. Assuming $\mathsf{dNTRU}_{\mathcal{R},q,t,k,\chi_f}$, then $\mathcal{E} := \left\{\mathbf{B} \mid (\mathbf{B}, \mathsf{td}) \leftarrow \mathsf{pTrapGen}(1^t, 1^k, q)\right\}$ is s-equivocal.

Theorem 3 (Induced Distribution of Equivocate). *Let $\mathcal{R}, q, t, k, \chi_f, s, \beta_\Lambda$ be parametrised by λ, where q is a rational prime unramified in $\mathcal{R}$ and splitting into g prime ideals. Denote $\kappa := \frac{1}{2} \cdot q^{\frac{1}{g}} \cdot \varphi^{\frac{1}{4}}/(\varphi+1)^{1+\frac{2}{\varphi}}$. Let $(\mathsf{pTrapGen}, \mathsf{Equivocate})$ be defined as in Fig. 3.*

Let $\varphi \geq \lambda$, $q > 4\beta_\Lambda \chi_f \sqrt{\varphi}$, $\beta_\Lambda \geq s(\sqrt{k\varphi}+1)$, $s \geq \chi_f k\varphi$, and $k \geq 2(t-1)\log q + 1$. If $t > 1$ then further let $\kappa > 2\beta_\Lambda \chi_f \sqrt{(k-1)\varphi}$ and $q^{\frac{1}{g}} > \chi_f\sqrt{\varphi}$. With probability $1 - \mathsf{negl}(\lambda)$ over the randomness of $(\mathbf{B}, \mathsf{td}) \leftarrow \mathsf{pTrapGen}(1^t, 1^k, q)$, for all $\mathbf{r} \in \mathcal{R}_q^t$ and $\mathbf{c} \in \mathcal{R}^k$ satisfying $\|\mathbf{c}^{\mathsf{T}} - \mathbf{r}^{\mathsf{T}}\mathbf{B} \bmod q\| \leq s$,

$$\left\{\tilde{\mathbf{r}}\mathbf{B} \bmod q \;\middle|\; \tilde{\mathbf{r}} \leftarrow \mathsf{Equivocate}(\mathsf{td}, \mathbf{r}, \mathbf{c}, s)\right\} \approx_s \mathcal{D}_{\Lambda_q(\mathbf{B}),s,\mathbf{c}} \bmod q.$$

5 Equivocal LWE Assumption

In Section 5.1 we define the Equivocal LWE assumption sketched in Section 1.3. In Appendix C we provide the details of our cryptanalytic efforts sketched in Section 1.4, where we investigate potential attacks against the assumption and reason why each of them seems either infeasible or ineffective. Summarising from the cryptanalysis, in Section 5.2 we suggest parameters of Equivocal LWE which we believe to be plausible. In Section 5.3 we compare Equivocal LWE with some prior assumptions used to build obfuscators, in particular under the BDGM framework, and explain why existing counterexamples to prior assumptions should not apply to the Equivocal LWE assumption.

5.1 Assumption Statement

We state the Equivocal LWE assumption, followed by a discussion on some details on its design choices.

Definition 9 (Equivocal LWE Assumption). *Let the parameters*

$$\mathsf{params} = (\mathcal{R}, q, h, k, d, p, t, s, \chi, \mathsf{SKE}, \mathsf{Encode}, \mathcal{E})$$

be parametrised by λ, where d is a divisor of p and

- $\mathsf{SKE} = (\mathsf{Enc}, \mathsf{Dec})$ *is a tagged-SKE (Definition 5), with key space $\mathcal{S}$, tag space $\mathcal{T}$, message spaces $\{\mathcal{M}_{\mathsf{tag}}\}_{\mathsf{tag}\in\mathcal{T}}$ and ciphertext spaces $\{\mathcal{C}_{\mathsf{tag}}\}_{\mathsf{tag}\in\mathcal{T}}$, such that there is a $\mathsf{tag}_{\mathsf{pad}} \in \mathcal{T}$ with $\mathcal{C}_{\mathsf{tag}_{\mathsf{pad}}} = \mathcal{R}_q^{(p/d)\times h}$.*
- $\mathsf{Encode} : \mathcal{S} \to \mathcal{R}_q^{p\lceil \log q\rceil \times k}$ *is a deterministic encoding algorithm,*
- $\mathcal{E}$ *is a distribution over $\mathcal{R}_q^{t\times k}$.*

The $\mathsf{EqLWE}_{\mathsf{params}}$ assumption states the following: For any two-stage PPT algorithm $\mathcal{A} = (\mathcal{A}_1, \mathcal{A}_2)$ where $\mathcal{A}_1$ on input 1^λ outputs:

- *a tag $\mathsf{tag}^* \in \mathcal{T} \setminus \{\mathsf{tag}_{\mathsf{pad}}\}$ and two messages $\mathsf{msg}_0, \mathsf{msg}_1 \in \mathcal{M}_{\mathsf{tag}^*}$,*
- *a message $\mathsf{msg}_{\mathsf{pad}} \in \mathcal{M}_{\mathsf{tag}_{\mathsf{pad}}}$, and*
- *an efficiently computable circuit $\Phi : \mathcal{C}_{\mathsf{tag}^*} \to \mathcal{R}_q^{p\times h} \times \mathcal{R}_q^{h\times k}$,*

it holds with the experiments $\mathsf{Exp}^b_{\mathsf{params},\mathcal{A}}$ defined in Fig. 4 that

$$\left|\Pr\left[\mathsf{Exp}^0_{\mathsf{params},\mathcal{A}}(1^\lambda) = 1\right] - \Pr\left[\mathsf{Exp}^1_{\mathsf{params},\mathcal{A}}(1^\lambda) = 1\right]\right| \le \mathsf{negl}(\lambda).$$

Necessary Conditions for Definition 9. We state the necessary conditions on the choice of params for $\mathsf{EqLWE}_{\mathsf{params}}$ to be plausible:

- SKE is IND-\$-CPA-secure (Definition 15),
- $\mathcal{E} \sim \mathcal{R}_q^{t\times k}$ is an s-equivocal distribution (Definition 7),
- $\mathsf{dLWE}_{\mathcal{R},q,t,k,\chi}$ is hard (Definition 13),
- $\mathsf{SIS}_{\mathcal{R},q,p/d,h,q/(s\sqrt{hk})}$ is hard (Definition 12).

In the rest of the section we explain why the first three are required for the assumption to be sound, and we defer the last one to Appendix C.

Remark 4 (Padding Term in Fig. 4). The experiment in Fig. 4 implicitly defines an encoding $\mathbf{P} \mapsto \mathbf{1}_d \otimes \mathbf{P}$ mapping an SKE ciphertext $\mathbf{P}$ to a padding matrix $\mathbf{1}_d \otimes \mathbf{P}$. Based on understanding from Section 1.4 and Appendix C, we believe this can generally be an arbitrary fixed encoding that maps the uniform distribution over $\mathcal{R}_q^{(p/d)\times h}$ to a matrix over $\mathcal{R}_q^{p\times h}$ such that at least p/d rows of the latter are uniformly distributed, without harming plausibility of the assumption. The mapping to $\mathbf{1}_d \otimes \mathbf{P}$ is a special case which we adopt for our XiO construction in Section 6, we put this in the formal assumption statement for simplicity.

To understand the Equivocal LWE assumption, we first examine the components of a problem instance generated by Exp^b:

- Two SKE ciphertexts, $\mathbf{P}$ encrypting $\mathsf{msg}_{\mathsf{pad}}$ and ctxt_b encrypting msg_b, both under secret key sk.
- An equivocal matrix $\mathbf{B}$ and its LWE samples $\mathbf{C} = \mathbf{R}\mathbf{B} + \mathbf{E} + \mathsf{Encode}(\mathsf{sk}) \bmod q$ masking the (encoding of) the secret key sk.
- A hint $\mathsf{hint}_b = \tilde{\mathbf{R}}_b \bmod q$ being the equivocated solution of some LWE sample $\mathbf{W}_b$ (which will become apparent short below), the latter correlated to the evaluation of an (adversarially chosen) circuit Φ on ctxt_b but additively masked by (an encoding of) $\mathbf{P}$.

$\mathsf{Exp}^b_{\mathsf{params},\mathcal{A}}(1^\lambda)$

$(\mathsf{tag}^*, \mathsf{msg}_0, \mathsf{msg}_1, \mathsf{msg}_{\mathsf{pad}}, \Phi) \leftarrow \mathcal{A}_1(1^\lambda)$
$\mathsf{sk} \leftarrow\!\!\$\ \mathcal{S}$
$\mathbf{P} \leftarrow \mathsf{SKE.Enc}(\mathsf{sk}, \mathsf{tag}_{\mathsf{pad}}, \mathsf{msg}_{\mathsf{pad}})$
$\mathbf{B} \leftarrow\!\!\$\ \mathcal{E} \sim \mathcal{R}_q^{t\times k}$
$\mathbf{R} \leftarrow\!\!\$\ \mathcal{R}_q^{p\lceil \log q\rceil \times t};\ \mathbf{E} \leftarrow\!\!\$\ \mathcal{D}_{\mathcal{R},\chi}^{p\lceil \log q\rceil \times k}$
$\mathbf{C} \coloneqq \mathbf{R}\mathbf{B} + \mathbf{E} + \mathsf{Encode}(\mathsf{sk}) \bmod q$
for $i \in \{0,1\}$ **do**
  $\mathsf{ctxt}_i \leftarrow \mathsf{SKE.Enc}(\mathsf{sk}, \mathsf{tag}^*, \mathsf{msg}_i)$
$(\mathbf{W}_0, \mathbf{W}_1) \leftarrow \mathsf{TargetGen}(\Phi, \mathbf{R}, \mathbf{B}, \mathbf{C}, \mathbf{P}, \mathsf{ctxt}_0, \mathsf{ctxt}_1)$
$\mathsf{hint}_b \leftarrow \mathsf{HintGen}(\mathbf{W}_b, \mathbf{B})$
return $b' \leftarrow \mathcal{A}_2(\mathbf{B}, \mathbf{C}, \mathbf{P}, \mathsf{ctxt}_b, \mathsf{hint}_b)$

$\mathsf{TargetGen}(\Phi, \mathbf{R}, \mathbf{B}, \mathbf{C}, \mathbf{P}, \mathsf{ctxt}_0, \mathsf{ctxt}_1)$

for $i \in \{0,1\}$ **do**
  $(\hat{\mathbf{H}}_i, \mathbf{M}_i) \coloneqq \Phi(\mathsf{ctxt}_i)$
  $\mathbf{L}_i^{\mathsf{T}} \coloneqq \mathbf{G}^{-1}(\hat{\mathbf{H}}_i + \mathbf{1}_d \otimes \mathbf{P})$
  $\mathbf{W}_i \coloneqq \mathbf{L}_i\mathbf{C} + \mathbf{M}_i \bmod q$
  assert $(\|\mathbf{W}_i - \mathbf{L}_i\mathbf{R}\mathbf{B} \bmod q\| \le s)$
assert $\mathbf{M}_0 = \mathbf{M}_1 \bmod q$
return $(\mathbf{W}_0, \mathbf{W}_1)$

$\mathsf{HintGen}(\mathbf{W}_b, \mathbf{B})$

$(\mathbf{w}_1\ \ldots\ \mathbf{w}_h) \coloneqq \mathbf{W}_b^{\mathsf{T}}$
$\tilde{\mathbf{R}}_i\mathbf{B} \leftarrow\!\!\$\ (\mathcal{D}_{\Lambda_q(\mathbf{B}), s, \mathbf{w}_j})_{j\in[h]}$
return $\mathsf{hint}_i \coloneqq \tilde{\mathbf{R}}_i \bmod q$

Fig. 4. Experiments for the Equivocal LWE assumption. $\mathbf{G}^{-1}(\cdot)$ denotes the binary decomposition operation.

LWE and IND-\$-CPA implies Indistinguishability without Hint. Suppose that a problem instance does not contain hint_b, then the claim $(\mathbf{B}, \mathbf{C}, \mathbf{P}, \mathsf{ctxt}_0) \approx_c (\mathbf{B}, \mathbf{C}, \mathbf{P}, \mathsf{ctxt}_1)$ is provably implied by the LWE assumption and that SKE is IND-\$-CPA secure. Indeed, pseudorandomness would follow from the following standard hybrid arguments:

– First, swap $\mathbf{B}$ to uniformly random, which follows from the guarantee of equivocal distribution (Definition 7), more precisely, its pseudorandomness (with or without leakage) property.
– Swap $\mathbf{C} = \mathbf{R}\mathbf{B} + \mathbf{E} + \mathsf{Encode}(\mathsf{sk}) \bmod q$ to uniformly random, which follows from the standard $\mathsf{dLWE}_{\mathcal{R},q,t,k,\chi}$ assumption.
– At this point, sk is already information-theoretically hidden in $(\mathbf{B}, \mathbf{C}, \mathbf{P}, \mathsf{ctxt}_b)$. Thus, by the IND-\$-CPA security of SKE, it holds that $(\mathbf{P}, \mathsf{ctxt}_0) \approx_c (\mathbf{P}, \mathsf{ctxt}_1)$.

The above signals that, under the standard LWE assumption and the requirement that SKE is IND-\$-CPA secure, the only way to distinguish $(\mathbf{B}, \mathbf{C}, \mathbf{P}, \mathsf{ctxt}_b)$ is to somehow make use of hint_b.

Necessity of Admissibility Check. The experiment Exp^b checks that $\mathbf{M}_0 = \mathbf{M}_1 \bmod q$ and for both $i \in \{0,1\}$ that $\|\mathbf{W}_i - \mathbf{L}_i\mathbf{R}\mathbf{B} \bmod q\| \le s$. These guarantee that, in some sense, the hints hint_0 and hint_1 cannot be too different. In more detail, recall that in Exp^b the matrix $\mathbf{W}_b$ can be expressed as

$$\mathbf{W}_b = \mathbf{L}_b\mathbf{C} + \mathbf{M}_b = \mathbf{L}_b\mathbf{R}\mathbf{B} + \mathbf{L}_b \cdot (\mathbf{E} + \mathsf{Encode}(\mathsf{sk})) + \mathbf{M}_b \bmod q, \tag{7}$$

where $(\hat{\mathbf{H}}_b, \mathbf{M}_b) = \Phi(\mathsf{ctxt}_b)$ and $\mathbf{L}_b^{\mathsf{T}} = \mathbf{G}^{-1}(\hat{\mathbf{H}}_b + \mathbf{1}_d \otimes \mathbf{P})$. That is, $\mathbf{L}_i$ is a low-norm matrix and $\mathbf{W}_i$ is s-close to the lattice points of $\Lambda_q(\mathbf{B})$ given by the rows of $\mathbf{L}_i\mathbf{R}\mathbf{B}$. Consequently, it must hold that

$$\mathbf{L}_b \cdot (\mathbf{E} + \mathsf{Encode}(\mathsf{sk})) + \mathbf{M}_b \approx \mathbf{0} \bmod q, \tag{8}$$

and thus $\mathbf{W}_i$ can be seen as LWE samples w.r.t. $\mathbf{B}$, with secret $\mathbf{L}_i\mathbf{R}$ and (structured) short error $\mathbf{L}_b \cdot (\mathbf{E} + \mathsf{Encode}(\mathsf{sk})) + \mathbf{M}_b$. This guarantees that the next step on running HintGen is meaningful.

We note that since SKE is assumed to be IND-\$-CPA secure, (the marginal distributions of) $(\hat{\mathbf{H}}_0, \mathbf{M}_0) = \Phi(\mathsf{ctxt}_0)$ and $(\hat{\mathbf{H}}_1, \mathbf{M}_1) = \Phi(\mathsf{ctxt}_1)$ are computationally indistinguishable. Consequently, so are (the marginal distributions of) $\mathbf{W}_0$ and $\mathbf{W}_1$. We also note that it is necessary to verify $\mathbf{W}_i$ for both $i \in \{0,1\}$, instead of only that of the challenge b. To see why, consider a circuit Φ such that $\mathbf{W}_0$ passes the norm-check but $\mathbf{W}_1$

not[16], then a distinguisher $\mathcal{A}$ would trivially succeed by always outputting 0, the case that the experiment does not abort. Finally, to see why the check $\mathbf{M}_0 = \mathbf{M}_1 \bmod q$ is necessary, we observe that the distinguisher $\mathcal{A}$ could recover

$$\tilde{\mathbf{R}}_b\mathbf{B} - \mathbf{L}_b\mathbf{C} \approx \mathbf{M}_b \bmod q$$

where $\mathbf{M}_0$ and $\mathbf{M}_1$ are potentially known to $\mathcal{A}$. If $\mathbf{M}_0 \neq \mathbf{M}_1 \bmod q$, then $\mathcal{A}$ can distinguish the two cases by checking whether the above is close to $\mathbf{M}_0$ or $\mathbf{M}_1$.

Necessity of Equivocal Distribution. We require the LWE matrix $\mathbf{B}$ to be sampled from an s-equivocal distribution $\mathcal{E}$, i.e. for any $\mathbf{x} \in \Lambda_q(\mathbf{B})$, there exists super-polynomially many other points in $\Lambda_q(\mathbf{B})$ close to $\mathbf{x}$. This ensures that the equivocation $\tilde{\mathbf{R}}_b \neq \mathbf{L}_b\mathbf{R} \bmod q$ with overwhelming probability, where we recall $\mathbf{R}$ is the LWE secret in $\mathbf{C} = \mathbf{R}\mathbf{B} + \mathbf{E} + \mathsf{Encode}(\mathsf{sk}) \bmod q$. Indeed, if $\tilde{\mathbf{R}}_b = \mathbf{L}_b\mathbf{R} \bmod q$ were to hold, then under the mild assumption that the rows of $\mathbf{L}_b$ generate $\mathcal{R}_q^p$, the problem instance can be easily distinguished: Recover $\mathbf{R}_b$ from $\tilde{\mathbf{R}}_b$, then recover sk from $\mathsf{Encode}(\mathsf{sk}) \approx \mathbf{C} - \mathbf{R}\mathbf{B} \bmod q$, and decrypt to msg_b.

Intended Use of Hint reveals only Public Information. Finally, we highlight that upon straightforward use of the hint $\tilde{\mathbf{R}}$ – left-multiplying which to $\mathbf{B}$, only public information is revealed.

Fact 1. *For both $b \in \{0,1\}$ and any $(\mathsf{tag}^*, \mathsf{msg}_0, \mathsf{msg}_1, \mathsf{msg}_{\mathsf{pad}}, \Phi)$, it holds that*

$$(\mathbf{B},\ \mathbf{C},\ \mathbf{P},\ \mathsf{ctxt}_b,\ \tilde{\mathbf{R}}_b\mathbf{B} \bmod q) \equiv (\mathbf{B},\ \mathbf{C},\ \mathbf{P},\ \mathsf{ctxt}_b,\ \mathcal{D}_{\Lambda_q(\mathbf{B}),s,\mathbf{W}_b}),$$

where $\mathbf{B}, \mathbf{C}, \mathbf{P}, \mathsf{ctxt}_b, \mathsf{hint}_b = \tilde{\mathbf{R}}_b$ are the challenge instance in Fig. 4.

The above fact is immediate from the design of the Equivocal LWE experiment. In particular, even when conditioned on all other components in the challenge instance, the product $\tilde{\mathbf{R}}_b\mathbf{B} \bmod q$ follows the *public* distribution $\mathcal{D}_{\Lambda_q(\mathbf{B}),s,\mathbf{W}_b}$, hence no non-trivial information can be learnt just from observing the product.[17] Therefore, to distinguish the challenge bit b, one must be able to use $\tilde{\mathbf{R}}_b$ in other non-trivial ways, but which seems difficult.

5.2 Suggested Parameters

Based on the discussions in Section 5.1 and Appendix C (the latter is overviewed in Section 1.4 but gives a much more careful treatment on distributions and parameters), we suggest what we believe to be plausible parameters $\mathsf{params} = (\mathcal{R}, q, h, k, d, p, t, s, \chi, \mathsf{SKE}, \mathsf{Encode}, \mathcal{E})$ for the Equivocal LWE assumption, with particular attention to s and χ.

Ring, SIS and LWE Parameters. There is no individual constraint on $\mathcal{R}, q$ (and implicitly φ), except that they should be compatible with the constraints below. Clearly, we require $(\mathcal{R}, q, t, k, \chi)$ to be such that $\mathsf{dLWE}_{\mathcal{R},q,t,k,\chi}$ is plausibly hard. Among others, this means $t\varphi \geq \omega(\log \lambda)$ and $q/\chi \leq 2^{o(\lambda)}$. For some of the "zero-ising from the left" attacks to be plausibly hard, we also require $(\mathcal{R}, q, d, h, k, p, s)$ to be such that $\mathsf{SIS}_{\mathcal{R},q,p/d,h,q/(s\sqrt{hk})}$ is hard (see Appendix C Case L1 for details), for which we require the SIS norm bound to satisfy $q \geq c \cdot s\sqrt{hk}$ for some $c > 2$.

IND-\$-CPA Security of SKE*.* Given the above discussion, we require the tagged-SKE to satisfy IND-\$-CPA security (Definition 5). In our XiO construction in Section 6, SKE will be instantiated with a tagged-variant of the GSW encryption scheme [GSW13]. The construction is described in Fig. 5. In Lemma 20 we show that assuming $\mathsf{dLWE}_{\mathcal{R},q,n-1,\chi}$ is sufficient to prove the required security of SKE.

[16] This is possible even if the marginal distributions of $\mathbf{W}_0$ and $\mathbf{W}_1$ are also computationally indistinguishable, since the norm-checking also depends on $(\mathbf{R}, \mathbf{B})$, given which sk can be recovered from $\mathbf{C}$.

[17] Although the distribution $\mathcal{D}_{\Lambda_q(\mathbf{B}),s,\mathbf{W}_b}$ is public, it is not publicly efficiently sampleable, this is why the assumption is not implied by that without the hint $\tilde{\mathbf{R}}$.

Equivocal Parameter s. From the discussion in Appendix C Case L2, based on heuristics borrowed from the ILWE problem, we would like $\|\tilde{\mathbf{E}}_b\|/\|\mathbf{L}_b\| \geq \sqrt{h\varphi}$ to hold. From the setting of the assumption, we have $\|\tilde{\mathbf{E}}_b\| \geq s$ with overwhelming probability and $\|\mathbf{L}_b\| = \|\mathbf{G}^{-1}(\hat{\mathbf{H}}_b + \mathbf{1}_d \otimes \mathbf{P})\| \leq \varphi\sqrt{hp\lceil\log q\rceil}$. Therefore, we suggest to set

$$s/\varphi\sqrt{hp\lceil\log q\rceil} \geq \sqrt{h\varphi} \qquad \iff \qquad s \geq \varphi^{3/2}h\sqrt{p\lceil\log q\rceil}. \tag{9}$$

Looking ahead, for our XiO construction in Section 6, we are able to pick parameters such that all of the above conditions are satisfied (together with other constraints, e.g. imposed by correctness and primal trapdoors). In particular, we can flexibly set a large enough s so that the above conditions are satisfied, while maintaining a polynomial-size modulus q.

5.3 Core Differences relative to Prior Assumptions

We believe the Equivocal LWE assumption has qualitative advantages over prior assumptions used for constructing lattice-based obfuscators, including those of [GP21, WW21, DQV+21, BDGM22][18]. We summarise the core differences below.

We first recall the 2-circular shielded randomness leakage (SRL) assumption of [GP21], which explicitly involves two public-key encryption schemes, one being an FHE and another a PKE, and takes the following form:

- An adversary $\mathcal{A}$ declares $\mathsf{msg}_0, \mathsf{msg}_1$, and receives an FHE ciphertext ctxt^* encrypting msg_b.
- Additionally $\mathcal{A}$ receives two ciphertexts, $\mathsf{ctxt}_{\mathsf{FHE}}$ and $\mathsf{ctxt}_{\mathsf{PKE}}$ under FHE and PKE respectively, each encrypting the secret key of the other.
- $\mathcal{A}$ is given access to a two-round interactive oracle: On the j-th query, in the first round, $\mathcal{A}$ receives an FHE ciphertext ctxt_j of $\mathbf{0}$. Then, in the second round, it provides a circuit Φ_j which the oracle homomorphically evaluates on the challenge ciphertext ctxt^*. In return, $\mathcal{A}$ receives the difference between the encryption randomness of ctxt_j and that of $\Phi_j(\mathsf{ctxt}^*)$.
- $\mathcal{A}$ decides if it is interacting with experiment $b = 0$ or 1.

The counterexample realised by [HJL21] crucially relies on these ingredients: the adaptivity on oracle queries (Step 3), the circularly encrypted secret keys (Step 2), and the determinism of second-round oracle answer conditioned on ctxt_j. In brief, the counterexample works by (1) computing an FHE ciphertext $\mathsf{ctxt}_{\mathsf{sk}}$ of its own secret key sk using the circularly encrypted keys, then (2) upon receiving ctxt_j, craft a query Φ_j dependent on both ctxt_j and $\mathsf{ctxt}_{\mathsf{sk}}$, so that the second-round oracle answer has non-trivial correlation to the challenge bit. In contrast, the Equivocal LWE assumption is non-interactive. In particular, it has no oracle available to the adversaries, and is not a circular assumption[19]. Moreover, decryption hints are generated probabilistically with independent randomness. Thus we expect attack strategies in the above flavour would not apply.

The key-randomness SRL assumption of [BDGM22] can be seen as a variant of the 2-circular SRL assumption above. In addition to explicitly requiring FHE to be the GSW FHE and PKE the packed dual-Regev PKE, it further comes with the following high-level differences:

- In Step 2, the experiment again computes a PKE ciphertext $\mathsf{ctxt}_{\mathsf{PKE}}$ of the secret key sk of FHE, but the FHE ciphertext $\mathsf{ctxt}_{\mathsf{FHE}}$ instead encrypts the encryption randomness of $\mathsf{ctxt}_{\mathsf{PKE}}$. Notably, at this point these two ciphertexts are withheld from $\mathcal{A}$.
- Step 3 is replaced by a once-off query, where $\mathcal{A}$ declares multiple queries query_j all at once. For each query_j, similar to the above, the experiment first computes an FHE ciphertext ctxt_j of $\mathbf{0}$. Then, it homomorphically evaluates a circuit Φ_j on $\mathsf{ctxt}_{\mathsf{FHE}}$, where Φ_j is specific the GSW FHE and non-trivially depends on all of ctxt_j, query_j, $\mathsf{ctxt}_{\mathsf{PKE}}$ and some other freshly sampled randomness r_j. Finally, $\mathcal{A}$ receives the difference between the encryption randomness of ctxt_j and that of $\Phi_j(\mathsf{ctxt}_{\mathsf{FHE}})$, and additionally some information on the randomness r_j used.

[18] The work of [BDGM20] has not provided an assumption which the security of their obfuscator can be based on, so we omit it in this discussion.

[19] In the sense that we do not involve multiple secret keys of schemes that are encrypted under each other.

– Eventually, $\mathsf{ctxt}_{\mathsf{PKE}}$ is released to $\mathcal{A}$, who then decides $b = 0$ or 1.

Due to these differences, in particular, no circularly encrypted secret keys nor $\mathsf{ctxt}_{\mathsf{FHE}}$ which is being evaluated on in Step 3 is available to $\mathcal{A}$, we may expect that the same attack strategy of [HJL21] sketched above would not directly apply to [BDGM22]. However, the above severe restrictions, including both the specific encryption schemes and the specific Φ_j tailor-made based on their obfuscator construction, also makes analysis of the assumption much more difficult. In contrast, we believe that the Equivocal LWE assumption is conceptually simpler to analyse, be it positively or negatively. Moreover, it involves only generic operations such as norm-checking and Gaussian-sampling.

Finally, turning to the work of [DQV+21][20], a subspace flooding assumption [DQV+21, Eprint P.9, Eq.(6)] was proposed, which is structurally similar to our Equivocal LWE assumption in multiple ways: It is non-interactive, involves explicitly some LWE samples and a hint correlated to these samples. Indeed, their hint takes a form very similar to the leakage term Eq. (6) that we inspected in Section 1.4 (and elaborated in Appendix C), as follows:

$$\mathbf{L}_b\mathbf{E} - \mathbf{E}^* + \mathbf{E}_b \tag{10}$$

where (1) $\mathbf{L}_b$ is a low-norm[21] and tall matrix known to the adversary, and (2) $\mathbf{E}, \mathbf{E}^*, \mathbf{E}_b$ are distinct error terms where $\mathbf{E}$ is wide and $\mathbf{E}_0 = \mathbf{0}$. Similar to Eq. (6), Eq. (10) holds without reduction modulo q since all operands are low-norm. The subspace flooding assumption of [DQV+21] essentially asserts that the supposedly hidden error $\mathbf{E}^*$ computationally hides $\mathbf{L}_b\mathbf{E} - \mathbf{E}_b$. This claim is later shown to be false in [JLLS23] with an attack which cleverly recovers $\mathbf{E}^*$ from other (structured) LWE samples given in the problem instance. This allows the attacker to recover $\mathbf{L}_b\mathbf{E} - \mathbf{E}_b$, from which b can be distinguished since $\mathbf{L}_0\mathbf{E}$ is rank-deficient (recall $\mathbf{E}_0 = \mathbf{0}$) while $\mathbf{L}_1\mathbf{E} - \mathbf{E}_1$ is likely not. A crucial distinction between Equivocal LWE and subspace flooding is that in the latter the error terms $\mathbf{E}, \mathbf{E}^*, \mathbf{E}_b$ are parallelly used as LWE errors in other LWE samples of the problem instance. In Equivocal LWE, $\mathbf{E}^*$ is replaced by a freshly sampled $\tilde{\mathbf{E}}_b$ which never reused elsewhere, which sets us away from the attack strategy of [JLLS23] against [DQV+21].

6 Obfuscator Construction

We formally present our obfuscator construction, which will be parametrised by the following objects and quantities, which are themselves parametrised by λ:

– (lattice parameters) a ring $\mathcal{R}$ of degree φ, a modulus q, dimensions n, m, t where $m = n\lceil \log q \rceil$,
– (circuit parameters) $\ell, h, k, N = \mathsf{poly}(\lambda)$, where ℓ denotes the description size, $N = hk$ is the size of the input space and the truth table, the latter of which is arranged into an h-by-k matrix $\mathbf{\Gamma} \in \{0,1\}^{h \times k}$, $h = o(N)$, and $k = o(\sqrt{N})$,[22]
– a universal circuit U which inputs the description $\boldsymbol{\gamma} \in \{0,1\}^\ell$ of a circuit $\Gamma : [h] \times [k] \to \{0,1\}$ and an input $(i,j) \in [h] \times [k]$, and outputs $\Gamma(i,j)$. We denote by d the depth of U.[23]

6.1 Construction

In Fig. 6, we construct an indistinguishability obfuscation $\mathsf{iO} = (\mathsf{Obf}, \mathsf{Eval})$ for the class $\mathcal{C}$ consisting of circuits $\Gamma : [h] \times [k] \to \{0,1\}$ with description of at most ℓ bits and of depth at most $O(\ell)$.

[20] Since [DQV+21] improved upon both the assumption and construction of [WW21], we focus on [DQV+21] while we believe the discussion also applies conceptually to [WW21].
[21] Indeed $\mathbf{L}_b$ in both works are binary decomposition of some matrix.
[22] For concreteness, take $N \gg n^3$, $h \approx N^{2/3}$, and $k \approx N^{1/3}$. When analysing efficiency, we treat N as a free variable.
[23] By [Val76], we know that the depth d of U can be $O(\ell \log \ell)$.

$\underline{\mathsf{sk} \leftarrow \mathsf{KeyGen}(1^\lambda)}$

$\textbf{return } \mathsf{sk} \coloneqq \mathbf{s}_d \leftarrow\!\$\ \mathcal{R}_q^{n-1}$

$\underline{\mathbf{V} \leftarrow \mathsf{GSW}(\mathbf{s}, \mathbf{x})}$

$p \coloneqq \mathsf{nrow}(\mathbf{x})$

$\mathbf{e} \leftarrow\!\$\ \mathcal{D}_{\mathcal{R},\chi}^{pm}$

$\mathbf{A} \leftarrow\!\$\ \mathcal{R}_q^{(n-1)\times pm}$

$\mathbf{K} \coloneqq \begin{pmatrix} \mathbf{A} \\ \mathbf{s}^\mathsf{T}\mathbf{A} + \mathbf{e}^\mathsf{T} \end{pmatrix} \bmod q$

$\mathbf{V} \coloneqq \mathbf{K} + \mathbf{x}^\mathsf{T} \otimes \mathbf{G} \bmod q$

$\quad /\!/\ \mathbf{V} \in \mathcal{R}_q^{n\times pm}$

$\textbf{return } \mathbf{V}$

$\underline{\mathsf{ctxt} \leftarrow \mathsf{Enc}(\mathsf{sk}, \mathsf{tag} \neq \mathsf{tag}_{\mathsf{pad}}, \mathsf{msg})}$

$\mathbf{s}_0 \leftarrow\!\$\ \mathcal{R}_q^{n-1}$

$\mathbf{V}_0 \leftarrow \mathsf{GSW}(\mathbf{s}_0, \mathsf{msg})$

$\textbf{for } i \in [d-1]$

$\quad \mathbf{s}_i \leftarrow\!\$\ \mathcal{R}_q^{n-1}$

$\quad \tilde{\mathbf{s}}_{i-1} = \mathbf{G}^{-1}\left(\left[-\mathbf{s}_{i-1}\ 1\right]^\mathsf{T}\right)$

$\quad \mathbf{V}_i \leftarrow \mathsf{GSW}(\mathbf{s}_i, \tilde{\mathbf{s}}_{i-1})$

$\tilde{\mathbf{s}}_{d-1} = \mathbf{G}^{-1}\left(\left[-\mathbf{s}_{d-1}\ 1\right]^\mathsf{T}\right)$

$\mathbf{V}_d \leftarrow \mathsf{GSW}(\mathbf{s}_d, \tilde{\mathbf{s}}_{d-1})$

$\mathsf{ctxt} \coloneqq (\mathbf{V}_0 \| \mathbf{V}_1 \| \dots \| \mathbf{V}_d)$

$\textbf{return } \mathsf{ctxt}$

$\underline{\mathsf{ctxt} \leftarrow \mathsf{Enc}(\mathsf{sk}, \mathsf{tag} = \mathsf{tag}_{\mathsf{pad}}, \mathsf{msg})}$

$\ell \coloneqq \mathsf{nrow}(\mathsf{msg})$

$\textbf{for } i \in [\ell]$

$\quad \mathbf{P}_i \leftarrow \mathsf{GSW}(\mathbf{s}_d, \mathsf{msg}_i)$

$\quad \mathbf{p}_i \coloneqq \mathbf{P}_i\mathbf{G}^{-1}(\boldsymbol{\tau})$

$\mathsf{ctxt} \coloneqq \mathbf{P} \coloneqq (\mathbf{p}_1 || \dots || \mathbf{p}_\ell)$

$\textbf{return } \mathsf{ctxt}$

Fig. 5. SKE, tagged symmetric-key variant of the GSW FHE scheme with $\boldsymbol{\tau} \coloneqq \left(0 \dots 0\ \lfloor q/2 \rceil\right)^\mathsf{T} \in \mathcal{R}_q^n$.

$\underline{\tilde{\Gamma} \leftarrow \mathsf{Obf}(\Gamma)}$

$\textbf{parse } \Gamma = \boldsymbol{\gamma} \in \{0,1\}^\ell$

$\mathbf{s}_d \leftarrow\!\$\ \mathcal{R}_q^{n-1}$

$\mathsf{ctxt} \leftarrow \mathsf{SKE.Enc}(\mathbf{s}_d, \mathsf{tag}^*, \boldsymbol{\gamma})$

$\mathbf{P} \leftarrow \mathsf{SKE.Enc}(\mathbf{s}_d, \mathsf{tag}_{\mathsf{pad}}, \mathbf{0}_h)$

$(\mathbf{B}, \mathsf{td}) \leftarrow \mathsf{pTrapGen}(1^t, 1^k, q)$

$\mathbf{R} \leftarrow\!\$\ \mathcal{R}_q^{km\times t}$

$\mathbf{E} \leftarrow\!\$\ \mathcal{D}_{\mathcal{R},\chi}^{km\times k}$

$\mathbf{C} \coloneqq \mathbf{R}\mathbf{B} + \mathbf{E} + \mathbf{I}_k \otimes ((-\mathbf{s}_d^\mathsf{T} \parallel 1)\mathbf{G})^\mathsf{T} \bmod q \in \mathcal{R}_q^{km\times k}$

$\textbf{for } (i,j) \in [h] \times [k] \textbf{ do}$

$\quad \mathbf{H}_{i,j} \leftarrow \mathsf{Evaluate}(\mathsf{ctxt}, U(\cdot, (i,j))) \in \mathcal{R}^{n\times m}$

$\quad \mathbf{h}_{i,j} \coloneqq \mathbf{H}_{i,j}\mathbf{G}^{-1}(\boldsymbol{\tau}) \in \mathcal{R}^n$

$\textbf{for } i \in [h] \textbf{ do}$

$\quad \mathbf{l}_i \leftarrow \mathsf{Linear}(\mathbf{P}, (\mathbf{h}_{i,j})_{j\in[k]}) \in \mathcal{R}^{km}$

$\quad \tilde{\mathbf{r}}_i \leftarrow \mathsf{Equivocate}(\mathsf{td}, \mathbf{l}_i^\mathsf{T}\mathbf{R}, \mathbf{l}_i^\mathsf{T}\mathbf{C} + \mathbf{m}_i^\mathsf{T}, s) \in \mathcal{R}^t$

$\mathsf{hint} \coloneqq \tilde{\mathbf{R}} = \begin{pmatrix} \tilde{\mathbf{r}}_1^\mathsf{T} \\ \vdots \\ \tilde{\mathbf{r}}_h^\mathsf{T} \end{pmatrix} \in \mathcal{R}_q^{h\times t}$

$\textbf{return } \tilde{\Gamma} \coloneqq (\mathbf{B}, \mathbf{C}, \mathbf{P}, \mathsf{ctxt}, \mathsf{hint})$

$\underline{y \leftarrow \mathsf{Eval}(\tilde{\Gamma}, (i,j))}$

$\textbf{parse}$

$\quad \tilde{\Gamma} = (\mathbf{B}, \mathbf{C}, \mathbf{P}, \mathsf{ctxt}, \mathsf{hint})$

$\quad \mathsf{hint} = \begin{pmatrix} \tilde{\mathbf{r}}_1^\mathsf{T} \\ \vdots \\ \tilde{\mathbf{r}}_h^\mathsf{T} \end{pmatrix} \in (\mathcal{R}_q^{1\times t})^h$

$\quad \mathbf{B} = \left(\mathbf{b}_1 \dots \mathbf{b}_k\right) \in (\mathcal{R}_q^t)^k$

$\quad \mathbf{C} = \left(\mathbf{c}_1 \dots \mathbf{c}_k\right) \in (\mathcal{R}_q^{km})^k$

$\textbf{for } j' \in [k] \textbf{ do}$

$\quad \mathbf{H}_{i,j'} \leftarrow \mathsf{Evaluate}(\mathsf{ctxt}, U(\cdot, (i,j'))) \in \mathcal{R}^{n\times m}$

$\quad \mathbf{h}_{i,j'} \coloneqq \mathbf{H}_{i,j'}\mathbf{G}^{-1}(\boldsymbol{\tau}) \in \mathcal{R}^n$

$\mathbf{l}_i \leftarrow \mathsf{Linear}(\mathbf{P}, (\mathbf{h}_{i,j'})_{j'\in[k]})$

$\tilde{\gamma}_{i,j} \coloneqq \mathbf{l}_i^\mathsf{T}\mathbf{c}_j - \tilde{\mathbf{r}}_i^\mathsf{T}\mathbf{b}_j \bmod q$

$\textbf{return } \lfloor (2/q)\tilde{\gamma}_{i,j} \rceil \bmod 2$

$\underline{\mathbf{l}_i \leftarrow \mathsf{Linear}(\mathbf{P}, (\mathbf{h}_{i,j})_{j\in[k]})}$

$\textbf{parse } \mathbf{P} = (\mathbf{p}_1 \| \mathbf{p}_2 \| \dots \| \mathbf{p}_h) \in \mathcal{R}_q^{n\times h}$

$\hat{\mathbf{h}}_i^\mathsf{T} \coloneqq \left(\mathbf{h}_{i,1}^\mathsf{T} \dots \mathbf{h}_{i,k}^\mathsf{T}\right) \in \mathcal{R}^{kn}$

$\mathbf{l}_i \coloneqq (\mathbf{I}_k \otimes \mathbf{G})^{-1}(\hat{\mathbf{h}}_i + \mathbf{1}_k \otimes \mathbf{p}_i) \in \mathcal{R}^{km}$

$\textbf{return } \mathbf{l}_i$

Fig. 6. The XiO construction. Algorithm $\mathsf{Evaluate}$ is defined in Theorem 7, $\boldsymbol{\tau} \coloneqq \left(0 \dots 0\ \lfloor q/2 \rceil\right)^\mathsf{T} \in \mathcal{R}_q^n$, and $\mathbf{m}_i^\mathsf{T}$ denotes the i-th row of $\boldsymbol{\Gamma}$ multiplied by $-\lfloor q/2 \rfloor$.

Parameters. The parameters are described in Table 1 in Appendix D, where q, s, β_Λ, and k are chosen as to satisfied various required lower-bounds and upper-bounds. The third column of the table describes the constraints satisfied. In particular:

– the constraints on q are due to Theorems 2 and 3, together with Theorem 4 for correctness and Section 5.2 for dLWE and SIS hardness,
– the constraints on s are due to Theorems 2 and 3 and the suggested parameter regime for the Equivocal LWE assumption in Section 5.2,
– the constraints on β_Λ, k are due to Theorems 2 and 3.

Non-Trivial Efficiency. From the above derivation, we observe that an obfuscated circuit $\tilde{\Gamma}$ is of size $(kt + k^2m + nh + n\ell m + ndm^2 + ht)\varphi \log q$ bits. Since $\varphi, n, m, t, d, \log q$ are fixed $\mathsf{poly}(\ell, \lambda)$ values independent of h and k, and $h = o(N)$ and $k = o(\sqrt{N})$ respectively, we conclude that $\tilde{\Gamma}$ is of size $o(N)\mathsf{poly}(\ell, \lambda)$. Furthermore, the obfuscator is shallow, since the depth of its computation does not depend on N (as the computation of $\mathbf{L}$ and $\mathbf{R}$ as well as their product) and can be trivially parallelized, whereas the output of the $\mathsf{Equivocate}$ algorithm can be computed row-by-row in depth independently of N. Thus, our construction satisfies the non-trivial efficiency requirement of Theorem 1.

6.2 Correctness and Security

The following theorems state the correctness and security of our obfuscator. The proofs can be found in Appendix D.2 and Appendix D.3 respectively.

Theorem 4. *Let $\mathcal{C}$ be the class of circuits $\Gamma : [h] \times [k] \to \{0, 1\}$ with description size at most ℓ and of depth at most $O(\ell)$. For the XiO construction in Fig. 6 with parameters as in Table 1, for all circuit $\Gamma \in \mathcal{C}$, all inputs $(i, j) \in [h] \times [k]$, with overwhelming probability in λ, it holds that*

$$\mathsf{Eval}(\mathsf{Obf}(\Gamma), (i, j)) = \Gamma(i, j).$$

Theorem 5. *Let parameters be specified as in Table 1. Further let* $\mathsf{params} = (\mathcal{R}, q, h, k, k, kn, t, s, \chi, \mathsf{SKE}, \mathsf{Encode}, \mathcal{E})$ *where*

– SKE *is the tagged-SKE described in Fig. 5,*
– Encode *is defined as* $\mathbf{s}_d \mapsto \mathsf{Encode}(\mathbf{s}_d) := \mathbf{I}_k \otimes ((-\mathbf{s}_d^{\mathtt{T}} \parallel 1)\mathbf{G})^{\mathtt{T}} \bmod q$,
– $\mathcal{E} := \left\{\mathbf{B} \;\middle|\; (\mathbf{B}, \mathsf{td}) \leftarrow \mathsf{pTrapGen}(1^t, 1^k, q)\right\}$.

Under the (resp. sub-exponential) $\mathsf{EqLWE}_{\mathsf{params}}$ *assumption, the XiO construction in Fig. 6 is (resp. sub-exponentially) secure.*

Collecting the primal trapdoor constructed from dNTRU in Theorems 2 and 3, the instantiation of SKE as the GSW FHE scheme from dLWE (see Lemma 20) and the generic obfuscator construction from EqLWE in Theorem 5, we arrive at the following corollary. The proof can be found in Appendix D.4.

Corollary 1. *Let parameters be specified as in Theorem 5. Then, under*

– *the (resp. sub-exponential)* $\mathsf{dLWE}_{\mathcal{R},q,n-1,\chi}$ *assumption,*
– *the (resp. sub-exponential)* $\mathsf{dNTRU}_{\mathcal{R},q,k,\chi_f}$ *assumption, and*
– *the (resp. sub-exponential)* $\mathsf{EqLWE}_{\mathsf{params}}$ *assumption,*

the XiO construction in Fig. 6 is (resp. sub-exponential) secure.

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A Extended Preliminaries

We use bold lower case letters (e.g. $\mathbf{x}$) to denote column vectors and bold upper case letters (e.g. $\mathbf{X}$) to denote matrices. For a complex number $a \in \mathbb{C}$, we denote by $\bar{a}$ its complex conjugate. Over the Euclidean space $\mathbb{R}^m$, the closed ℓ_2-ball of radius 1 centred at zero is denoted by $\mathcal{B} \coloneqq \{\mathbf{x} \in \mathbb{R}^m : \|\mathbf{x}\| \leq 1\}$.

To avoid writing very long tuples repeatedly, for any tuple $(x_1, x_2, \ldots, x_m)$ over any domain, we write $x_i \leftarrow (x_1, x_2, \ldots, x_m)$ for parsing the entry x_i from the tuple. Similarly, we write $(x_{i_1}, \ldots, x_{i_n}) \leftarrow (x_1, x_2, \ldots, x_m)$ for parsing a sub-tuple.

Let $\lambda \in \mathbb{N}$ denote the security parameter, and $\mathsf{poly}(\lambda)$ and $\mathsf{negl}(\lambda)$ the set of all polynomials and negligible functions in λ respectively. By convention, all PPT algorithms that we construct are uniform, while all "PPT adversaries" or "PPT distinguishers" refer to non-uniform algorithms. For any computational assumption which states that the advantage of a PPT adversary is upper-bounded, its sub-exponential version states that there exists a constant $\epsilon \in (0, 1)$ such that the advantage is upper-bounded by $2^{-\lambda^\epsilon}$ for all adversaries of size 2^{λ^ϵ}.

For discrete distributions $\mathcal{X}$ and $\mathcal{Y}$ over a countable set A, the statistical distance between $\mathcal{X}$ and $\mathcal{Y}$ is defined as

$$\Delta(\mathcal{X}, \mathcal{Y}) = \frac{1}{2} \sum_{a \in A} |\mathcal{X}(a) - \mathcal{Y}(a)|.$$

Two ensembles of distributions $\{\mathcal{X}_\lambda\}_{\lambda \in \mathbb{N}}$ and $\{\mathcal{Y}_\lambda\}_{\lambda \in \mathbb{N}}$ are said to be statistically close (in λ) if $\Delta(\mathcal{X}_\lambda, \mathcal{Y}_\lambda) \leq \mathsf{negl}(\lambda)$, denoted by $\mathcal{X} \approx_s \mathcal{Y}$.

A.1 Algebraic Number Theory Background

Let $\mathcal{K} = \mathbb{Q}(\zeta)$ be a cyclotomic field, where $\zeta = \zeta_\mathfrak{f} \in \mathbb{C}$ is a root of the $\mathfrak{f}$-th cyclotomic polynomial of degree $\varphi = \varphi(\mathfrak{f})$, and $\mathcal{R} = \mathbb{Z}[\zeta]$ be its ring of integers. For $q \in \mathbb{N}$, define the quotient ring $\mathcal{R}_q \coloneqq \mathcal{R}/q\mathcal{R}$. We denote by $\mathcal{R}^\times$ and $\mathcal{R}_q^\times$ the sets of units in $\mathcal{R}$ and $\mathcal{R}_q$ respectively. We write $\langle a_1, \ldots, a_n \rangle \subseteq \mathcal{R}$ for the ideal generated by $a_1, \ldots, a_n \in \mathcal{R}$. For $\mathbf{a}^\mathsf{T} = (a_1, \ldots, a_n) \in \mathcal{R}^n$, both $\langle a_1, \ldots, a_n \rangle = \mathcal{R}$ and $\mathbf{a}^\mathsf{T}\mathcal{R}^n = \mathcal{R}$ denote that $\mathbf{a}$ generates $\mathcal{R}$. Similarly, for $\mathbf{a}^\mathsf{T} = (a_1, \ldots, a_n) \in \mathcal{R}_q^n$, $\mathbf{a}^\mathsf{T}\mathcal{R}_q^n = \mathcal{R}_q$ denotes that $\mathbf{a}^\mathsf{T}$ is primitive over $\mathcal{R}_q$.

By default we assume that q is a rational prime not dividing $\mathfrak{f}$ with multiplicative order φ/g modulo $\mathfrak{f}$. We have the prime ideal factorisation $q\mathcal{R} = \prod_{j \in [g]} \mathfrak{q}_j$, where each prime ideal $\mathfrak{q}_j$ has norm $\mathcal{N}(\mathfrak{q}_j) = q^{\varphi/g} \geq \lambda^{\omega(1)}$. In other words, $\mathcal{R}_q$ splits into fields $\mathcal{R}_{\mathfrak{q}_j} \coloneqq \mathcal{R}/\mathfrak{q}_j$ with $|\mathcal{R}_{\mathfrak{q}_j}| \geq \lambda^{\omega(1)}$. We make use of this assumption so that matrices $\mathbf{A}$ sampled at random over $\mathcal{R}_q$ are primitive with high probability. The same could be achieved e.g. by conditioning on that a sampled $\mathbf{A}$ contains invertible submatrices; the assumption is so that we can omit conditional probability terms unnecessary for the sake of this work.

We consider the Minkowski space $\mathbb{H} \subseteq \mathbb{C}^{\mathbb{Z}_\mathfrak{f}^\times}$ defined as

$$\mathbb{H} \coloneqq \left\{ \mathbf{x} \in \mathbb{C}^{\mathbb{Z}_\mathfrak{f}^\times} : x_i = \overline{x_{\varphi-i}} \ \forall i \in \mathbb{Z}_\mathfrak{f}^\times \right\} \cong \mathbb{R}^\varphi.$$

An element $a = \sum_{i=0}^{\varphi-1} a_i \zeta^i \in \mathcal{R}$ (or $\mathcal{R}_q$) is represented by its coefficient embedding $(a_i)_{i \in \mathbb{Z}_\varphi} \in \mathbb{Z}^\varphi$ (or $\mathbb{Z}_q^\varphi$), or canonical embedding $\sigma(a) \coloneqq (\sigma_j(a))_{j \in \mathbb{Z}_\mathfrak{f}^\times} \in \mathbb{H}$ where σ_j runs through all φ ring homomorphisms from $\mathcal{K}$ to $\mathbb{C}$ fixing elements of $\mathbb{Q}$. For a vector $\mathbf{a} = (a_1, \ldots, a_m) \in \mathcal{K}^m$, we abuse notation slightly and define $\sigma(\mathbf{a})^\mathsf{T} \coloneqq (\sigma(a_1)^\mathsf{T} \| \ldots \| \sigma(a_m)^\mathsf{T}) \in \mathbb{H}^m$. The trace of an element $a \in \mathcal{K}$ is defined as $\mathrm{Tr}(a) \coloneqq \sum_{j \in \mathbb{Z}_\mathfrak{f}^\times} \sigma_j(a)$. The ($\ell_2$ canonical) norm of $a \in \mathcal{K}$ is taken as

$$\|a\| \coloneqq \|\sigma(a)\|_2 = \sqrt{\overline{\sigma(a)}^\mathsf{T} \sigma(a)} = \sqrt{\mathrm{Tr}(\bar{a}a)}.$$

For a vector $\mathbf{a} = (a_1, \ldots, a_m) \in \mathcal{K}^m$, its ($\ell_2$ canonical) norm is defined as

$$\|\mathbf{a}\| \coloneqq \sqrt{\sum_{i=1}^m \|a_i\|^2} = \sqrt{\overline{\sigma(\mathbf{a})}^\mathsf{T} \sigma(\mathbf{a})} = \sqrt{\mathrm{Tr}(\bar{\mathbf{a}}^\mathsf{T} \mathbf{a})}.$$

For a matrix $\mathbf{A} \in \mathcal{K}^{n \times m}$ with the i-th row given by $\mathbf{a}_i^{\mathsf{T}} \in \mathcal{K}^m$, we write

$$\|\mathbf{A}\| = \max_{i \in [n]} \|\mathbf{a}_i\|.$$

Similarly, we denote by $\|a\|_\infty$ and $\|\mathbf{a}\|_\infty$ the ℓ_∞ canonical norm of an element $a \in \mathcal{K}$ and a vector $\mathbf{a} \in \mathcal{K}^m$ respectively, and $\|\mathbf{A}\|_\infty$ for a matrix $\mathbf{A} \in \mathcal{K}^{n \times m}$ is defined analogously.

We recall an upper bound for the number of bounded norm ring elements.

Lemma 2 (Specialisation of e.g. [FHM20, Theorem 2]). *Let $\mathcal{R}$ be the ring of integers of a cyclotomic field $\mathcal{K}$ of degree φ and let $\mathcal{C}_\beta \coloneqq \{x \in \mathcal{R} : \|x\|_\infty \leq \beta\}$ with norm taken over the canonical embedding. It holds that*

$$|\mathcal{C}_\beta| \leq (\varphi + 1)! \cdot \frac{(2\beta)^\varphi}{\Delta^{1/2}}$$

where Δ is the discriminant of $\mathcal{K}$.

Taking the upper bound $n! \leq \frac{n^{n+1}}{e^{n-1}}$ and the lower bound $\Delta \geq \varphi^{\varphi/2}$ for cyclotomic fields, we obtain the following corollary.

Corollary 2. *Let $\mathcal{R}$ be a cyclotomic ring of degree φ and let $\mathcal{C}_\beta \coloneqq \{x \in \mathcal{R} : \|x\|_\infty \leq \beta\}$ with norm taken over the canonical embedding. It holds that*

$$|\mathcal{C}_\beta| \leq \frac{(\varphi + 1)^{\varphi+2}}{\varphi^{\varphi/4}} \cdot \beta^\varphi.$$

The following lemma characterises the distribution $\mathbf{s}^{\mathsf{T}}\mathbf{A}$ for any fixed $\mathbf{s}$ and uniformly random $\mathbf{A}$.

Lemma 3 (Adapted from [Mic07, Lemma 4.4]). *Let $\mathcal{R}_q$ be a cyclotomic ring modulo q (or any finite ring). Fix $\mathbf{s} \in \mathcal{R}_q^n$ and let $I = \mathcal{R}_q^n \cdot \mathbf{s}$ be the ideal generated by $\mathbf{s}$. The following distributions are identical:*

$$\mathcal{U}(I^m) \qquad \text{and} \qquad \mathbf{s}^{\mathsf{T}}\mathcal{U}(\mathcal{R}_q^{n \times m}).$$

We recall a lemma from [ACX21] which states that short elements are invertible modulo q.

Lemma 4 ([ACX21, Specialisation of Theorem 2]). *Let $\mathcal{R}$ be a cyclotomic ring of conductor $\mathfrak{f}$, integer z be a factor of $\mathfrak{f}$, and q be a rational prime with $q = 1 \bmod z$ and multiplicative order $\varphi(\mathfrak{f})/\varphi(z)$ modulo $\mathfrak{f}$. An element $x \in \mathcal{R}$ is invertible modulo q if*

$$0 < \|x\| < q^{1/\varphi(z)}.$$

We note that [ACX21, Theorem 2] stated the above result for the ℓ_2-norm over the coefficient embedding. However, the result holds for the ℓ_2-norm over the canonical embedding as well since the latter is not smaller.

A.2 Module Lattices

We recall some necessary definitions of module lattices. For a more comprehensive overview of the preliminaries, we refer for example to [LPR10, LPR13, LS15].

We define a module lattice to be a discrete additive subgroup of $\mathbb{H}^m \cong \mathbb{R}^{\varphi m}$ for some $m \in \mathbb{N}$. This is consistent with the standard definition of a lattice since $\mathbb{H}$ is isomorphic to $\mathbb{R}^\varphi$ as an inner product space over $\mathbb{R}$, with the inner product $(\mathbf{x}, \mathbf{y}) \mapsto \bar{\mathbf{x}}^{\mathsf{T}}\mathbf{y}$ inherited from $\mathbb{C}^m$.

In more detail, we say that a lattice $\mathcal{L} \subseteq \mathbb{H}^m$ is generated by a basis of linearly-independent (over $\mathcal{K}$) vectors $\{\mathbf{f}_j\}_{j \in [d]} \subseteq \mathcal{K}^m$, if for $\mathbf{F}^{\mathsf{T}} \coloneqq \begin{pmatrix}\mathbf{f}_1 \ldots \mathbf{f}_d\end{pmatrix} \in \mathcal{K}^{m \times d}$,

$$\mathcal{L} = \mathcal{L}(\mathbf{F}) = \left\{\sigma(\mathbf{x}) \in \mathbb{H}^m : \exists \mathbf{r} \in \mathcal{R}^d,\ \mathbf{r}^{\mathsf{T}}\mathbf{F} = \mathbf{x}^{\mathsf{T}}\right\}.$$

In other words, $\mathcal{L}$ is the additive subgroup of $\mathbb{H}$ obtained by applying σ on the $\mathcal{R}$-module spanned by the rows of $\mathbf{F}$. As a $\mathbb{R}$-lattice, $\mathcal{L}$ has dimension φm and rank $\varphi d \leq \varphi m$, and is said to be full-rank if $d = m$. The span of $\mathcal{L}$ is defined as

$$\mathsf{span}(\mathcal{L}) \coloneqq \left\{\sigma(\mathbf{x}) \in \mathbb{H}^m : \exists \mathbf{k} \in \mathcal{K}^d,\ \mathbf{k}^\mathsf{T}\mathbf{F} = \mathbf{x}^\mathsf{T}\right\}$$

which is the subspace of $\mathbb{H}$ obtained by applying σ on the $\mathcal{K}$-row-span of $\mathbf{F}$. If $\mathbf{B} \in \mathbb{R}^{m\varphi \times d\varphi}$ is a basis of $\mathcal{L}$ (as a $\mathbb{R}$-lattice), then the determinant of $\mathcal{L}$ is defined as $\det(\mathcal{L}) \coloneqq \sqrt{\det(\mathbf{B}^\mathsf{T}\mathbf{B})}$. However, in this work we will never need to compute the determinant explicitly from a basis. For $1 \leq i \leq \varphi d$, the i-th (ℓ_2-)successive minimum of $\mathcal{L}$, denoted by λ_i, is

$$\lambda_i \coloneqq \inf\{r \in \mathbb{R} : \mathsf{rank}(\mathsf{span}(r\mathcal{B} \cap \mathcal{L})) \geq i\}.$$

In particular, $\lambda_1(\mathcal{L})$ is length of the shortest non-zero vector(s) in $\mathcal{L}$, and $\lambda_{\varphi d}(\mathcal{L})$ is the minimum length such that there exists a set of φd linearly independent vectors in $\mathcal{L}$ all bounded from above by $\lambda_{\varphi d}(\mathcal{L})$. Similarly, the i-th ℓ_∞-successive minimum of $\mathcal{L}$ is denoted by λ_i^∞. For any $\mathbf{c} \in \mathbb{R}^{\varphi m}$, the lattice coset $\mathcal{L} + \mathbf{c}$ is

$$\mathcal{L} + \mathbf{c} \coloneqq \{\mathbf{x} + \mathbf{c} : \mathbf{x} \in \mathcal{L}\}.$$

The dual lattice of a lattice $\mathcal{L}$ is defined as

$$\mathcal{L}^\vee \coloneqq \left\{\sigma(\mathbf{y}) \in \mathbb{H}^m :\ \forall \sigma(\mathbf{x}) \in \mathcal{L},\ \mathrm{Tr}(\bar{\mathbf{x}}^\mathsf{T}\mathbf{y}) \in \mathbb{Z}\right\}.$$

It holds that $(\mathcal{L}^\vee)^\vee = \mathcal{L}$, and $\det(\mathcal{L}^\vee) = \det(\mathcal{L})^{-1}$.

For a matrix $\mathbf{B} \in \mathcal{R}_q^{t\times k}$, we define the q-ary lattice

$$\varLambda_q(\mathbf{B}) \coloneqq \left\{\sigma(\mathbf{x}) \in \mathbb{H}^k : \exists \mathbf{r} \in \mathcal{R}_q^t,\ \mathbf{r}^\mathsf{T}\mathbf{B} = \mathbf{x}^\mathsf{T} \bmod q\right\}.$$

In other words, $\varLambda_q(\mathbf{B})$ is the additive subgroup of $\mathbb{H}$ obtained by applying σ on the $\mathcal{R}_q$-module spanned by the rows of $\mathbf{B}$. For a lattice $\mathcal{L}$ and vector $\mathbf{x} \in \mathcal{R}^k$, we often simply write $\mathbf{x} \in \mathcal{L}$ for $\sigma(\mathbf{x}) \in \mathcal{L}$, which should not cause confusion.

Definition 1 (Distance to Lattice). *For $\beta > 0$, a lattice $\mathcal{L}$, and a matrix $\mathbf{W}$ with the i-th row given by $\mathbf{w}_i^\mathsf{T}$, we write $\mathsf{Dist}(\mathbf{W}, \mathcal{L}) \leq \beta$ if, for each row $\mathbf{w}_i^\mathsf{T}$, it holds that $\min_{\mathbf{x}\in\mathcal{L}} \|\mathbf{x}^\mathsf{T} - \mathbf{w}_i^\mathsf{T}\| \leq \beta$.*

The following lemma adapts [GPV08, Lemma 5.3 (eprint)] to the ring setting.

Lemma 5. *Let $\mathcal{R}$ be a cyclotomic ring of degree φ. Let q be a rational prime unramified in $\mathcal{R}$ and splitting into g prime ideals. Let $m \geq 2n \log q$ and $\beta \leq \frac{1}{2} \cdot q^{\frac{1}{g}} \cdot \varphi^{\frac{1}{4}}/(\varphi+1)^{1+\frac{2}{\varphi}}$. With probability overwhelming in φn over $\mathbf{A} \hookleftarrow\!\$\ \mathcal{R}_q^{n\times m}$, $\lambda_1^\infty(\varLambda_q(\mathbf{A})) > \beta$.*

Proof. Let $\mathcal{C}_\beta \coloneqq \{x \in \mathcal{R} : \|x\|_\infty \leq \beta\}$. Fix any non-zero $\mathbf{s} \in \mathcal{R}_q^n$. Then $I = \mathcal{R}_q^n \cdot \mathbf{s}$ satisfies $|I| \geq q^{\varphi/g}$. By Lemma 3, for uniformly random $\mathbf{A} \hookleftarrow\!\$\ \mathcal{R}_q^{n\times m}$, the probability that $\mathbf{s}^\mathsf{T}\mathbf{A} = \mathbf{t}^\mathsf{T} \bmod q$ for some $\mathbf{t} \in \mathcal{C}_\beta^m$ is at most $|\mathcal{C}_\beta|^m/|I|^m$. By Corollary 2 and $|I| \geq q^{\varphi/g}$, this quantity is at most

$$\left(\frac{(\varphi+1)^{\varphi+2}}{\varphi^{\varphi/4}} \cdot \frac{\beta^\varphi}{q^{\varphi/g}}\right)^m.$$

Taking a union bound over all possible $\mathbf{s} \in \mathcal{R}_q^n$, we have that for uniformly random $\mathbf{A} \hookleftarrow\!\$\ \mathcal{R}_q^{n\times m}$ the probability of $\mathbf{s}^\mathsf{T}\mathbf{A} = \mathbf{t}^\mathsf{T} \bmod q$ for some $\mathbf{s} \in \mathcal{R}_q^n$ and $\mathbf{t} \in \mathcal{C}_\beta^m$ is at most

$$\left(\frac{(\varphi+1)^{\varphi+2}}{\varphi^{\varphi/4}} \cdot \frac{\beta^\varphi}{q^{\varphi/g}}\right)^m \cdot q^{\varphi n}.$$

Substituting the upper bound of β, the above quantity is at most $q^{\varphi n}/2^{\varphi m} \leq q^{\varphi n}/q^{2\varphi n} = q^{-\varphi n}$. □

Definition 10 (Orthogonal Projection). *Let $\mathbb{L} \subseteq \mathbb{H}^m$ and write $\langle \mathbf{x}, \mathbf{y}\rangle = \bar{\mathbf{x}}^\mathsf{T}\mathbf{y}$ for $\mathbf{x}, \mathbf{y} \in \mathbb{H}^m$. Let $\left(\mathbf{f}_1 \ldots \mathbf{f}_d\right) \in \mathbb{H}^{m\times d}$ be a basis of $\mathbb{L}$. For a vector $\mathbf{x} \in \mathbb{H}^m$, the orthogonal projection of $\mathbf{x}$ onto $\mathbb{L}$ is defined as*

$$\mathsf{proj}_{\mathbb{L}}(\mathbf{x}) \coloneqq \sum_{j\in[d]} \mathbf{f}_j \cdot \frac{\langle \mathbf{f}_j, \mathbf{x}\rangle}{\langle \mathbf{f}_j, \mathbf{f}_j\rangle} \in \mathbb{L}.$$

A.3 Discrete Gaussians over Module Lattices

We define the discrete Gaussian distributions over lattices in $\mathbb{H}^m$.

Definition 2 (Gaussian Function). *For a Gaussian (width) parameter* $\mathbf{s} \in (\mathbb{R}^+)^{\varphi m}$, $\mathbf{x} \in \mathbb{H}^m$, *centre* $\mathbf{c} \in \mathbb{H}^m$, *define the (elliptical) Gaussian function* $\rho_{\mathbf{s},\mathbf{c}}(\mathbf{x}) \coloneqq \exp(-\pi \|\mathbf{x} - \mathbf{c}\|_{\mathbf{s}}^2)$, *where* $\|\mathbf{x}\|_{\mathbf{s}} \coloneqq \left(\overline{\mathbf{x}}^{\mathsf{T}} \mathsf{diag}(\mathbf{s})^{-2} \mathbf{x}\right)^{1/2} \in \mathbb{R}_{\geq 0}$. *In the special case where* $\mathbf{s} = \mathbf{1}_{\varphi m} \cdot s$ *for some* $s > 0$, *where* $\mathbf{1}_{\varphi m}$ *denotes the all-one vector, we write* $\rho_{s,\mathbf{c}}(\mathbf{x}) \coloneqq \exp(-\pi \|\mathbf{x} - \mathbf{c}\|^2 / s^2)$. *We omit the subscripts* s *and* $\mathbf{c}$ *in cases* $s = 1$ *and* $\mathbf{c} = \mathbf{0}$ *respectively.*

Definition 11 (Discrete Gaussian Distribution). *The discrete Gaussian probability distribution* $\mathcal{D}_{\mathcal{L},\mathbf{s},\mathbf{c}}$ *over* $\mathcal{L}$ *is defined as*

$$\forall \mathbf{x} \in \mathcal{L},\ \mathcal{D}_{\mathcal{L},\mathbf{s},\mathbf{c}}(\mathbf{x}) \coloneqq \frac{\rho_{\mathbf{s},\mathbf{c}}(\mathbf{x})}{\rho_{\mathbf{s},\mathbf{c}}(\mathcal{L})}.$$

In the special case where $\mathbf{s} = \mathbf{1}_{\varphi m} s$ *for some* $s > 0$, *we write*

$$\forall \mathbf{x} \in \mathcal{L},\ \mathcal{D}_{\mathcal{L},s,\mathbf{c}}(\mathbf{x}) \coloneqq \frac{\rho_{s,\mathbf{c}}(\mathbf{x})}{\rho_{s,\mathbf{c}}(\mathcal{L})}.$$

We omit the subscripts s *and* $\mathbf{c}$ *in cases* $s = 1$ *and* $\mathbf{c} = \mathbf{0}$ *respectively.*

When $\mathcal{L} = \sigma(\mathcal{R}^m)$, we also write $\mathcal{D}_{\mathcal{R}^m,\mathbf{s},\mathbf{c}}$ for the discrete Guassian distribution over $\mathcal{L}$.

Lemma 6 ([MR04, Lemma 2.9]). *For any lattice* $\mathcal{L}$, *positive real* $s > 0$ *and vector* $\mathbf{c}$, $\rho_{s,\mathbf{c}}(\mathcal{L}) \leq \rho_s(\mathcal{L})$.

Lemma 7 ([Ban93, Lemma 1.5]). *For any* $c > 1/\sqrt{2\pi}$, m*-dimensional lattice* $\mathcal{L}$, *and vector* $\mathbf{v} \in \mathbb{R}^m$,

$$\rho(\mathcal{L} \setminus c\sqrt{m}\mathcal{B}) < C^m \cdot \rho(\mathcal{L})$$
$$\rho((\mathcal{L} + \mathbf{v}) \setminus c\sqrt{m}\mathcal{B}) < 2C^m \cdot \rho(\mathcal{L})$$

where $C = c\sqrt{2\pi e} \cdot e^{-\pi c^2} < 1$.

Lemma 8 ([HR14, Claim 2.10]). *For every* m*-dimensional lattice* $\mathcal{L}$, *real* $s > 0$ *and vector* $\mathbf{v} \in \mathbb{R}^m$,

$$\rho_s(\mathcal{L} + \mathbf{v}) \geq \rho_s(\mathbf{v}) \cdot \rho_s(\mathcal{L}).$$

The next two corollaries are direct from Lemmas 7 and 8.

Corollary 3. *For any* $c > 1/\sqrt{2\pi}$, m*-dimensional lattice* $\mathcal{L}$, *real* $s > 0$, *and vector* $\mathbf{v} \in \mathbb{R}^m$,

$$\rho_s(\mathcal{L} \setminus sc\sqrt{m}\mathcal{B}) < C^m \cdot \rho_s(\mathcal{L})$$
$$\rho_{s,-\mathbf{v}}(\mathcal{L} \setminus (sc\sqrt{m} + \|\mathbf{v}\|)\mathcal{B}) < 2C^m \cdot \rho_s(\mathcal{L})$$

where $C = c\sqrt{2\pi e} \cdot e^{-\pi c^2} < 1$.

Proof. For any set $A \subseteq \mathbb{R}^m$ it holds that $\rho_s(A) = \rho\left(\frac{1}{s}A\right)$. Since $\frac{1}{s}\left(\mathcal{L} \setminus c\sqrt{m}\mathcal{B}\right) = \frac{1}{s}\mathcal{L} \setminus \frac{c}{s}\sqrt{m}\mathcal{B}$, applying Lemma 7,

$$\rho_s(\mathcal{L} \setminus sc\sqrt{m}\mathcal{B}) = \rho\left(\frac{1}{s}\mathcal{L} \setminus c\sqrt{m}\mathcal{B}\right) < C^m \cdot \rho\left(\frac{1}{s}\mathcal{L}\right) = C^m \cdot \rho_s(\mathcal{L}).$$

For the second inequality, analogously

$$\rho_s((\mathcal{L} + \mathbf{v}) \setminus sc\sqrt{m}\mathcal{B}) < 2C^m \cdot \rho_s(\mathcal{L}),$$

and additionally we use

$$\rho_s((\mathcal{L} + \mathbf{v}) \setminus sc\sqrt{m}\mathcal{B}) = \rho_{s,-\mathbf{v}}(\mathcal{L} \setminus (sc\sqrt{m}\mathcal{B} - \mathbf{v})) > \rho_{s,-\mathbf{v}}(\mathcal{L} \setminus (sc\sqrt{m} + \|\mathbf{v}\|)\mathcal{B}),$$

where the inequality follows from $sc\sqrt{m}\mathcal{B} - \mathbf{v} \subseteq (sc\sqrt{m} + \|\mathbf{v}\|)\mathcal{B}$. $\square$

Corollary 4. *For any $c > 1/\sqrt{2\pi}$, m-dimensional lattice $\mathcal{L}$, real $s > 0$, and vector $\mathbf{v} \in \mathbb{R}^m$,*

$$\rho_s(\mathbf{v}) \cdot \rho_{s,-\mathbf{v}} \left(\mathcal{L} \setminus (sc\sqrt{m} + \|\mathbf{v}\|)\mathcal{B}\right) < 2C^m \rho_{s,-\mathbf{v}}(\mathcal{L})$$

where $C = c\sqrt{2\pi e} \cdot e^{-\pi c^2} < 1$. Equivalently,

$$\rho_{s,-\mathbf{v}} \left(\mathcal{L} \setminus (sc\sqrt{m} + \|\mathbf{v}\|)\mathcal{B}\right) < \frac{2C^m}{\rho_s(\mathbf{v}) - 2C^m} \cdot \rho_{s,-\mathbf{v}}(\mathcal{L} \cap (sc\sqrt{m} + \|\mathbf{v}\|)\mathcal{B}).$$

Proof.

$$2C^m \rho_{s,-\mathbf{v}}(\mathcal{L}) = 2C^m \rho_s(\mathcal{L} + \mathbf{v}) \geq 2C^m \rho_s(\mathbf{v}) \cdot \rho_s(\mathcal{L}) > \rho_s(\mathbf{v}) \cdot \rho_{s,-\mathbf{v}}(\mathcal{L} \setminus (sc\sqrt{m} + \|\mathbf{v}\|)\mathcal{B}),$$

where the two inequalities are respectively due to Lemma 8 and Corollary 3. The second expression follows from writing $\rho_{s,-\mathbf{v}}(\mathcal{L}) = \rho_{s,-\mathbf{v}}(\mathcal{L} \setminus (sc\sqrt{m} + \|\mathbf{v}\|)\mathcal{B}) + \rho_{s,-\mathbf{v}}(\mathcal{L} \cap (sc\sqrt{m} + \|\mathbf{v}\|)\mathcal{B})$. ◻

Definition 3 (Smoothing Parameter [MR04]). *Let $\mathbf{s} \in (\mathbb{R}^+)^{\varphi m}$, $\mathcal{L} \subseteq \mathbb{H}^m$ be a lattice, and $\epsilon \geq 0$. We say that $\mathbf{s} \geq \eta_\epsilon(\mathcal{L})$ if $\rho_{1/\mathbf{s}}(\mathcal{L}^\vee) \leq 1 + \epsilon$, where the inverse $1/\mathbf{s}$ is taken component-wise.*

The following lemma gives an upper bound on the smoothing parameter of a lattice, relative to the length of the longest vector in its shortest set of linearly independent vectors.

Lemma 9 ([MR04, PR06]). *For any φm-dimensional rank-φd lattice $\mathcal{L}$, any superlogarithmic function $f(\varphi m) = \omega(\log \varphi m)$, there exists a negligible function $\epsilon(\varphi m) \leq 1/2$ such that $\eta_\epsilon(\mathcal{L}) \leq \sqrt{f(\varphi m)} \cdot \lambda_{\varphi d}(\mathcal{L})$. In particular, when $f(\varphi m) = \varphi m$ we have $\eta_\epsilon(\mathcal{L}) \leq \sqrt{\varphi m} \cdot \lambda_{\varphi d}(\mathcal{L})$.*

For an m-dimensional full-rank lattice over $\mathbb{R}$, [MR04, Reg05] gave both lower and upper bounds on $\rho_{s,\mathbf{c}}(\mathcal{L})$ for any $\mathbf{c} \in \mathbb{R}^m$. These were generalised by [PR06] to the case of non-full-rank lattices with the Gaussian centre $\mathbf{c} \in \mathsf{span}(\mathcal{L})$, and their extensions to module lattices over $\mathbb{H}$ have been considered in e.g. [LPR10, LPR13]. The following lemmas are combinations of these results.

Lemma 10 ([MR04, PR06, LPR10]). *Let $\mathcal{L}$ be an φm-dimensional lattice such that $\mathbb{L} := \mathsf{span}(\mathcal{L}) \subseteq \mathbb{H}^m$ is of rank $\varphi d \leq \varphi m$. For any $\mathbf{c} \in \mathbb{L}$, $\epsilon > 0$, and $s \geq \eta_\epsilon(\mathcal{L})$, it holds that*

$$s^{\varphi d} \det(\mathcal{L}^\vee)(1 - \epsilon) \leq \rho_{s,\mathbf{c}}(\mathcal{L}) \leq s^{\varphi d} \det(\mathcal{L}^\vee)(1 + \epsilon).$$

Lemma 11 ([MR04, PR06, LPR10]). *Let $\mathcal{L}$ be an φm-dimensional lattice such that $\mathbb{L} := \mathsf{span}(\mathcal{L}) \subseteq \mathbb{H}^m$ is of rank $\varphi d \leq \varphi m$. For any $\mathbf{c} \in \mathbb{L}$, $\epsilon > 0$, $s \geq 2\eta_\epsilon(\mathcal{L})$, and $\mathbf{y} \in \mathcal{L}$, it holds that*

$$\mathcal{D}_{\mathcal{L},s,\mathbf{c}}(\mathbf{y}) \leq 2^{-\varphi d} \frac{1 + \epsilon}{1 - \epsilon}.$$

A.4 Lattice-based Computational Assumptions

We recall standard lattice-based computational assumptions, namely ISIS, LWE and NTRU, stated over a cyclotomic ring $\mathcal{R}$.

We give a slight variation of the standard ISIS assumption in that, instead of requiring the problem to be hard for average $\mathbf{t} \leftarrow\$ \mathcal{R}_q^n$, we require hardness for all $\mathbf{t} \in \mathcal{R}_q^t$.

Definition 12 ($\mathsf{ISIS}_{\mathcal{R},q,n,m,\alpha}$ Assumption (with worst case target) [Ajt96, LS15]). *Let $\mathsf{params} = (\mathcal{R}, q, n, m, \alpha)$ be parametrised by λ. The $\mathsf{ISIS}_{\mathsf{params}}$ assumption states that for any PPT adversary $\mathcal{A}$ and any target vector $\mathbf{t} \in \mathcal{R}_q^n$,*

$$\mathsf{Adv}^{\mathsf{isis}}_{\mathsf{params},\mathcal{A},\mathbf{t}}(\lambda) := \left| \Pr\left[\mathbf{A}\mathbf{x} = \mathbf{t} \bmod q \wedge 0 < \|\mathbf{x}\| \leq \alpha \;\middle|\; \begin{array}{l} \mathbf{A} \leftarrow\$ \mathcal{R}_q^{n \times m} \\ \mathbf{x} \leftarrow \mathcal{A}(\mathbf{A}) \end{array} \right] \right| \leq \mathsf{negl}(\lambda).$$

The $\mathsf{SIS}_{\mathsf{params}}$ assumption is identical to the above except that $\mathbf{t}$ is fixed to $\mathbf{t} = \mathbf{0}$.

Definition 13 ((Decisional) Learning with Errors (LWE) Assumption [Reg05, LPR10, LS15]). *Let* $\mathsf{params} = (\mathcal{R}, q, n, m, \chi)$ *be parametrised by* λ. *We say that the* $\mathsf{dLWE}_{\mathsf{params}}$ *assumption holds if for any PPT adversary* $\mathcal{A}$,

$$\left| \Pr\left[\mathcal{A}(\mathbf{A}, \mathbf{b}) = 1 \;\middle|\; \begin{array}{l} \mathbf{A} \leftarrow\!\!\$\; \mathcal{R}_q^{n \times m} \\ \mathbf{s} \leftarrow\!\!\$\; \mathcal{R}_q^n;\; \mathbf{e} \leftarrow\!\!\$\; \mathcal{D}_{\mathcal{R}^m, \chi} \\ \mathbf{b}^{\mathtt{T}} := \mathbf{s}^{\mathtt{T}} \mathbf{A} + \mathbf{e}^{\mathtt{T}} \bmod q \end{array} \right] - \Pr\left[\mathcal{A}(\mathbf{A}, \mathbf{b}) = 1 \;\middle|\; \begin{array}{l} \mathbf{A} \leftarrow\!\!\$\; \mathcal{R}_q^{n \times m} \\ \mathbf{b} \leftarrow\!\!\$\; \mathcal{R}_q^m \end{array} \right] \right| \leq \mathsf{negl}(\lambda).$$

If the $\mathsf{dLWE}_{\mathcal{R},q,n,m,\chi}$ *assumption holds for all* $m \in \mathsf{poly}(\lambda)$, *we say that the* $\mathsf{dLWE}_{\mathcal{R},q,n,\chi}$ *assumption holds, i.e. omitting* m.

We state a (weaker) variant of the decisional extended-NTRU assumption. Recall that the standard variant states that the distribution of vectors of the form $\mathbf{b} := \mathbf{f}/d \bmod q \in \mathcal{R}_q^m$, where $\mathbf{f} \in \mathcal{R}^m$ and $d \in \mathcal{R}$ are short Gaussians, is indistinguishable from the uniform distribution over $\mathcal{R}_q^m$.[24] We state a variant where d is instead sampled uniformly at random from $\mathcal{R}_q^\times$, and hence is unlikely to be short.

We observe that our variant is implied by the standard variant since, if $\mathbf{b} := \mathbf{f}/d \bmod q$ is indistinguishable from uniform, then $\mathbf{b}' := \mathbf{b} \cdot c = \mathbf{f} \cdot \frac{c}{d} \bmod q$, where c is Gaussian subject to being invertible over $\mathcal{R}_q$, is also indistinguishable from uniform, with a polynomial loss in distinguishing advantage due to the difference between $\mathcal{R}_q$ and $\mathcal{R}_q^\times$. Then, applying the NTRU assumption (for $m = 1$) on $c/d \bmod q$, $\mathbf{b}'$ is indistinguishable from $\mathbf{b}'' := \mathbf{f}/d'' \bmod q$ for $d'' \leftarrow\!\!\$\; \mathcal{R}_q^\times$, with a polynomial loss in distinguishing advantage due to the difference between $\mathcal{R}_q$ and $\mathcal{R}_q^\times$.

We also remark that extended-NTRU does not seem to be easier than NTRU (i.e. $m = 1$), since given one NTRU sample many can be simulated, although possibly with slightly different distributions (see e.g. [PS21, AFLN24]).

Definition 4 (Decisional (extended-)NTRU Assumption [SS11]). *Let the parameters* $\mathsf{params} = (\mathcal{R}, q, m, \chi)$ *be parametrised by* λ. *The* $\mathsf{dNTRU}_{\mathsf{params}}$ *assumption states that for any PPT adversary* $\mathcal{A}$ *it holds that*

$$\left| \Pr\left[\mathcal{A}(\mathbf{b}) = 1 \;\middle|\; \begin{array}{l} \mathbf{f} \leftarrow\!\!\$\; \mathcal{D}_{\mathcal{R}^m, \chi};\; d \leftarrow\!\!\$\; \mathcal{R}_q^\times \\ \mathbf{b} := \mathbf{f}/d \bmod q \end{array} \right] - \Pr\left[\mathcal{A}(\mathbf{b}) = 1 \;\middle|\; \mathbf{b} \leftarrow\!\!\$\; \mathcal{R}_q^m \right] \right| \leq \mathsf{negl}(\lambda).$$

We note that for $\mathcal{R} = \mathbb{Z}$, the NTRU problem as stated above is probably not hard, at least for polynomial-size q. To cover the case $\mathcal{R} = \mathbb{Z}$, we could consider the matrix variant of NTRU, which is less well-known but nevertheless used in constructions (e.g. [GGH+19, BIP+22, ADDG24]) and considered in cryptanalysis (e.g. [DvW21]).

Remark 5 (On Overstretched NTRU). As mentioned earlier, for our primal trapdoor and obfuscator constructions, we need to rely on the NTRU assumption with a large but polynomial-size modulus q. For q in the so-called "overstretched" regime or which exceeds a lower bound known as the "fatigue point", [ABD16, CJL16] showed that NTRU is solvable in sub-exponential time by a classical algorithm, and quantum computers do not seem to change the situation significantly. The fatigue point appears to depend on the degree φ of the ring $\mathcal{R}$ (or more accurately the lattice dimension) and the length of the short vector which generates the hidden dense sublattice, the latter in our case is measured by the Gaussian parameter χ_f. Subsequent works [KF17, DvW21] presented improved attacks and obtained better estimates of the fatigue point. (See also [AD21].) However, these works focused on NTRU with ternary secrets, while in this work we use NTRU with Gaussian secrets. The latest estimate of the fatigue point of NTRU with Gaussian secrets, given by [HSS23], is currently

$$0.0058 \cdot \chi_f^2 \cdot \varphi^{2.484}.$$

While it is plausible that, with some optimisation such as FHE ciphertext packing, our obfuscator can be instantiated with a modulus q below the fatigue point, the allowable parameters regime would be uncomfortably tight. Nevertheless, we emphasise that even though instantiating our obfuscator with q above the

[24] The term "extended" refers to $m > 1$.

fatigue point may lead to sub-exponential-time attacks, we may still conjecture sub-exponential security for a small enough sub-exponential function.

A.5 Cryptographic Primitives

We recall cryptographic notions relevant to this work.

Tagged Symmetric-Key Encryption. We define the notion of a tagged symmetric-key encryption (SKE) and its IND-CPA security, a generalisation of SKE where encryption additionally inputs a tag. By letting the tag space to be singleton, we recover the usual SKE notion.

Definition 5 (Tagged-SKE). *A tagged symmetric-key encryption (tagged-SKE) scheme for a key space $\mathcal{S}$, a tag space $\mathcal{T}$, message spaces $\{\mathcal{M}_{\mathsf{tag}}\}_{\mathsf{tag}\in\mathcal{T}}$ and ciphertext spaces $\{\mathcal{C}_{\mathsf{tag}}\}_{\mathsf{tag}\in\mathcal{T}}$ is a tuple of PPT algorithms* $(\mathsf{Enc}, \mathsf{Dec})$*:*

– $\mathsf{ctxt} \leftarrow \mathsf{Enc}(\mathsf{sk}, \mathsf{tag}, \mathsf{msg})$*: For any tag* $\mathsf{tag} \in \mathcal{T}$*, generate a ciphertext* $\mathsf{ctxt} \in \mathcal{C}_{\mathsf{tag}}$ *encrypting message* $\mathsf{msg} \in \mathcal{M}_{\mathsf{tag}}$ *under the secret key* $\mathsf{sk} \in \mathcal{S}$*.*
– $\mathsf{msg} \leftarrow \mathsf{Dec}(\mathsf{sk}, \mathsf{tag}, \mathsf{ctxt})$*: Decrypt a ciphertext* ctxt *using a secret key* sk *and a tag* tag *to obtain a message* $\mathsf{msg} \in \mathcal{M}$ *or a failure symbol* $\perp$*.*

Definition 14 (Correctness). *A tagged-SKE* $(\mathsf{Enc}, \mathsf{Dec})$ *for a key space $\mathcal{S}$, a tag space $\mathcal{T}$, message spaces $\{\mathcal{M}_{\mathsf{tag}}\}_{\mathsf{tag}\in\mathcal{T}}$ and ciphertext spaces $\{\mathcal{C}_{\mathsf{tag}}\}_{\mathsf{tag}\in\mathcal{T}}$ is said to be correct if for any* $\mathsf{sk} \in \mathcal{S}$, $\mathsf{tag} \in \mathcal{T}$*, and* $\mathsf{msg} \in \mathcal{M}_{\mathsf{tag}}$ *it holds that*

$$\Pr[\mathsf{Dec}(\mathsf{sk}, \mathsf{tag}, \mathsf{Enc}(\mathsf{sk}, \mathsf{tag}, \mathsf{msg})) = \mathsf{msg}] \geq 1 - \mathsf{negl}(\lambda).$$

The IND-\$-CPA security of a tagged-SKE says that, for all queried message-tag tuples under the coupling spaces, the resulting ciphertexts are jointly indistinguishable from random samples over the corresponding ciphertext spaces.

Definition 15 (IND-\$-CPA). *A tagged-SKE* $\mathsf{SKE} = (\mathsf{Enc}, \mathsf{Dec})$ *for a key space $\mathcal{S}$, a tag space $\mathcal{T}$, message spaces $\{\mathcal{M}_{\mathsf{tag}}\}_{\mathsf{tag}\in\mathcal{T}}$ and ciphertext spaces $\{\mathcal{C}_{\mathsf{tag}}\}_{\mathsf{tag}\in\mathcal{T}}$ is said to have indistinguishability (from random) under chosen plaintext attack (IND-\$-CPA security) if for any PPT stateful adversary $\mathcal{A}$ it holds that*

$$\left|\Pr\left[\mathsf{Exp\text{-}IND\text{-}\$\text{-}CPA}^{0}_{\mathsf{SKE},\mathcal{A}}(1^{\lambda}) = 1\right] - \Pr\left[\mathsf{Exp\text{-}IND\text{-}\$\text{-}CPA}^{1}_{\mathsf{SKE},\mathcal{A}}(1^{\lambda}) = 1\right]\right| \leq \mathsf{negl}(\lambda)$$

where the experiments $\mathsf{Exp\text{-}IND\text{-}\$\text{-}CPA}^{b}_{\mathsf{SKE},\mathcal{A}}$ *are defined in Fig. 7.*

$\mathsf{Exp\text{-}IND\text{-}\$\text{-}CPA}^{b}_{\mathsf{SKE},\mathcal{A}}(1^{\lambda})$	$\mathsf{Enc}\mathcal{O}^{b}(\mathsf{msg}, \mathsf{tag})$
$\mathsf{sk} \leftarrow\$ \mathcal{S}$	**assert** $\mathsf{tag} \in \mathcal{T} \wedge \mathsf{msg} \subseteq \mathcal{M}_{\mathsf{tag}}$
$b' \leftarrow \mathcal{A}^{\mathsf{Enc}\mathcal{O}^{b}}(1^{\lambda})$	**if** $b = 0$ **then** $\mathsf{ctxt} \leftarrow \mathsf{Enc}(\mathsf{sk}, \mathsf{tag}, \mathsf{msg})$
return b'	**if** $b = 1$ **then** $\mathsf{ctxt} \leftarrow\$ \mathcal{C}_{\mathsf{tag}}$
	return ctxt

Fig. 7. Security experiment for IND-\$-CPA.

Homomorphic Computation. In our obfuscator construction in Section 6, we will make non-black-box use of the fully homomorphic encryption (FHE) scheme of Gentry, Sahai, and Waters (GSW) [GSW13]. First, we recall the notation related to the "gadget matrix" commonly used in lattice-based homomorphic computation. For $\mathcal{R}, q, n \in \mathbb{N}$, let $m \coloneqq n\lceil \log q \rceil$. The gadget matrix $\mathbf{G} \in \mathcal{R}_q^{n\times m}$ is defined as

$$\mathbf{G} \coloneqq \mathbf{I}_n \otimes \left(1\ 2\ \ldots\ 2^{\lceil \log q\rceil -1}\right)$$

where $\mathbf{I}_n \in \mathcal{R}_q^{n\times n}$ denotes the identity matrix. Then, for any $\mathbf{a} = (a_1, \ldots, a_n) \in \mathcal{R}_q$, we write $\mathbf{G}^{-1}(\mathbf{a})$ for the "$\mathcal{R}_2$-decomposition" of $\mathbf{a}$. That is, $\mathbf{G}^{-1}(\mathbf{a})$ is the unique vector in $\mathcal{R}^m$ whose coefficient embedding consists of binary entries and satisfying $\mathbf{G}\mathbf{G}^{-1}(\mathbf{a}) = \mathbf{a}$. Note that $\mathbf{G}^{-1}(\cdot)$ is an operator but not an inverse of $\mathbf{G}$. The following fact is immediate.

Lemma 12. *For any $\mathbf{a} \in \mathcal{R}_q^n$, $\left\|\mathbf{G}^{-1}(\mathbf{a})\right\| \leq \varphi\sqrt{n\lceil \log q\rceil}$.*

Next, we summarise the properties of the GSW FHE scheme below.

Theorem 6 (Homomorphic Computation [GSW13, BGG+14]). *Let $\mathcal{R}, q, n, \ell, d, \beta$ be parametrised by λ, and $m = n\lceil \log q\rceil$. There exists deterministic polynomial-time algorithms $(\mathsf{EvalF}, \mathsf{EvalFX})$ with the following syntax and properties:*

- *Syntax:*
  - $\mathbf{K}_{\mathbf{V},C} \leftarrow \mathsf{EvalFX}(\mathbf{V}, C)$*: On input a matrix $\mathbf{V} \in \mathcal{R}_q^{n\times \ell m}$ and a Boolean circuit $C : \{0,1\}^\ell \to \{0,1\}$, output a matrix $\mathbf{K}_{\mathbf{V},C} \in \mathcal{R}^{\ell m\times m}$.*
  - $\mathbf{K}_{\mathbf{V},C,\mathbf{x}} \leftarrow \mathsf{EvalF}(\mathbf{V}, C, \mathbf{x})$*: On input a matrix $\mathbf{V} \in \mathcal{R}_q^{n\times \ell m}$, a Boolean circuit $C : \{0,1\}^\ell \to \{0,1\}$, and an input $\mathbf{x} \in \{0,1\}^\ell$, output a matrix $\mathbf{K}_{\mathbf{V},C,\mathbf{x}} \in \mathcal{R}^{\ell m\times m}$.*
- *Properties: Let $\mathbf{A} \in \mathcal{R}_q^{n\times \ell m}$, $\mathbf{V} \in \mathcal{R}_q^{n\times \ell m}$, $C : \{0,1\}^\ell \to \{0,1\}$ be a Boolean circuit, and $\mathbf{x} \in \{0,1\}^\ell$ be such that*
$$\mathbf{V} = \mathbf{A} + \mathbf{x}^\mathsf{T} \otimes \mathbf{G} \bmod q.$$
It holds that
$$\mathbf{V}\mathbf{K}_{\mathbf{V},C} = \mathbf{A}\mathbf{K}_{\mathbf{V},C,\mathbf{x}} + C(\mathbf{x}) \cdot \mathbf{G} \bmod q.$$

Theorem 7 (Bootstrapping Homomorphic Computation [GSW13, AP14, Hal17]). *Let $\mathcal{R}, q, n, \ell, d, \beta$ be parametrised by λ, and $m = n\lceil \log q\rceil$. There exists deterministic polynomial-time algorithm $\mathsf{Evaluate}$ with the following syntax and properties:*

- *Syntax:*
  - $\mathbf{H}_{(\mathbf{V}_i)_{i\in[0,d]},C} \leftarrow \mathsf{Evaluate}((\mathbf{V}_i)_{i\in[0,d]}, C)$*: On input matrices $\mathbf{V}_0 \in \mathcal{R}_q^{n\times \ell m}$ and $\mathbf{V}_i \in \mathcal{R}_q^{n\times m^2}$ for $i \in [d]$, and a Boolean circuit $C : \{0,1\}^\ell \to \{0,1\}$ of depth d, output a matrix $\mathbf{H}_{(\mathbf{V}_i)_{i\in[0,d]},C} \in \mathcal{R}^{n\times m}$.*
- *Properties: Let $\mathbf{A}_0 \in \mathcal{R}_q^{(n-1)\times \ell m}$, $\mathbf{V}_0 \in \mathcal{R}_q^{n\times \ell m}$, $\mathbf{s}_0 \in \mathcal{R}_q^{n-1}$, $\mathbf{e}_0 \in \mathcal{D}_{\mathcal{R},\chi}^{\ell m}$, and $\mathbf{A}_i \in \mathcal{R}_q^{(n-1)\times m^2}$, $\mathbf{V}_i \in \mathcal{R}_q^{n\times m^2}$, $\mathbf{s}_i \in \mathcal{R}_q^{n-1}$, $\mathbf{e}_i \in \mathcal{D}_{\mathcal{R},\chi}^{m^2}$, for $i \in [d]$, $C : \{0,1\}^\ell \to \{0,1\}$ be a Boolean circuit, and $\mathbf{x} \in \{0,1\}^\ell$ be such that*
$$\mathbf{V}_0 = \begin{pmatrix} \mathbf{A}_0 \\ \mathbf{s}_0^\mathsf{T}\mathbf{A}_0 + \mathbf{e}_0^\mathsf{T} \end{pmatrix} + \mathbf{x}^\mathsf{T} \otimes \mathbf{G} \bmod q,$$
$$\mathbf{V}_i = \begin{pmatrix} \mathbf{A}_i \\ \mathbf{s}_i^\mathsf{T}\mathbf{A}_i + \mathbf{e}_i^\mathsf{T} \end{pmatrix} + \left(\mathbf{G}^{-1}\begin{pmatrix} -\mathbf{s}_{i-1} \\ 1 \end{pmatrix}\right)^\mathsf{T} \otimes \mathbf{G} \bmod q, \ \text{for } i \in [d].$$
It holds that
$$\mathbf{H}_{(\mathbf{V}_i)_{i\in[0,d]},C} = \begin{pmatrix} \mathbf{A} \\ \mathbf{s}_d^\mathsf{T}\mathbf{A} + \mathbf{e}^\mathsf{T} \end{pmatrix} + C(\mathbf{x}) \cdot \mathbf{G} \bmod q,$$
where $\mathbf{A} \in \mathcal{R}_q^{(n-1)\times m}$, $\mathbf{e} \in \mathcal{R}^m$ with $\|\mathbf{e}\| \leq \sqrt{\varphi m}\eta\chi$, $\eta = \tilde{O}(\lambda^2)$.

Indistinguishability Obfuscation. We recall the notion of indistinguishability obfuscation (iO).

Definition 6 (Indistinguishability Obfuscation [GGH+13]). *An indistinguishability obfuscation for a circuit family $\mathcal{C} = \{\mathcal{C}_\lambda\}_{\lambda \in \mathbb{N}}$ is a pair of PPT algorithms $\mathsf{iO} = (\mathsf{Obf}, \mathsf{Eval})$ satisfying the following properties:*

– *Correctness: For all $\lambda \in \mathbb{N}$, all circuit $\Gamma \in \mathcal{C}_\lambda$, all inputs x it holds that $\mathsf{Eval}(\mathsf{Obf}(\Gamma), x) = \Gamma(x)$.*
– *Security: For any PPT adversary $\mathcal{A}$, any pair of circuits $(\Gamma_0, \Gamma_1) \in \mathcal{C}_\lambda$ such that $|\Gamma_0| = |\Gamma_1|$ and $\Gamma_0(x) = \Gamma_1(x)$ on all inputs x, it holds that*

$$|\mathsf{Pr}\left[\mathcal{A}(\mathsf{Obf}(\Gamma_0)) = 1\right] - \mathsf{Pr}\left[\mathcal{A}(\mathsf{Obf}(\Gamma_1)) = 1\right]| \leq \mathsf{negl}(\lambda).$$

The following theorem states that, under the assumption that the LWE problem is sub-exponentially hard, it suffices to consider circuits with a polynomial-size input domain and obfuscators that output obfuscated circuits of size slightly sublinear in size of the truth table of the circuit.

Theorem 1 ([LPST16, BDGM22]). *Assuming sub-exponentially hard LWE, if there exists a shallow sub-exponentially secure indistinguishability obfuscator for $\mathsf{P}^{\mathsf{log}}/\mathsf{poly}$ with non-trivial efficiency, then there exists an indistinguishability obfuscator for P/poly with sub-exponential security.*[25]

B Proofs for Section 4

B.1 Proof of Lemma 1

Proof. Let $\boldsymbol{\iota}_t \in \{0,1\}^t$ be the t-th (last) unit vector. The PPT algorithm Sample, on input $\mathbf{d} \in \mathcal{R}_q^t, \mathbf{f} \in \mathcal{R}_q^k$, does the following:

– Find any basis $\{\mathbf{w}_1, \ldots, \mathbf{w}_{t-1}\}$ of the right-kernel of $\mathbf{d}^\mathsf{T}$ modulo every prime ideal dividing q.
– Find $\mathbf{w}_t$ such that $\mathbf{d}^\mathsf{T}\mathbf{w}_t = 1 \bmod q$, which exists due to the condition that $\mathbf{d}^\mathsf{T}\mathcal{R}_q^t = \mathcal{R}_q$.
– Set $\mathbf{W} := \left(\mathbf{w}_1, \ldots, \mathbf{w}_t\right)$, which is invertible over $\mathcal{R}_q$ and $\mathbf{d}^\mathsf{T}\mathbf{W} = \boldsymbol{\iota}_t^\mathsf{T} \bmod q$ by construction.
– it samples $\bar{\mathbf{b}}_1^\mathsf{T}, \ldots, \bar{\mathbf{b}}_{t-1}^\mathsf{T} \leftarrow\$ \mathcal{R}_q^k$ uniformly at random, and sets $\bar{\mathbf{b}}_t^\mathsf{T} := \mathbf{f}^\mathsf{T}$,
– it sets $\bar{\mathbf{B}}^\mathsf{T} := \left(\bar{\mathbf{b}}_1 \ldots \bar{\mathbf{b}}_{t-1}\ \bar{\mathbf{b}}_t\right)$ with $\bar{\mathbf{B}} \in \mathcal{R}_q^{t \times k}$ and outputs $\mathbf{B} := \mathbf{W}\bar{\mathbf{B}} \bmod q$.

It remains to verify that i) $\mathbf{B} \in \mathcal{B}$, and ii) the distribution from which $\mathbf{B}$ is sampled is the uniform distribution over $\mathcal{B}$. Let

$$\bar{\mathcal{B}} = \bar{\mathcal{B}}_\mathbf{f} = \left\{\bar{\mathbf{B}} \in \mathcal{R}_q^{t \times k} : \boldsymbol{\iota}_t^\mathsf{T}\bar{\mathbf{B}} = \mathbf{f}^\mathsf{T} \bmod q\right\}.$$

We are going to show that $\mathbf{W}^{-1}\mathcal{B} = \bar{\mathcal{B}}$ by double inclusion, from which i) follows by definition of Sample, as Sample's output $\mathbf{B}$ is of the form $\mathbf{W}\bar{\mathbf{B}}$, for $\bar{\mathbf{B}} \in \bar{\mathcal{B}}$.

$\mathbf{W}^{-1}\mathcal{B} \subseteq \bar{\mathcal{B}}$. If $\mathbf{B} \in \mathcal{B}$, then

$$\boldsymbol{\iota}_t^\mathsf{T}\mathbf{W}^{-1}\mathbf{B} = \mathbf{d}^\mathsf{T}\mathbf{W}\mathbf{W}^{-1}\mathbf{B} = \mathbf{d}^\mathsf{T}\mathbf{B} = \mathbf{f}^\mathsf{T} \bmod q,$$

so $\mathbf{W}^{-1}\mathbf{B} \in \bar{\mathcal{B}}$.

[25] $\mathsf{P}^{\mathsf{log}}/\mathsf{poly}$ denotes the class of circuits with description size $\ell = \mathsf{poly}(\lambda)$ and input space size $N = \mathsf{poly}(\lambda)$. Non-trivial efficiency means that the size of the obfuscated circuit is bounded by $o(N)\mathsf{poly}(\ell, \lambda)$. The theorem poses no restriction on the runtime of the obfuscator, which can be as large as $N\mathsf{poly}(\ell, \lambda)$, except that its depth should be at most $\mathsf{poly}(\log N)$.

$\mathbf{W}^{-1}\mathcal{B} \supseteq \bar{\mathcal{B}}$. Conversely, if $\bar{\mathbf{B}} \in \bar{\mathcal{B}}$, then

$$\mathbf{d}^{\mathsf{T}}\mathbf{W}\bar{\mathbf{B}} = \boldsymbol{\iota}_t^{\mathsf{T}}\bar{\mathbf{B}} = \mathbf{f}^{\mathsf{T}} \bmod q,$$

so $\mathbf{W}\bar{\mathbf{B}} \in \mathcal{B}$.

It follows that $\mathbf{W}^{-1}\mathcal{B} = \bar{\mathcal{B}}$ as required. It remains to argue about the distribution of Sample's output. Let us examine the definition of $\bar{\mathcal{B}}$: let $\bar{\mathbf{B}} \in \bar{\mathcal{B}}$ with rows given by $\bar{\mathbf{b}}_1^{\mathsf{T}}, \ldots, \bar{\mathbf{b}}_t^{\mathsf{T}}$. Since $\boldsymbol{\iota}_t^{\mathsf{T}}\bar{\mathbf{B}} = \mathbf{f}^{\mathsf{T}} \bmod q$, we have $\bar{\mathbf{b}}_t = \mathbf{f}^{\mathsf{T}} \bmod q$. There are no other constraints on the first $t-1$ rows, i.e. $\bar{\mathbf{b}}_1^{\mathsf{T}}, \ldots, \bar{\mathbf{b}}_{t-1}^{\mathsf{T}}$. Therefore, the uniform distribution over $\bar{\mathcal{B}}$ can be equivalently described by the following sampling procedure: Sample $\bar{\mathbf{b}}_1^{\mathsf{T}}, \ldots, \bar{\mathbf{b}}_{t-1}^{\mathsf{T}} \leftarrow\$ \mathcal{R}_q^k$ uniformly at random, and set $\bar{\mathbf{b}}_t^{\mathsf{T}} \coloneqq \mathbf{f}^{\mathsf{T}}$. This means that $\bar{\mathbf{B}}$ computed by Sample is distributed uniformly random in $\bar{\mathcal{B}}$. Since $\mathbf{W}$ is invertible, it preserves uniform distributions, i.e. $\mathcal{U}(\mathcal{B}) = \mathbf{W} \cdot \mathcal{U}(\bar{\mathcal{B}})$, where $\mathcal{U}(\cdot)$ denotes the uniform distribution. Using that $\mathbf{B} \coloneqq \mathbf{W}\bar{\mathbf{B}}$ and that $\bar{\mathbf{B}} \sim \mathcal{U}(\bar{\mathcal{B}})$, we deduce that $\mathbf{B} \sim \mathcal{U}(\mathcal{B})$ as required. $\square$

B.2 Utility Lemmas

The goal of this subsection is to establish Lemma 16, which says the following:

> Let $\mathbf{B}$ and $\mathbf{f}$ be distributed as in our primal trapdoor scheme in Fig. 3, and let $\mathbf{c}$ be arbitrary vector that is offset by some short vector $\mathbf{e}$ from some lattice point in $\Lambda_q(\mathbf{B})$, i.e. $\mathbf{c} \in \Lambda_q(\mathbf{B}) + \mathbf{e}$. For a Gaussian parameter s mildly larger than $\|\mathbf{f}\|$ and $\|\mathbf{e}\|$, the distributions $\mathcal{D}_{\Lambda_q(\mathbf{B})+\mathbf{c},s}$ and $\mathcal{D}_{\mathcal{L}(\mathbf{f}^{\mathsf{T}})+\mathbf{e},s}$ are statistically close. That is, the Gaussian weight of the (coset of) the hidden dense sublattice $\mathcal{L}(\mathbf{f}^{\mathsf{T}})$ dominates that of the ambient lattice $\Lambda_q(\mathbf{B})$.

Lemma 16 will prove to be useful in Appendix B.3 for showing that Π_{PTD} in Fig. 3 satisfies the two desired properties stated in Definition 8.

Towards proving Lemma 16, we will establish a sequence of lemmas concerning the lattices $\Lambda(\mathbf{B})$ and $\mathcal{L}(\mathbf{f}^{\mathsf{T}})$. In all theorem and lemma statements in the remaining of this section, we will assume that $\mathcal{R}$ is the $\mathfrak{f}$-th cyclotomic ring where $\mathfrak{f} \geq 4$ is a power of 2, and $\chi_f \geq 7(\mathfrak{f} \ln \mathfrak{f})^{1.5}$, without further explicitly mentioning them. This is so that we can invoke [SS11, Lemma 4.2, 4.4 (eprint version)] to argue that the probability, of rejecting an $\mathbf{f} \leftarrow\$ \mathcal{D}_{\mathcal{R}^k,\chi_f}$ due to $\langle f_1, \ldots, f_h \rangle \neq \mathcal{R}$, is at most a constant smaller than 1 even for $k = 2$. We believe that analogous lemma statements should hold for general cyclotomic rings $\mathcal{R}$ and should be negligible in k. However, proving such statements involves analysing the Dedekind zeta function of the corresponding cyclotomic field, which we leave as a future work.

Lemma 13 below says that, for $\mathbf{U}$ being the right-kernel-basis of $\mathbf{f}^{\mathsf{T}}$, the lattice $\Lambda(\mathbf{B}\mathbf{U})$ likely contains no short vectors.

Lemma 13. *Let $q, t, k \in \mathbb{N}$, where q is a rational prime unramified in $\mathcal{R}$ and splitting into g prime ideals, $k \geq 2(t-1)\log q + 1$ and $t > 1$.*

Fix arbitrary $\mathbf{d} \in \mathcal{R}_q^t, \mathbf{f} = (f_1, \ldots, f_k)^{\mathsf{T}} \in \mathcal{R}_q^k$, where at least one entry of $\mathbf{f}$ is in $\mathcal{R}_q^{\times}$. Let $\mathcal{B} = \mathcal{B}_{\mathbf{d},\mathbf{f}} = \left\{\mathbf{B} \in \mathcal{R}_q^{t\times k} : \mathbf{d}^{\mathsf{T}}\mathbf{B} = \mathbf{f}^{\mathsf{T}} \bmod q\right\}$. W.l.o.g. let $f_1 \in \mathcal{R}_q^{\times}$, and define

$$\mathbf{U} \coloneqq \begin{pmatrix} f_2 & f_3 & \cdots & f_k \\ -f_1 & & & \\ & -f_1 & & \\ & & \ddots & \\ & & & -f_1 \end{pmatrix} \in \mathcal{R}_q^{k\times(k-1)}. \tag{11}$$

For $\mathbf{B} \leftarrow\$ \mathcal{B}$, the lattice $\Lambda_q(\mathbf{B}\mathbf{U})$ contains no non-zero vector of norm $< \frac{1}{2} \cdot q^{\frac{1}{g}} \cdot \varphi^{\frac{1}{4}}/(\varphi+1)^{1+\frac{2}{\varphi}}$ with overwhelming probability in $(t-1)\varphi$.

Proof. Let $\boldsymbol{\iota}_t \in \{0,1\}^t$ be the t-th (last) unit vector, and $\bar{\mathcal{B}} = \left\{\bar{\mathbf{B}} \in \mathcal{R}_q^{t\times k} : \boldsymbol{\iota}_t^{\mathrm{T}}\bar{\mathbf{B}} = \mathbf{f}^{\mathrm{T}} \bmod q\right\}$. From the proof of Lemma 1, there exists a square $\mathbf{W} \in \mathcal{R}_q^{t\times t}$ invertible over $\mathcal{R}_q$ such that $\mathbf{W}^{-1}\mathcal{B} = \bar{\mathcal{B}}$. Notice also that $\mathbf{f}^{\mathrm{T}}\mathbf{U} = \mathbf{0}^{\mathrm{T}}$. Define the following families of lattices:

$$\mathcal{M} \coloneqq \Lambda_q(\mathcal{B}) = \{\Lambda_q(\mathbf{B}) : \mathbf{B} \in \mathcal{B}\}, \qquad \text{and} \qquad \bar{\mathcal{M}} \coloneqq \Lambda_q(\bar{\mathcal{B}}) = \left\{\Lambda_q(\bar{\mathbf{B}}) : \bar{\mathbf{B}} \in \bar{\mathcal{B}}\right\}.$$

Since $\mathbf{W}^{-1}\mathcal{B} = \bar{\mathcal{B}}$, we have

$$\mathcal{M} = \mathbf{W}^{-1}\Lambda_q(\mathcal{B}) = \Lambda_q(\mathbf{W}^{-1}\mathcal{B}) = \Lambda_q(\bar{\mathcal{B}}) = \bar{\mathcal{M}}.$$

In other words, the set of lattices spanned by the individual elements in $\mathcal{B}$ is identical to the set of lattices spanned by the individual elements in $\bar{\mathcal{B}}$.

Then, consider the following families of lattices obtained by right-multiply $\mathbf{U}$ to the lattices in $\mathcal{M}$ and $\bar{\mathcal{M}}$:

$$\mathcal{M}\mathbf{U} = \Lambda_q(\mathcal{B}\mathbf{U}) = \{\Lambda_q(\mathbf{B}\mathbf{U}) : \mathbf{B} \in \mathcal{B}\}, \qquad \text{and} \qquad \bar{\mathcal{M}}\mathbf{U} = \Lambda_q(\bar{\mathcal{B}}\mathbf{U}) = \left\{\Lambda_q(\bar{\mathbf{B}}\mathbf{U}) : \bar{\mathbf{B}} \in \bar{\mathcal{B}}\right\}.$$

Since $\mathcal{M} = \bar{\mathcal{M}}$, we have $\mathcal{M}\mathbf{U} = \bar{\mathcal{M}}\mathbf{U}$.

For any element $\bar{\mathbf{B}} = \begin{pmatrix}\mathbf{A}\\ \mathbf{f}^{\mathrm{T}}\end{pmatrix} \in \bar{\mathcal{B}}$, note that

$$\bar{\mathbf{B}}\mathbf{U} = \begin{pmatrix}\mathbf{AU}\\ \mathbf{0}^{\mathrm{T}}\end{pmatrix} \bmod q,$$

$$\Lambda_q(\bar{\mathbf{B}}\mathbf{U}) = \Lambda_q\left(\begin{pmatrix}\mathbf{AU}\\ \mathbf{0}^{\mathrm{T}}\end{pmatrix}\right) = \Lambda_q(\mathbf{AU}).$$

That is, the $\mathcal{R}_q$-row-span of $\bar{\mathbf{B}}\mathbf{U}$ is identical to the $\mathcal{R}_q$-row-span of $\mathbf{AU}$. If $\bar{\mathbf{B}} \leftarrow\$ \bar{\mathcal{B}}$ is uniformly random, hence $\mathbf{A} \leftarrow\$ \mathcal{R}_q^{(t-1)\times k}$ is uniformly random, then $\mathbf{AU} \bmod q$ is also uniformly random over $\mathcal{R}_q^{(t-1)\times(k-1)}$, since $f_1 \in \mathcal{R}_q^{\times}$. It follows that $\bar{\mathcal{M}}\mathbf{U}$, and hence $\mathcal{M}\mathbf{U}$, can be equivalently described by the following sampling procedure: Sample $\mathbf{M} \leftarrow\$ \mathcal{R}_q^{(t-1)\times(k-1)}$ and output $\Lambda_q(\mathbf{M})$. By Lemma 5, the $\mathcal{R}_q$-submodule spanned by the rows of $\mathbf{M}$ has no non-zero vector of norm $< \frac{1}{2} \cdot q^{\frac{1}{g}} \cdot \varphi^{\frac{1}{4}}/(\varphi+1)^{1+\frac{2}{\varphi}}$ with overwhelming probability in $(t-1)\varphi$. ◻

In Lemma 14 below we argue that, with overwhelming probability, short vectors in $\Lambda_q(\mathbf{B})$ have to lie in the sublattice $\mathcal{L}(\mathbf{f}^{\mathrm{T}}) \subset \Lambda_q(\mathbf{B})$ spanned by the short vector $\mathbf{f}^{\mathrm{T}}$.

Lemma 14. *Let $\mathcal{R}, q, t, k, \chi_f, \beta_\Lambda$ be parametrised by λ, where q is a rational prime unramified in $\mathcal{R}$ and splitting into g prime ideals. Denote $\kappa \coloneqq \frac{1}{2} \cdot q^{\frac{1}{g}} \cdot \varphi^{\frac{1}{4}}/(\varphi+1)^{1+\frac{2}{\varphi}}$. Let* $\mathsf{pTrapGen}$ *be defined as in Fig. 3.*

Suppose $\varphi \geq \lambda$, $q > 4\beta_\Lambda\chi_f\sqrt{\varphi}$, and $k \geq 2(t-1)\log q + 1$. If $t > 1$ then further suppose $\kappa > 2\beta_\Lambda\chi_f\sqrt{(k-1)\varphi}$, and $q^{\frac{1}{g}} > \chi_f\sqrt{\varphi}$. With probability $1 - \mathsf{negl}(\lambda)$ over the randomness of $(\mathbf{B}, (\mathbf{f}, \mathbf{d})) \leftarrow \mathsf{pTrapGen}(1^t, 1^k, q)$, for any $\mathbf{x} = (x_1, \ldots, x_k) \in \Lambda_q(\mathbf{B})$ satisfying $\|x_i\| \leq \beta_\Lambda$ for all $i \in [k]$, it holds that $\mathbf{x} \in \mathcal{L}(\mathbf{f}^{\mathrm{T}})$.

Proof. Let $\mathbf{x}^{\mathrm{T}} = (x_1, \ldots, x_k) = \mathbf{r}^{\mathrm{T}}\mathbf{B} \bmod q$ be arbitrary in $\Lambda_q(\mathbf{B})$ such that $\|x_i\| \leq \beta_\Lambda$ for all $i \in [k]$. Write $\mathbf{f} = (f_1, \ldots, f_k)$. We first claim that $\mathbf{x}$ is a $\mathcal{K}$-multiple of $\mathbf{f}$ with overwhelming probability in λ. To show this, we distinguish two cases.

<u>Case 1: $t = 1$</u>: In this case, $\mathbf{B} = \mathbf{b}^{\mathrm{T}} = (b_1, \ldots, b_k)$ is a row vector, and $\mathbf{d} = d$ is a single invertible ring element. Then, for all $i \in [k]$, since $b_i = d^{-1}f_i \bmod q$, we have

$$b_1 f_i - b_i f_1 = 0 \bmod q$$
$$r b_1 f_i - r b_i f_1 = x_1 f_i - x_i f_1 = 0 \bmod q.$$

By Corollary 3 and [SS11, Lemma 4.2, 4.4 (eprint version)], with overwhelming probability in $\varphi \geq \lambda$, we have $\|f_i\| \leq \chi_f\sqrt{\varphi}$, in which case $\|x_1 f_i - x_i f_1\| \leq \|x_1 f_i\| + \|x_i f_1\| \leq 2\beta_\Lambda \chi_f \sqrt{\varphi} < q/2$, and thus

$$\begin{aligned} x_1 f_i - x_i f_1 &= 0 \\ x_i &= \frac{x_1}{f_1} f_i \end{aligned}$$

where in the last line division is over $\mathcal{K}$. This shows that $\mathbf{x}$ is a $\mathcal{K}$-multiple of $\mathbf{f}$ with overwhelming probability.

Case 2: $t > 1$: Define the matrix $\mathbf{T} \coloneqq \begin{pmatrix} \mathbf{A} \\ \mathbf{d}^{\mathsf{T}} \end{pmatrix} \in \mathcal{R}_q^{t \times t}$, where $\mathbf{A} \in \mathcal{R}_q^{(t-1)\times t}$ is arbitrary s.t. $\mathbf{T}$ is invertible over $\mathcal{R}_q$. Such $\mathbf{A}$ exists with overwhelming probability, concretely, as long as $\mathbf{d} \neq \mathbf{0} \bmod q$. Then it holds that

$$\mathbf{T}\mathbf{B} = \begin{pmatrix} \mathbf{A}\mathbf{B} \\ \mathbf{f}^{\mathsf{T}} \end{pmatrix} \bmod q.$$

Write $\mathbf{r}^{\mathsf{T}}\mathbf{T}^{-1} = \begin{pmatrix} \mathbf{r}_L^{\mathsf{T}} & r_R \end{pmatrix}$. We have

$$\begin{aligned} \mathbf{r}^{\mathsf{T}}\mathbf{B} &= \mathbf{x}^{\mathsf{T}} \bmod q \\ \mathbf{r}^{\mathsf{T}}\mathbf{T}^{-1}\mathbf{T}\mathbf{B} &= \mathbf{x}^{\mathsf{T}} \bmod q \\ \mathbf{r}_L^{\mathsf{T}}\mathbf{A}\mathbf{B} + r_R \mathbf{f}^{\mathsf{T}} &= \mathbf{x}^{\mathsf{T}} \bmod q \\ \mathbf{r}_L^{\mathsf{T}}\mathbf{A}\mathbf{B} &= \mathbf{x}^{\mathsf{T}} - r_R\mathbf{f}^{\mathsf{T}} \bmod q. \end{aligned}$$

Let $\mathbf{U}$ be same as that defined in Eq. (11). Notice $\mathbf{f}^{\mathsf{T}}\mathbf{U} = \mathbf{0}^{\mathsf{T}}$, hence

$$\mathbf{r}_L^{\mathsf{T}}\mathbf{A}\mathbf{B}\mathbf{U} = \mathbf{x}^{\mathsf{T}}\mathbf{U} \bmod q.$$

Next, observe that

- By Corollary 3 and [SS11, Lemma 4.2, 4.4 (eprint version)], with overwhelming probability in $\varphi \geq \lambda$, each entry of $\mathbf{f}$, and therefore also each non-zero entry of $\mathbf{U}$, has norm $\leq \chi_f\sqrt{\varphi}$.
- From the first item, we have $\|\mathbf{x}^{\mathsf{T}}\mathbf{U}\| \leq 2\beta_\Lambda\chi_f\sqrt{(k-1)\varphi} < \kappa$, and $\mathbf{x}^{\mathsf{T}}\mathbf{U} \bmod q$ equals $\mathbf{x}^{\mathsf{T}}\mathbf{U}$ (without modding).
- Using the first item, the constraint that $\chi_f\sqrt{\varphi} < q^{\frac{1}{g}}$, and Lemma 4, any entry of $\mathbf{f}$ is invertible over $\mathcal{R}_q$ with overwhelming probability in λ.
- Because of the third item, we can apply Lemma 13 to conclude that, $\Lambda_q(\mathbf{B}\mathbf{U})$ has no non-zero vector of norm $< \kappa$ with overwhelming probability in $(t-1)\varphi \geq \lambda$, Since $\Lambda_q(\mathbf{A}\mathbf{B}\mathbf{U}) \subseteq \Lambda_q(\mathbf{B}\mathbf{U})$, the same holds for $\Lambda_q(\mathbf{A}\mathbf{B}\mathbf{U})$.

The second item implies the short vector $\mathbf{x}^{\mathsf{T}}\mathbf{U}$ is contained in $\Lambda_q(\mathbf{A}\mathbf{B}\mathbf{U})$, and therefore the last item forces $\mathbf{r}_L^{\mathsf{T}}\mathbf{A}\mathbf{B}\mathbf{U} = \mathbf{x}^{\mathsf{T}}\mathbf{U} = \mathbf{0}^{\mathsf{T}}$ (without modding). That $\mathbf{x}^{\mathsf{T}}\mathbf{U} = \mathbf{0}^{\mathsf{T}}$ can equivalently be written as $x_i = \frac{x_1}{f_1} f_i$ for all $i \in [k]$, where the division is over $\mathcal{K}$.

The remaining is conditioned on that $\mathbf{x}$ is a $\mathcal{K}$-multiple of $\mathbf{f}$. For all $i \in [k]$, since $x_i \in \mathcal{R}$, we have $f_i \in \langle f_1/x_1 \rangle$, which implies $\langle f_i \rangle \subseteq \langle f_1/x_1 \rangle$, and hence $\sum_{i=1}^k \langle f_i \rangle \subseteq \langle f_1/x_1 \rangle$. Since $\sum_{i=1}^k \langle f_i \rangle = \mathcal{R}$ by construction, we have $\mathcal{R} \subseteq \langle f_1/x_1 \rangle$, which implies $x_1 \in \langle f_1 \rangle$. In other words, $\mathbf{x}$ is an $\mathcal{R}$-multiple of $\mathbf{f}$, or $\mathbf{x} \in \mathcal{L}(\mathbf{f}^{\mathsf{T}})$. $\square$

Remark 6. Conditioned on Lemma 14, the remaining results in this subsection could be generalised to cover the case where $\Lambda_q(\mathbf{B})$ contains a hidden dense sublattice $\mathcal{L}(\mathbf{F})$ of module rank $\mathsf{nrows}_{\mathcal{R}}(\mathbf{F}) > 1$, e.g. when $\mathbf{D} \cdot \mathbf{B} = \mathbf{F} \bmod q$ for some uniformly random matrix $\mathbf{D}$ invertible modulo q and a low-norm matrix $\mathbf{F}$. This would allow, for example, instantiating $\mathcal{R}$ with $\mathcal{R} = \mathbb{Z}$, and obtaining the adapted Theorem 3 in such setting. However, we are unable to extend Lemmas 13 and 14 in the above to the setting of $\mathsf{nrows}_{\mathcal{R}}(\mathbf{F}) > 1$. The difficulty, roughly, is that for the former we do not know how to construct a short $\mathbf{U}$ satisfying $\mathbf{F}\mathbf{U} = \mathbf{0}$, while for the latter we could not impose reasonable conditions to rule out the possibility that a linear combination of two long vectors results in a short vector after reduction modulo q.

The next lemma says that the Gaussian function over $\Lambda_q(\mathbf{B})$ and $\mathcal{L}(\mathbf{f}^{\mathsf{T}})$ are close. Intuitively this is due to most of the probability mass being concentrated at lattice points close to zero, which are precisely those in $\mathcal{L}(\mathbf{f}^{\mathsf{T}})$.

Lemma 15. *Let $\mathcal{R}, q, t, k, \chi_f, s, \beta_\Lambda$ be parametrised by λ, where q is a rational prime unramified in $\mathcal{R}$ and splitting into g prime ideals. Denote $\kappa := \frac{1}{2} \cdot q^{\frac{1}{g}} \cdot \varphi^{\frac{1}{4}}/(\varphi+1)^{1+\frac{2}{\varphi}}$. Let* pTrapGen *be defined as in Fig. 3.*

Suppose $\varphi \geq \lambda$, $q > 4\beta_\Lambda \chi_f \sqrt{\varphi}$, $\beta_\Lambda \geq s(\sqrt{k\varphi}+1)$, $s \geq \chi_f k\varphi$, and $k \geq 2(t-1)\log q + 1$. If $t > 1$ then further suppose $\kappa > 2\beta_\Lambda \chi_f \sqrt{(k-1)\varphi}$, and $q^{\frac{1}{g}} > \chi_f \sqrt{\varphi}$. With probability $1 - \mathsf{negl}(\lambda)$ over the randomness of $(\mathbf{B}, (\mathbf{f}, \mathbf{d})) \leftarrow \mathsf{pTrapGen}(1^t, 1^k, q)$, for any $\mathbf{e} \in \mathcal{R}^m$ satisfying $\|\mathbf{e}\| \leq s$, it holds that

$$\rho_{s,\mathbf{e}}(\mathcal{L}(\mathbf{f}^{\mathsf{T}}))/\rho_{s,\mathbf{e}}(\Lambda_q(\mathbf{B})) \geq 1 - \mathsf{negl}(\lambda).$$

Proof. For brevity, below write $\mathcal{L} = \mathcal{L}(\mathbf{f}^{\mathsf{T}})$ and $\Lambda = \Lambda_q(\mathbf{B})$.

Using Lemma 9, there exists $\epsilon \leq \mathsf{negl}(\lambda)$ such that

$$\eta_\epsilon(\mathcal{L}) \leq \sqrt{k\varphi}\lambda_\varphi(\mathcal{L}) \leq \sqrt{k\varphi}\|\mathbf{f}\|,$$

where

$$\|\mathbf{f}\| \leq \chi_f \sqrt{k\varphi}$$

with overwhelming probability in $k\varphi \geq \lambda$ by Corollary 3 and [SS11, Lemma 4.2, 4.4 (eprint version)], in which case $s \geq \chi_f k\varphi \geq \eta_\epsilon(\mathcal{L})$. The analysis below is conditioned on this event.

Let $\mathbb{L} = \mathsf{Span}(\mathcal{L})$ and $\mathbb{P} \subset \mathbb{H}^k$ be the subspace orthogonal to $\mathbb{L}$. Decompose $\mathbf{e} = \mathbf{e}_\mathbb{L} + \mathbf{e}_\mathbb{P}$ where $\mathbf{e}_\mathbb{L} = \mathsf{proj}_\mathbb{L}(\mathbf{e})$ and $\mathbf{e}_\mathbb{P} = \mathsf{proj}_\mathbb{P}(\mathbf{e})$. The numerator $\rho_{s,\mathbf{e}}(\mathcal{L})$ equals

$$\begin{aligned}
\rho_{s,\mathbf{e}}(\mathcal{L}) &= \sum_{\mathbf{x} \in \mathcal{L}} \exp(-\pi\|\mathbf{x}-\mathbf{e}\|^2/s^2) \\
&= \sum_{\mathbf{x} \in \mathcal{L}} \exp(-\pi\|\mathbf{x}-\mathbf{e}_\mathbb{L}-\mathbf{e}_\mathbb{P}\|^2/s^2) \\
&= \sum_{\mathbf{x} \in \mathcal{L}} \exp(-\pi\|\mathbf{x}-\mathbf{e}_\mathbb{L}\|^2/s^2) \cdot \exp(-\pi\|\mathbf{e}_\mathbb{P}\|^2/s^2) \\
&= \rho_{s,\mathbf{e}_\mathbb{L}}(\mathcal{L}) \cdot \exp(-\pi\|\mathbf{e}_\mathbb{P}\|^2/s^2).
\end{aligned}$$

where the third equality is due to $\mathbf{x}-\mathbf{e}_\mathbb{L}$ being orthogonal to $\mathbf{e}_\mathbb{P}$. To analyse the denominator $\rho_{s,\mathbf{e}}(\Lambda)$, consider the lattice $\Lambda/\mathcal{L}$, where we identify each equivalence class in $\Lambda/\mathcal{L}$ by the shortest vector $\mathbf{k}$ in the class (e.g. the representative of $\mathcal{L}$ is $\mathbf{0}$). Express

$$\rho_{s,\mathbf{e}}(\Lambda) = \sum_{\mathbf{k} \in \Lambda/\mathcal{L}} \rho_{s,\mathbf{e}}(\mathcal{L}+\mathbf{k}).$$

For any $\mathbf{k} \in \Lambda/\mathcal{L}$, decompose $\mathbf{k} = \mathbf{k}_\mathbb{L} + \mathbf{k}_\mathbb{P}$, where $\mathbf{k}_\mathbb{L} = \mathsf{proj}_\mathbb{L}(\mathbf{k})$ and $\mathbf{k}_\mathbb{P} = \mathsf{proj}_\mathbb{P}(\mathbf{k})$. Then

$$\begin{aligned}
\rho_{s,\mathbf{e}}(\mathcal{L}+\mathbf{k}) &= \sum_{\mathbf{x} \in \mathcal{L}} \exp(-\pi\|\mathbf{x}+\mathbf{k}-\mathbf{e}\|^2/s^2) \\
&= \sum_{\mathbf{x} \in \mathcal{L}} \exp(-\pi\|\mathbf{x}+\mathbf{k}_\mathbb{L}-\mathbf{e}_\mathbb{L}+\mathbf{k}_\mathbb{P}-\mathbf{e}_\mathbb{P}\|^2/s^2) \\
&= \sum_{\mathbf{x} \in \mathcal{L}} \exp(-\pi\|\mathbf{x}+\mathbf{k}_\mathbb{L}-\mathbf{e}_\mathbb{L}\|^2/s^2) \cdot \exp(-\pi\|\mathbf{k}_\mathbb{P}-\mathbf{e}_\mathbb{P}\|^2/s^2) \\
&= \rho_{s,\mathbf{e}_\mathbb{L}-\mathbf{k}_\mathbb{L}}(\mathcal{L}) \cdot \exp(-\pi\|\mathbf{k}_\mathbb{P}-\mathbf{e}_\mathbb{P}\|^2/s^2) \\
&\leq (1+\epsilon) \cdot \rho_{s,\mathbf{e}_\mathbb{L}}(\mathcal{L}) \cdot \exp(-\pi\|\mathbf{k}_\mathbb{P}-\mathbf{e}_\mathbb{P}\|^2/s^2),
\end{aligned}$$

where the third equality is due to $\mathbf{x} + \mathbf{k}_{\mathbb{L}} - \mathbf{e}_{\mathbb{L}}$ being orthogonal to $\mathbf{k}_{\mathbb{P}} - \mathbf{e}_{\mathbb{P}}$, and the last inequality follows from $s \geq \eta_\epsilon(\mathcal{L})$ and Lemma 10. Therefore,

$$\begin{aligned}
\rho_{s,\mathbf{e}}(\Lambda) &\leq \sum_{\mathbf{k}\in\Lambda/\mathcal{L}} (1+\epsilon)\cdot\rho_{s,\mathbf{e}_{\mathbb{L}}}(\mathcal{L})\cdot\exp(-\pi\|\mathbf{k}_{\mathbb{P}}-\mathbf{e}_{\mathbb{P}}\|^2/s^2)\\
&= (1+\epsilon)\cdot\rho_{s,\mathbf{e}_{\mathbb{L}}}(\mathcal{L})\cdot\left(\exp(-\pi\|\mathbf{e}_{\mathbb{P}}\|^2/s^2) + \sum_{\mathbf{k}\in(\Lambda/\mathcal{L})\setminus\{\mathbf{0}\}}\exp(-\pi\|\mathbf{k}_{\mathbb{P}}-\mathbf{e}_{\mathbb{P}}\|^2/s^2)\right)\\
&= (1+\epsilon)\cdot\rho_{s,\mathbf{e}_{\mathbb{L}}}(\mathcal{L})\cdot\exp(-\pi\|\mathbf{e}_{\mathbb{P}}\|^2/s^2)\cdot\left(1+\exp(\pi\|\mathbf{e}_{\mathbb{P}}\|^2/s^2)\sum_{\mathbf{k}\in(\Lambda/\mathcal{L})\setminus\{\mathbf{0}\}}\exp(-\pi\|\mathbf{k}_{\mathbb{P}}-\mathbf{e}_{\mathbb{P}}\|^2/s^2)\right).
\end{aligned}$$

Now recall $\|\mathbf{e}_{\mathbb{P}}\| \leq \|\mathbf{e}\| \leq s$. Therefore, $\exp(\pi\|\mathbf{e}_{\mathbb{P}}\|^2/s^2) \leq \exp(\pi) = O(1)$, and

$$\rho_{s,\mathbf{e}}(\Lambda) \leq (1+\epsilon)\cdot\rho_{s,\mathbf{e}_{\mathbb{L}}}(\mathcal{L})\cdot\exp(-\pi\|\mathbf{e}_{\mathbb{P}}\|^2/s^2)\left(1+O(1)\sum_{\mathbf{k}\in(\Lambda/\mathcal{L})\setminus\{\mathbf{0}\}}\cdot\exp(-\pi\|\mathbf{k}_{\mathbb{P}}-\mathbf{e}_{\mathbb{P}}\|^2/s^2)\right).$$

In the remaining we show

$$\sum_{\mathbf{k}\in(\Lambda/\mathcal{L})\setminus\{\mathbf{0}\}}\exp(-\pi\|\mathbf{k}_{\mathbb{P}}-\mathbf{e}_{\mathbb{P}}\|^2/s^2) \leq \mathsf{negl}(\lambda)$$

with overwhelming probability in λ (over the randomness of $(\mathbf{B},\mathsf{td}) \leftarrow \mathsf{pTrapGen}(1^t, 1^k, q)$), in which case it follows that

$$\frac{\rho_{s,\mathbf{e}}(\mathcal{L})}{\rho_{s,\mathbf{e}}(\Lambda)} \geq \frac{1}{(1+\epsilon)(1+O(1)\cdot\mathsf{negl}(\lambda))} \geq 1-\mathsf{negl}(\lambda),$$

completing the proof.

Let $\Lambda_{\mathbb{P}} := \left\{\mathbf{k}_{\mathbb{P}} \in \mathbb{H}^k : \exists \mathbf{k}\in\Lambda, \mathsf{proj}_{\mathbb{P}}(\mathbf{k}) = \mathbf{k}_{\mathbb{P}}\right\}$, i.e. the lattice containing all vectors $\mathbf{k}_{\mathbb{P}}$ which are the component of some $\mathbf{k}\in\Lambda/\mathcal{L}$ orthogonal to $\mathcal{L}$. For any $\mathbf{k}_{\mathbb{P}} \in \Lambda_{\mathbb{P}}\setminus\{\mathbf{0}\}$, since $\mathbf{k}_{\mathbb{P}} \notin \mathcal{L}$ by definition, by Lemma 14 we have $\|\mathbf{k}_{\mathbb{P}}\| > \beta_\Lambda$ with overwhelming probability in λ.[26] Conditioned on this event, it holds that

$$\|\mathbf{k}_{\mathbb{P}}\| > s\sqrt{k\varphi} + s$$

by the choice of β_Λ, hence $\Lambda_{\mathbb{P}} \cap (s\sqrt{k\varphi}+s)\mathcal{B} = \{\mathbf{0}\}$. Moreover, since $\|\mathbf{e}_{\mathbb{P}}\| \leq \|\mathbf{e}\| \leq s$, this implies $\Lambda_{\mathbb{P}} \cap (s\sqrt{k\varphi}+\|\mathbf{e}_{\mathbb{P}}\|)\mathcal{B} = \{\mathbf{0}\}$. We thus obtain

$$\begin{aligned}
\sum_{\mathbf{k}\in(\Lambda/\mathcal{L})\setminus\{\mathbf{0}\}}\exp(-\pi\|\mathbf{k}_{\mathbb{P}}-\mathbf{e}_{\mathbb{P}}\|^2/s^2) &= \rho_{s,\mathbf{e}_{\mathbb{P}}}(\Lambda_{\mathbb{P}}\setminus\{\mathbf{0}\})\\
&= \rho_{s,\mathbf{e}_{\mathbb{P}}}(\Lambda_{\mathbb{P}}\setminus(s\sqrt{k\varphi}+\|\mathbf{e}_{\mathbb{P}}\|)\mathcal{B})\\
&\leq \frac{2C^{k\varphi}}{\rho_s(\mathbf{e}_{\mathbb{P}})-2C^{k\varphi}}\cdot\rho_{s,\mathbf{e}_{\mathbb{P}}}(\Lambda_{\mathbb{P}}\cap(s\sqrt{k\varphi}+\|\mathbf{e}_{\mathbb{P}}\|)\mathcal{B})\\
&\leq \frac{2C^{k\varphi}}{\exp(-\pi)-2C^{k\varphi}}\cdot\rho_{s,\mathbf{e}_{\mathbb{P}}}(\Lambda_{\mathbb{P}}\cap(s\sqrt{k\varphi}+\|\mathbf{e}_{\mathbb{P}}\|)\mathcal{B})
\end{aligned}$$

for some $C<1$, where the first equality is by definition, the second due to $\Lambda_{\mathbb{P}} \cap (s\sqrt{k\varphi}+\|\mathbf{e}_{\mathbb{P}}\|)\mathcal{B} = \{\mathbf{0}\}$, the first inequality follows from applying Corollary 4 and letting $c=1$, and the last line by $\rho_s(\mathbf{e}_{\mathbb{P}}) \geq \rho_s(\mathbf{e}) \geq \exp(-\pi\frac{s^2}{s^2}) = \exp(-\pi)$. Finally, since $\Lambda_{\mathbb{P}} \cap (s\sqrt{k\varphi}+\|\mathbf{e}_{\mathbb{P}}\|)\mathcal{B} = \{\mathbf{0}\}$,

$$\sum_{\mathbf{k}\in(\Lambda/\mathcal{L})\setminus\{\mathbf{0}\}}\exp(-\pi\|\mathbf{k}_{\mathbb{P}}-\mathbf{e}_{\mathbb{P}}\|^2/s^2) \leq \frac{2C^{k\varphi}}{\exp(-\pi)-2C^{k\varphi}}\cdot 1 = \mathsf{negl}(\lambda). \qquad \square$$

[26] Indeed Lemma 14 implies the stronger statement that at least one entry of $\mathbf{k}_{\mathbb{P}}$ has norm $> \beta_\Lambda$.

We are ready to prove Lemma 16.

Lemma 16. *Let $\mathcal{R}, q, t, k, \chi_f, s, \beta_\Lambda$ be parametrised by λ, where q is a rational prime unramified in $\mathcal{R}$ and splitting into g prime ideals. Denote $\kappa := \frac{1}{2} \cdot q^{\frac{1}{g}} \cdot \varphi^{\frac{1}{4}} / (\varphi+1)^{1+\frac{2}{\varphi}}$. Let* pTrapGen *be defined as in Fig. 3.*

Suppose $\varphi \geq \lambda$, $q > 4\beta_\Lambda \chi_f \sqrt{\varphi}$, $\beta_\Lambda \geq s(\sqrt{k\varphi}+1)$, $s \geq \chi_f k \varphi$, and $k \geq 2(t-1)\log q + 1$. If $t > 1$ then further suppose $\kappa > 2\beta_\Lambda \chi_f \sqrt{(k-1)\varphi}$, and $q^{\frac{1}{g}} > \chi_f \sqrt{\varphi}$. With probability $1 - \mathsf{negl}(\lambda)$ over the randomness of $(\mathbf{B}, (\mathbf{f}, \mathbf{d})) \leftarrow \mathsf{pTrapGen}(1^t, 1^k, q)$, for any $\mathbf{e} \in \mathcal{R}^k$ with $\|\mathbf{e}\| \leq s$, and any $\mathbf{c} \in \mathcal{R}^k$ such that there exists $\mathbf{r} \in \mathcal{R}^t$, $\mathbf{c}^{\mathrm{T}} = \mathbf{r}^{\mathrm{T}}\mathbf{B} + \mathbf{e}^{\mathrm{T}} \bmod q$, it holds:

$$\mathcal{D}_{\Lambda_q(\mathbf{B}),s,\mathbf{e}} \approx_s \mathcal{D}_{\mathcal{L}(\mathbf{f}^{\mathrm{T}}),s,\mathbf{e}} \qquad \text{and} \qquad \mathcal{D}_{\Lambda_q(\mathbf{B})+\mathbf{c},s} \approx_s \mathcal{D}_{\mathcal{L}(\mathbf{f}^{\mathrm{T}})+\mathbf{e},s}.$$

Proof. Fix any $(\mathbf{B}, (\mathbf{f}, \mathbf{d})) \in \mathsf{pTrapGen}(1^t, 1^k, q)$. For the first statement, by directly inspecting the statistical distance, we have

$$\begin{aligned}
&\Delta\left(\mathcal{D}_{\Lambda_q(\mathbf{B}),s,\mathbf{e}}, \mathcal{D}_{\mathcal{L}(\mathbf{f}^{\mathrm{T}}),s,\mathbf{e}}\right)\\
&= \frac{1}{2}\sum_{\mathbf{x}\in\mathcal{L}(\mathbf{f}^{\mathrm{T}})} \left|\mathcal{D}_{\mathcal{L}(\mathbf{f}^{\mathrm{T}}),s,\mathbf{e}}(\mathbf{x}) - \mathcal{D}_{\Lambda_q(\mathbf{B}),s,\mathbf{e}}(\mathbf{x})\right| + \frac{1}{2}\sum_{\mathbf{x}\in\Lambda_q(\mathbf{B})\setminus\mathcal{L}(\mathbf{f}^{\mathrm{T}})} \left|\mathcal{D}_{\Lambda_q(\mathbf{B}),s,\mathbf{e}}(\mathbf{x})\right|\\
&= \frac{1}{2}\sum_{\mathbf{x}\in\mathcal{L}(\mathbf{f}^{\mathrm{T}})} \left|\frac{\rho_{s,\mathbf{e}}(\mathbf{x})}{\rho_{s,\mathbf{e}}(\mathcal{L}(\mathbf{f}^{\mathrm{T}}))} - \frac{\rho_{s,\mathbf{e}}(\mathbf{x})}{\rho_{s,\mathbf{e}}(\Lambda_q(\mathbf{B}))}\right| + \frac{1}{2}\sum_{\mathbf{x}\in\Lambda_q(\mathbf{B})\setminus\mathcal{L}(\mathbf{f}^{\mathrm{T}})} \frac{\rho_{s,\mathbf{e}}(\mathbf{x})}{\rho_{s,\mathbf{e}}(\Lambda_q(\mathbf{B}))}\\
&= \frac{1}{2}\sum_{\mathbf{x}\in\mathcal{L}(\mathbf{f}^{\mathrm{T}})} \left|\left(1 - \frac{\rho_{s,\mathbf{e}}(\mathcal{L}(\mathbf{f}^{\mathrm{T}}))}{\rho_{s,\mathbf{e}}(\Lambda_q(\mathbf{B}))}\right)\frac{\rho_{s,\mathbf{e}}(\mathbf{x})}{\rho_{s,\mathbf{e}}(\mathcal{L}(\mathbf{f}^{\mathrm{T}}))}\right| + \frac{1}{2}\frac{\rho_{s,\mathbf{e}}(\Lambda_q(\mathbf{B})\setminus\mathcal{L}(\mathbf{f}^{\mathrm{T}}))}{\rho_{s,\mathbf{e}}(\Lambda_q(\mathbf{B}))}\\
&= \frac{1}{2}\left(1 - \frac{\rho_{s,\mathbf{e}}(\mathcal{L}(\mathbf{f}^{\mathrm{T}}))}{\rho_{s,\mathbf{e}}(\Lambda_q(\mathbf{B}))}\right) + \frac{1}{2}\left(1 - \frac{\rho_{s,\mathbf{e}}(\mathcal{L}(\mathbf{f}^{\mathrm{T}}))}{\rho_{s,\mathbf{e}}(\Lambda_q(\mathbf{B}))}\right)\\
&= 1 - \frac{\rho_{s,\mathbf{e}}(\mathcal{L}(\mathbf{f}^{\mathrm{T}}))}{\rho_{s,\mathbf{e}}(\Lambda_q(\mathbf{B}))}.
\end{aligned}$$

The claim then follows by applying Lemma 15, which says that, with probability $1 - \mathsf{negl}(\lambda)$ over the randomness of $(\mathbf{B}, (\mathbf{f}, \mathbf{d})) \leftarrow \mathsf{pTrapGen}(1^t, 1^k, q)$, it holds that

$$\frac{\rho_{s,\mathbf{e}}(\mathcal{L}(\mathbf{f}^{\mathrm{T}}))}{\rho_{s,\mathbf{e}}(\Lambda_q(\mathbf{B}))} \geq 1 - \mathsf{negl}(\lambda).$$

For the second statement, observe that

$$\mathcal{D}_{\Lambda_q(\mathbf{B})+\mathbf{c},s} = \mathcal{D}_{\Lambda_q(\mathbf{B})+\mathbf{e},s} = \mathbf{e} + \mathcal{D}_{\Lambda_q(\mathbf{B}),s,-\mathbf{e}}$$

and similarly

$$\mathcal{D}_{\mathcal{L}(\mathbf{f}^{\mathrm{T}})+\mathbf{e},s} = \mathbf{e} + \mathcal{D}_{\mathcal{L}(\mathbf{f}^{\mathrm{T}}),s,-\mathbf{e}},$$

hence it suffices to consider the statistical distance between $\mathcal{D}_{\Lambda_q(\mathbf{B}),s,-\mathbf{e}}$ and $\mathcal{D}_{\mathcal{L}(\mathbf{f}^{\mathrm{T}}),s,-\mathbf{e}}$. We apply the same argument as for the first statement, with the only difference being replacing $\mathbf{e}$ by $-\mathbf{e}$, whose norm is unchanged. $\square$

B.3 Proofs of Primal Trapdoor Properties

In this section we establish that the primal lattice trapdoor construction Π_{PTD} in Fig. 3 satisfy the two properties specified in Definition 8, namely (1) equivocality of $\mathbf{B}$ sampled by pTrapGen and (2) a Gaussian distribution of $\tilde{\mathbf{r}}^{\mathrm{T}}\mathbf{B} \bmod q$ induced by Equivocate. These will be summarised by Theorems 2 and 3 respectively.

Recalling from Definition 7, to show that $\mathbf{B}$ sampled by pTrapGen is equivocal, we need to show two properties: (a) $\Lambda_q(\mathbf{B})$ admits a dense sublattice, and (b) the matrix $\mathbf{B}$ is pseudorandom given leakage. To show (a), we establish two results on the entropy of Gaussian over lattices: The first one, given by Lemma 17, relates the entropy of Gaussian over a lattice to that over its sublattice; The second one, given by Lemma 18, says that Gaussian over a lattice centred at any $\mathbf{c}$ (possibly not in the span of the lattice) is identical to Gaussian centred at the projection of $\mathbf{c}$ onto the lattice.

Lemma 17. *Let $s > 0$, $\mathcal{L} \subseteq \Lambda \subset \mathbb{H}^m$ be lattices, $\mathbf{c} \in \mathbb{H}^m$ and $\mathbf{c}_\Lambda$ be the shortest vector in $\Lambda + \mathbf{c}$. It holds that*

$$H_\infty(\mathcal{D}_{\Lambda,s,\mathbf{c}}) \geq H_\infty(\mathcal{D}_{\mathcal{L},s,\mathbf{c}_\Lambda}).$$

Proof. Note that $H_\infty(\mathcal{D}_{\Lambda,s,\mathbf{c}}) = H_\infty(\mathcal{D}_{\Lambda,s,\mathbf{c}_\Lambda})$ since shifting the centre $\mathbf{c}$ by $\mathbf{c}_\Lambda - \mathbf{c} \in \Lambda$ only permutes the probability masses and in particular does not change the min-entropy.

Next, we show that

$$\max_{\mathbf{x}\in\Lambda} \mathcal{D}_{\Lambda,s,\mathbf{c}_\Lambda}(\mathbf{x}) = \max_{\mathbf{x}\in\mathcal{L}} \mathcal{D}_{\Lambda,s,\mathbf{c}_\Lambda}(\mathbf{x}). \tag{12}$$

The "$\geq$" direction is trivial since the L.H.S. is maximising over a superset of that in the R.H.S. For the "$\leq$" direction, suppose towards a contradiction that Eq. (12) is false, i.e. there exists $\mathbf{x} \in \Lambda$ such that for all $\mathbf{y} \in \mathcal{L}$ it holds that

$$\mathcal{D}_{\Lambda,s,\mathbf{c}_\Lambda}(\mathbf{x}) > \mathcal{D}_{\Lambda,s,\mathbf{c}_\Lambda}(\mathbf{y}).$$

Writing $\mathcal{D}_{\Lambda,s,\mathbf{c}_\Lambda}(\cdot) = \rho_{s,\mathbf{c}_\Lambda}(\cdot)/\rho_{s,\mathbf{c}}(\Lambda)$ and examining $\rho_{s,\mathbf{c}_\Lambda}(\cdot)$, we conclude that

$$\|\mathbf{c}_\Lambda - \mathbf{x}\| < \|\mathbf{c}_\Lambda - \mathbf{y}\|.$$

Since this holds for all $\mathbf{y} \in \mathcal{L}$, we can substitute $\mathbf{y} = \mathbf{0}$ and conclude that

$$\|\mathbf{c}_\Lambda - \mathbf{x}\| < \|\mathbf{c}_\Lambda\|.$$

However, this means that there exists a vector $\mathbf{c}' := \mathbf{c}_\Lambda - \mathbf{x} \in \Lambda + \mathbf{c}$ which is shorter than $\mathbf{c}_\Lambda$, contradicting the fact that $\mathbf{c}_\Lambda$ is the shortest vector in $\Lambda + \mathbf{c}$.

It remains to show that, for any $\mathbf{x} \in \mathcal{L}$, it holds that

$$\mathcal{D}_{\Lambda,s,\mathbf{c}_\Lambda}(\mathbf{x}) \leq \mathcal{D}_{\mathcal{L},s,\mathbf{c}_\Lambda}(\mathbf{x}).$$

This can be seen by a direct calculation:

$$\mathcal{D}_{\Lambda,s,\mathbf{c}_\Lambda}(\mathbf{x}) = \frac{\rho_{s,\mathbf{c}_\Lambda}(\mathbf{x})}{\rho_{s,\mathbf{c}}(\Lambda)} = \frac{\rho_{s,\mathbf{c}_\Lambda}(\mathbf{x})}{\rho_{s,\mathbf{c}}(\mathcal{L}) + \rho_{s,\mathbf{c}}(\Lambda \setminus \mathcal{L})} \leq \frac{\rho_{s,\mathbf{c}_\Lambda}(\mathbf{x})}{\rho_{s,\mathbf{c}}(\mathcal{L})} = \mathcal{D}_{\mathcal{L},s,\mathbf{c}_\Lambda}(\mathbf{x})$$

where the inequality holds since $\rho_{s,\mathbf{c}_\Lambda}(\Lambda \setminus \mathcal{L}) \geq 0$.

Putting everything together, we have

$$\begin{aligned}
H_\infty(\mathcal{D}_{\Lambda,s,\mathbf{c}}) = H_\infty(\mathcal{D}_{\Lambda,s,\mathbf{c}_\Lambda}) &= -\log \max_{\mathbf{x}\in\Lambda} \mathcal{D}_{\Lambda,s,\mathbf{c}_\Lambda}(\mathbf{x}) \\
&= -\log \max_{\mathbf{x}\in\mathcal{L}} \mathcal{D}_{\Lambda,s,\mathbf{c}_\Lambda}(\mathbf{x}) \\
&\geq -\log \max_{\mathbf{x}\in\mathcal{L}} \mathcal{D}_{\mathcal{L},s,\mathbf{c}_\Lambda}(\mathbf{x}) = H_\infty(\mathcal{D}_{\mathcal{L},s,\mathbf{c}_\Lambda}).
\end{aligned}$$ ◻

Lemma 18. *Let $\mathcal{L} \subset \mathbb{H}^m$ be a lattice, $\mathbb{L} = \mathsf{Span}(\mathcal{L})$, $s > 0$, $\mathbf{c} \in \mathbb{H}^m$ and $\mathbf{c}_\mathbb{L} = \mathsf{proj}_\mathbb{L}(\mathbf{c})$. It holds that*

$$\mathcal{D}_{\mathcal{L},s,\mathbf{c}} \equiv \mathcal{D}_{\mathcal{L},s,\mathbf{c}_\mathbb{L}}.$$

Proof. The statement is trivial for $\mathbf{c} \in \mathsf{Span}(\mathcal{L})$. In the following, suppose $\mathbf{c} \notin \mathsf{Span}(\mathcal{L})$. Write $\mathbf{c} = \mathbf{c}_\mathbb{L} + \mathbf{c}_\mathbb{P}$, where $\mathbf{c}_\mathbb{P} \in \mathbb{R}^n \setminus \mathbb{L}$ is orthogonal to $\mathbb{L}$. By definition of the Gaussian function, for any $\mathbf{x} \in \mathcal{L}$,

$$\rho_{\mathcal{L},s,\mathbf{c}}(\mathbf{x}) = \exp\left(-\pi \frac{\|\mathbf{x} - \mathbf{c}\|^2}{s^2}\right)$$

$$= \exp\left(-\pi \frac{\|\mathbf{x} - \mathbf{c}_{\mathbb{L}}\|^2 + \|\mathbf{c}_{\mathbb{P}}\|^2}{s^2}\right)$$

$$= \exp\left(-\pi \frac{\|\mathbf{x} - \mathbf{c}_{\mathbb{L}}\|^2}{s^2}\right) \cdot \exp\left(-\pi \frac{\|\mathbf{c}_{\mathbb{P}}\|^2}{s^2}\right) \quad = \rho_{\mathcal{L},s,\mathbf{c}_{\mathbb{L}}}(\mathbf{x}) \cdot \exp\left(-\pi \frac{\|\mathbf{c}_{\mathbb{P}}\|^2}{s^2}\right).$$

where the second equality is due to $\mathbf{c}_{\mathbb{P}}$ being orthogonal to $\mathbb{L}$. Then by definition of the Gaussian distribution, for any $\mathbf{x} \in \mathcal{L}$,

$$\mathcal{D}_{\mathcal{L},s,\mathbf{c}}(\mathbf{x}) = \frac{\rho_{s,\mathbf{c}}(\mathbf{x})}{\sum_{\mathbf{z}\in\mathcal{L}} \rho_{s,\mathbf{c}}(\mathbf{z})} = \frac{\rho_{s,\mathbf{c}_{\mathbb{L}}}(\mathbf{x}) \cdot \exp\left(-\pi \frac{\|\mathbf{c}_{\mathbb{P}}\|^2}{s^2}\right)}{\sum_{\mathbf{z}\in\mathcal{L}} \rho_{s,\mathbf{c}_{\mathbb{L}}}(\mathbf{z}) \cdot \exp\left(-\pi \frac{\|\mathbf{c}_{\mathbb{P}}\|^2}{s^2}\right)} = \frac{\rho_{s,\mathbf{c}_{\mathbb{L}}}(\mathbf{x})}{\sum_{\mathbf{z}\in\mathcal{L}} \rho_{s,\mathbf{c}_{\mathbb{L}}}(\mathbf{z})} = \mathcal{D}_{\mathcal{L},s,\mathbf{c}_{\mathbb{L}}}(\mathbf{x}). \qquad \square$$

We are now ready to prove that $\Lambda_q(\mathbf{B})$ admits a dense sublattice.

Theorem 8 (Dense Sublattice). *Let $\mathcal{R}, q, t, k, \chi_f, s$ be parametrised by λ. Let $\mathsf{pTrapGen}$ be defined as in Fig. 3. Define $\mathcal{E} := \{\mathbf{B} \mid (\mathbf{B}, \mathsf{td} = (\mathbf{B}, \mathbf{f}, \mathbf{d})) \leftarrow \mathsf{pTrapGen}(1^t, 1^k, q)\}$.*

Suppose $\varphi \geq \lambda$ and $s \geq 2\chi_f k\varphi$. With probability $1 - \mathsf{negl}(\lambda)$ over the randomness of $\mathbf{B} \leftarrow\!\$\ \mathcal{E}$, for any $\mathbf{c} \in \mathcal{R}^k$, it holds that

$$H_\infty(\mathcal{D}_{\Lambda_q(\mathbf{B}),s,\mathbf{c}}) \geq \omega(\log \lambda).$$

Proof. Denote by the shorthands $\Lambda := \Lambda_q(\mathbf{B})$ and $\mathcal{L} := \mathcal{L}(\mathbf{f}^{\mathsf{T}})$. Denote by $\mathbb{L} := \mathsf{span}(\mathcal{L})$ which is of rank φ.

First we note that, by the choice of parameters, with overwhelming probability it holds that $s \geq 2\eta_\epsilon(\mathcal{L})$ for some $\epsilon \in \mathsf{negl}(\lambda)$. This follows from Lemma 9 which says that $\eta_\epsilon(\mathcal{L}) \leq \sqrt{k\varphi}\lambda_\varphi(\mathcal{L}) \leq \sqrt{k\varphi}\|\mathbf{f}\|$, and from Corollary 3 and [SS11, Lemma 4.2, 4.4 (eprint version)], which says $\|\mathbf{f}\| \leq \chi_f\sqrt{k\varphi}$ with overwhelming probability in $k\varphi \geq \lambda$.

For any $\mathbf{c} \in \mathbb{H}^m$, since $\mathcal{L} \subseteq \Lambda$, by Lemmas 17 and 18, we have

$$H_\infty(\mathcal{D}_{\Lambda,s,\mathbf{c}}) \geq H_\infty(\mathcal{D}_{\mathcal{L},s,\mathbf{c}_\Lambda}) = H_\infty(\mathcal{D}_{\mathcal{L},s,\mathbf{c}_{\mathbb{L}}})$$

where $\mathbf{c}_\Lambda \in \mathbb{H}^m$ is the shortest representative of $\mathbf{c}$ modulo Λ and $\mathbf{c}_{\mathbb{L}} = \mathsf{proj}_{\mathbb{L}}(\mathbf{c}_\Lambda) \in \mathbb{L}$. Then, by Lemma 11, for any $\epsilon > 0$ and $\mathbf{c} \in \mathbb{H}^m$,

$$\mathcal{D}_{\mathcal{L},s,\mathbf{c}_{\mathbb{L}}}(\mathbf{y}) \leq 2^{-\varphi} \cdot \frac{1+\epsilon}{1-\epsilon}$$

for any $\mathbf{y} \in \mathcal{L}$. Therefore

$$H_\infty(\mathcal{D}_{\mathcal{L},s,\mathbf{c}_{\mathbb{L}}}) \geq \varphi - \log\frac{1+\epsilon}{1-\epsilon} \geq \varphi - \log 3,$$

where the last inequality follows from $\varphi \geq \lambda$ and $\epsilon \leq 1/2$. $\square$

Next we argue that $\mathbf{B}$ is pseudorandom given leakage. For this we first prove Lemma 19 below, which guarantees the invertibility of the coefficient for a Gaussian-sample from $\mathcal{L}(\mathbf{f}^{\mathsf{T}})$. This will be useful in the subsequent Theorem 9, where we show that $\mathbf{B}$ is pseudorandom given leakage under the decisional NTRU assumption.

Lemma 19. *Let $\mathcal{R}, q, k, s$ be paramterised by λ, and let g be the number of (potentially not distinct) prime ideal factors of $\langle q \rangle$. Fix any $\mathbf{f} \in \mathcal{R}^k$.*

Suppose $\varphi \geq \lambda$, $q^{\frac{1}{g}} \geq s\sqrt{k}$, and $s \geq 2\eta_\epsilon(\mathcal{L}(\mathbf{f}^{\mathsf{T}}))$ for some $\epsilon \leq \mathsf{negl}(\lambda)$. Then for any $\mathbf{e} \in \mathcal{R}^k$ satisfying $\|\mathbf{e}\| \leq s$, it holds that

$$\Pr\left[p \notin \mathcal{R}_q^\times \mid p \cdot \mathbf{f}^{\mathsf{T}} \leftarrow\!\$\ \mathcal{D}_{\mathcal{L}(\mathbf{f}^{\mathsf{T}}),s,\mathbf{e}}\right] \leq \mathsf{negl}(\lambda).$$

Proof. Let $\langle q \rangle$ be factored into (potentially not distinct) prime ideals as $\langle q \rangle = \prod_{j=1}^{g} \mathfrak{q}_j$, where $\mathcal{N}(\mathfrak{q}_j) = q^{\varphi/g}$.

Consider $p \cdot \mathbf{f}^{\mathrm{T}}$ sampled from $\mathcal{D}_{\mathcal{L}(\mathbf{f}^{\mathrm{T}}),s,\mathbf{e}}$. Suppose $p \notin \mathcal{R}_q^\times$, then it must hold that $p \in \mathfrak{q}_j$ for some j, and therefore $p \cdot \mathbf{f} \in \mathfrak{q}_j^k$. The last event is equivalent to $p \cdot \mathbf{f} = \mathbf{0} \bmod \mathfrak{q}_j^k$. We conclude

$$\begin{aligned}
&\Pr\left[p \notin \mathcal{R}_q^\times \;\middle|\; p \cdot \mathbf{f}^{\mathrm{T}} \leftarrow\$\, \mathcal{D}_{\mathcal{L}(\mathbf{f}^{\mathrm{T}}),s,\mathbf{e}}\right] \\
\leq\; &\Pr\left[p \cdot \mathbf{f} = \mathbf{0} \bmod \mathfrak{q}_j^k \;\middle|\; p \cdot \mathbf{f}^{\mathrm{T}} \leftarrow\$\, \mathcal{D}_{\mathcal{L}(\mathbf{f}^{\mathrm{T}}),s,\mathbf{e}}\right] \\
=\; &\Pr\left[p \cdot \mathbf{f} = \mathbf{0} \;\middle|\; p \cdot \mathbf{f}^{\mathrm{T}} \leftarrow\$\, \mathcal{D}_{\mathcal{L}(\mathbf{f}^{\mathrm{T}}),s,\mathbf{e}}\right] + \Pr\left[p \cdot \mathbf{f} \neq \mathbf{0} \wedge p \cdot \mathbf{f} = \mathbf{0} \bmod \mathfrak{q}_j \;\middle|\; p \cdot \mathbf{f}^{\mathrm{T}} \leftarrow\$\, \mathcal{D}_{\mathcal{L}(\mathbf{f}^{\mathrm{T}}),s,\mathbf{e}}\right],
\end{aligned}$$

where in the last line we distinguishes the cases (1) $p \cdot \mathbf{f} = \mathbf{0}$ (without modding), and (2) $p \cdot \mathbf{f} \neq \mathbf{0}$ (without modding) but $p \cdot \mathbf{f} = \mathbf{0} \bmod \mathfrak{q}_j$. We upper-bound the probability of each of these cases. Note that $\exp\left(\pi \|\mathbf{e}\|^2 / s^2\right) \leq \exp(\pi)$ in below holds, since $s \geq \|\mathbf{e}\|$ by choice of parameters.

For the first term: Since $s \geq 2\eta_\epsilon(\mathcal{L}(\mathbf{f}^{\mathrm{T}}))$ by the choice of parameters, applying Lemmas 11 and 18, we obtain

$$\begin{aligned}
\Pr\left[p \cdot \mathbf{f}^{\mathrm{T}} = \mathbf{0} \;\middle|\; p \cdot \mathbf{f}^{\mathrm{T}} \leftarrow\$\, \mathcal{D}_{\mathcal{L}(\mathbf{f}^{\mathrm{T}}),s,\mathbf{e}}\right] &= \mathcal{D}_{\mathcal{L}(\mathbf{f}^{\mathrm{T}}),s,\mathbf{e}}(\mathbf{0}) \\
&= \mathcal{D}_{\mathcal{L}(\mathbf{f}^{\mathrm{T}}),s,\mathsf{proj}_{\mathcal{L}(\mathbf{f}^{\mathrm{T}})}(\mathbf{e})}(\mathbf{0}) \\
&\leq 2^{-\varphi} \frac{1+\epsilon}{1-\epsilon} \leq 3 \cdot 2^{-\varphi} \leq \mathsf{negl}(\lambda).
\end{aligned}$$

For the second term: We notice that the event $p \cdot \mathbf{f} \neq \mathbf{0} \wedge p \cdot \mathbf{f} = \mathbf{0} \bmod \mathfrak{q}_j$ implies

$$\|p \cdot \mathbf{f}\| > \lambda_1(\mathfrak{q}_j) \geq \sqrt{\varphi} \mathcal{N}(\mathfrak{q}_j)^{1/\varphi} = \sqrt{\varphi} q^{\frac{1}{g}} \geq s\sqrt{k\varphi}$$

where the second inequality follows from the inequality of the arithmetic and geometric means, and the last holds since $q^{\frac{1}{g}} \geq s\sqrt{k}$ by choice of parameters. Together with Corollary 4, we have

$$\begin{aligned}
&\Pr\left[p \cdot \mathbf{f} \neq \mathbf{0} \wedge p \cdot \mathbf{f} = \mathbf{0} \bmod \mathfrak{q}_j \;\middle|\; p \cdot \mathbf{f}^{\mathrm{T}} \leftarrow\$\, \mathcal{D}_{\mathcal{L}(\mathbf{f}^{\mathrm{T}}),s,\mathbf{e}}\right] \\
\leq\; &\Pr\left[\|p \cdot \mathbf{f}\| > s\sqrt{k\varphi} \;\middle|\; p \cdot \mathbf{f}^{\mathrm{T}} \leftarrow\$\, \mathcal{D}_{\mathcal{L}(\mathbf{f}^{\mathrm{T}}),s,\mathbf{e}}\right] \leq 2C^{k\varphi} / \rho_s(\mathbf{e})
\end{aligned}$$

for some $C < 1$. By direct calculation, $2C^{k\varphi}/\rho_s(\mathbf{e}) = 2C^{k\varphi} \cdot \exp(\pi \frac{\|\mathbf{e}\|}{s^2}) \leq 2C^{k\varphi} \cdot \exp(\pi) \leq \mathsf{negl}(\lambda)$.

Putting the above together yields the claim. ☐

Theorem 9 (Pseudorandom with Leakage). *Let $\mathcal{R}, q, t, k, \chi_f, s, \beta_\Lambda$ be parametrised by λ, where q is a rational prime unramified in $\mathcal{R}$ and splitting into g prime ideals. Denote $\kappa := \frac{1}{2} \cdot q^{\frac{1}{g}} \cdot \varphi^{\frac{1}{4}} / (\varphi+1)^{1+\frac{2}{\varphi}}$. Let $\mathsf{pTrapGen}$ be defined as in Fig. 3. Define $\mathcal{E} := \left\{\mathbf{B} \;\middle|\; (\mathbf{B}, \mathsf{td}) \leftarrow \mathsf{pTrapGen}(1^t, 1^k, q)\right\}$.*

Suppose $\varphi \geq \lambda$, $q > 4\beta_\Lambda \chi_f \sqrt{\varphi}$, $q^{\frac{1}{g}} \geq s\sqrt{k}$, $\beta_\Lambda \geq s(\sqrt{k\varphi}+1)$, $s \geq 2\chi_f k\varphi$, and $k \geq 2(t-1)\log q + 1$. If $t > 1$ then further suppose $\kappa > 2\beta_\Lambda \chi_f \sqrt{(k-1)\varphi}$. Under the $\mathsf{dNTRU}_{\mathcal{R},q,k,\chi_f}$ assumption, for any PPT $\mathcal{A}$ and $Q \in \mathsf{poly}(\lambda)$, any $\mathbf{e}_1, \ldots, \mathbf{e}_Q \in \mathcal{R}^k$ with $\|\mathbf{e}_i\| \leq s$ for all $i \in [Q]$, it holds that

$$\left|\Pr\left[\mathsf{ExpPRwL}^0_{\mathcal{E},\mathcal{A},1,(\mathbf{e}_i)_{i\in[Q]}}(1^\lambda) = 1\right] - \Pr\left[\mathsf{ExpPRwL}^1_{\mathcal{E},\mathcal{A},1,(\mathbf{e}_i)_{i\in[Q]}}(1^\lambda) = 1\right]\right| \leq \mathsf{negl}(\lambda)$$

where the experiments $\mathsf{ExpPRwL}^b_{\mathcal{E},\mathcal{A},1,(\mathbf{e}_i)_{i\in[Q]}}$ are defined in Fig. 2.

Proof. We define a sequence of hybrids $\mathsf{Hyb}_0, \ldots, \mathsf{Hyb}_6$, whose formal descriptions are given in Fig. 8. Hyb_0 and Hyb_6 are statistically close to $\mathsf{ExpPRwL}^0_{\mathcal{E},\mathcal{A},1,(\mathbf{e}_i)_{i\in[Q]}}$ and $\mathsf{ExpPRwL}^1_{\mathcal{E},\mathcal{A},1,(\mathbf{e}_i)_{i\in[Q]}}$ respectively, with the only difference that $\mathbf{B}$ is not subject to being primitive[27], and the closeness follows from the setting that $\mathcal{R}_q$ splits into super-polynomial-size subfields. Below we show that Hyb_{i-1} is (at least) computationally indistinguishable from Hyb_i for $i \in [6]$, from which the theorem follows.

[27] This is purely for the sake of easing notation.

$\mathsf{Hyb}_0(1^\lambda)$

$\mathbf{d} \leftarrow\$\ \mathcal{R}_q^t : \langle d_1, \ldots, d_t \rangle_{\mathcal{R}_q} = \mathcal{R}_q$
$\mathbf{f} \leftarrow\$\ \mathcal{D}_{\mathcal{R}^k, \chi_f} : \langle f_1, \ldots, f_k \rangle = \mathcal{R}$
$\mathbf{B} \leftarrow\$\ \mathcal{R}_q^{t \times k} : \mathbf{d}^\mathsf{T}\mathbf{B} = \mathbf{f}^\mathsf{T} \bmod q$
for $i \in [Q]$ **do**
  $\tilde{\mathbf{r}}_i^\mathsf{T}\mathbf{B} \leftarrow\$\ \mathcal{D}_{\Lambda_q(\mathbf{B}), s, \mathbf{e}_i}$
  $x_i \leftarrow\$\ \mathcal{R}_q$
  $\mathbf{l}_i^\mathsf{T} \coloneqq x_i \tilde{\mathbf{r}}_i^\mathsf{T} \bmod q$
$b' \leftarrow \mathcal{A}(\mathbf{B}, \mathbf{l}_1, \ldots, \mathbf{l}_Q)$
return b'

$\mathsf{Hyb}_1(1^\lambda)$

$\mathbf{d} \leftarrow\$\ \mathcal{R}_q^t : \langle d_1, \ldots, d_t \rangle_{\mathcal{R}_q} = \mathcal{R}_q$
$\mathbf{f} \leftarrow\$\ \mathcal{D}_{\mathcal{R}^k, \chi_f} : \langle f_1, \ldots, f_k \rangle = \mathcal{R}$
$\mathbf{B} \leftarrow\$\ \mathcal{R}_q^{t \times k} : \mathbf{d}^\mathsf{T}\mathbf{B} = \mathbf{f}^\mathsf{T} \bmod q$
for $i \in [Q]$ **do**
  $p_i \cdot \mathbf{f}^\mathsf{T} \leftarrow\$\ \mathcal{D}_{\mathcal{L}(\mathbf{f}^\mathsf{T}), s, \mathbf{e}_i}$
  $x_i \leftarrow\$\ \mathcal{R}_q$
  $\mathbf{l}_i^\mathsf{T} \coloneqq x_i \cdot p_i \cdot \mathbf{d}^\mathsf{T} \bmod q$
$b' \leftarrow \mathcal{A}(\mathbf{B}, \mathbf{l}_1, \ldots, \mathbf{l}_Q)$
return b'

$\mathsf{Hyb}_2(1^\lambda)$

$\mathbf{d} \leftarrow\$\ \mathcal{R}_q^t : \langle d_1, \ldots, d_t \rangle_{\mathcal{R}_q} = \mathcal{R}_q$
$\mathbf{W} \leftarrow\$\ \mathcal{R}_q^{t \times t} : \det(\mathbf{W}) \in \mathcal{R}_q^\times, \mathbf{d}^\mathsf{T}\mathbf{W} = d \cdot \boldsymbol{\iota}_t^\mathsf{T} \bmod q$
$\mathbf{f} \leftarrow\$\ \mathcal{D}_{\mathcal{R}^k, \chi_f} : \langle f_1, \ldots, f_k \rangle = \mathcal{R}$
$\underline{\mathbf{a}}^\mathsf{T} \coloneqq d^{-1} \cdot \mathbf{f}^\mathsf{T} \bmod q$; $\overline{\mathbf{A}} \leftarrow\$\ \mathcal{R}_q^{(t-1) \times k}$
$\mathbf{B} \coloneqq \mathbf{W} \begin{pmatrix} \overline{\mathbf{A}} \\ \underline{\mathbf{a}}^\mathsf{T} \end{pmatrix} \bmod q$
for $i \in [Q]$ **do**
  $p_i \cdot \mathbf{f}^\mathsf{T} \leftarrow\$\ \mathcal{D}_{\mathcal{L}(\mathbf{f}^\mathsf{T}), s, \mathbf{e}_i}$
  $x_i \leftarrow\$\ \mathcal{R}_q$
  $\mathbf{l}_i^\mathsf{T} \coloneqq x_i \cdot p_i \cdot \mathbf{d}^\mathsf{T} \bmod q$
$b' \leftarrow \mathcal{A}(\mathbf{B}, \mathbf{l}_1, \ldots, \mathbf{l}_Q)$
return b'

$\mathsf{Hyb}_3(1^\lambda)$

$d \leftarrow\$\ \mathcal{R}_q^\times$; $\mathbf{W} \leftarrow\$\ \mathcal{R}_q^{t \times t} : \det(\mathbf{W}) \in \mathcal{R}_q^\times$
$\mathbf{d}^\mathsf{T} \coloneqq (0, \ldots, 0, d) \cdot \mathbf{W}^{-1} \bmod q$
$\mathbf{f} \leftarrow\$\ \mathcal{D}_{\mathcal{R}^k, \chi_f} : \langle f_1, \ldots, f_k \rangle = \mathcal{R}$
$\underline{\mathbf{a}}^\mathsf{T} \coloneqq d^{-1} \cdot \mathbf{f}^\mathsf{T} \bmod q$; $\overline{\mathbf{A}} \leftarrow\$\ \mathcal{R}_q^{(t-1) \times k}$
$\mathbf{B} \coloneqq \mathbf{W} \begin{pmatrix} \overline{\mathbf{A}} \\ \underline{\mathbf{a}}^\mathsf{T} \end{pmatrix} \bmod q$
for $i \in [Q]$ **do**
  $p_i \cdot \mathbf{f}^\mathsf{T} \leftarrow\$\ \mathcal{D}_{\mathcal{L}(\mathbf{f}^\mathsf{T}), s, \mathbf{e}_i}$
  $x_i \leftarrow\$\ \mathcal{R}_q$
  $\mathbf{l}_i^\mathsf{T} \coloneqq x_i \cdot p_i \cdot \mathbf{d}^\mathsf{T} \bmod q$
$b' \leftarrow \mathcal{A}(\mathbf{B}, \mathbf{l}_1, \ldots, \mathbf{l}_Q)$
return b'

$\mathsf{Hyb}_4(1^\lambda)$

$d \leftarrow\$\ \mathcal{R}_q^\times$; $\mathbf{W} \leftarrow\$\ \mathcal{R}_q^{t \times t} : \det(\mathbf{W}) \in \mathcal{R}_q^\times$
$\mathbf{f} \leftarrow\$\ \mathcal{D}_{\mathcal{R}^k, \chi_f} : \langle f_1, \ldots, f_k \rangle = \mathcal{R}$
$\underline{\mathbf{a}}^\mathsf{T} \coloneqq d^{-1} \cdot \mathbf{f}^\mathsf{T} \bmod q$; $\overline{\mathbf{A}} \leftarrow\$\ \mathcal{R}_q^{(t-1) \times k}$
$\mathbf{B} \coloneqq \mathbf{W} \begin{pmatrix} \overline{\mathbf{A}} \\ \underline{\mathbf{a}}^\mathsf{T} \end{pmatrix} \bmod q$
for $i \in [Q]$ **do**
  $x_i \leftarrow\$\ \mathcal{R}_q$
  $\mathbf{l}_i^\mathsf{T} \coloneqq (0, \ldots, 0, x_i) \cdot \mathbf{W}^{-1} \bmod q$
$b' \leftarrow \mathcal{A}(\mathbf{B}, \mathbf{l}_1, \ldots, \mathbf{l}_Q)$
return b'

$\mathsf{Hyb}_5(1^\lambda)$

$\mathbf{W} \leftarrow\$\ \mathcal{R}_q^{t \times t} : \det(\mathbf{W}) \in \mathcal{R}_q^\times$
$\underline{\mathbf{a}} \leftarrow\$\ \mathcal{R}_q^k$; $\overline{\mathbf{A}} \leftarrow\$\ \mathcal{R}_q^{(t-1) \times k}$
$\mathbf{B} \coloneqq \mathbf{W} \begin{pmatrix} \overline{\mathbf{A}} \\ \underline{\mathbf{a}}^\mathsf{T} \end{pmatrix} \bmod q$
for $i \in [Q]$ **do**
  $x_i \leftarrow\$\ \mathcal{R}_q$
  $\mathbf{l}_i^\mathsf{T} \coloneqq (0, \ldots, 0, x_i) \cdot \mathbf{W}^{-1} \bmod q$
$b' \leftarrow \mathcal{A}(\mathbf{B}, \mathbf{l}_1, \ldots, \mathbf{l}_Q)$
return b'

$\mathsf{Hyb}_6(1^\lambda)$

$\mathbf{B} \leftarrow\$\ \mathcal{R}_q^{t \times k}$
$\hat{\mathbf{r}} \leftarrow\$\ \mathcal{R}_q^t : \mathbf{l}^\mathsf{T}\mathcal{R}_q^t = \mathcal{R}_q$
for $i \in [Q]$ **do**
  $x_i \leftarrow\$\ \mathcal{R}_q$
  $\mathbf{l}_i^\mathsf{T} \coloneqq x_i \hat{\mathbf{r}}^\mathsf{T} \bmod q$
$b' \leftarrow \mathcal{A}(\mathbf{B}, \mathbf{l}_1, \ldots, \mathbf{l}_Q)$
return b'

Fig. 8. Hybrid experiments for proof of Theorem 9.

$\mathsf{Hyb}_0 \approx_s \mathsf{Hyb}_1$. This follows from Lemma 16 which says that $\mathcal{D}_{\Lambda_q(\mathbf{B}), s, \mathbf{e}_i} \approx_s \mathcal{D}_{\mathcal{L}(\mathbf{f}^\mathsf{T}), s, \mathbf{e}_i}$, and by realising that $p_i \cdot \mathbf{f}^\mathsf{T} = p_i \cdot \mathbf{d}^\mathsf{T}\mathbf{B} \bmod q$, therefore $\tilde{\mathbf{r}}_i^\mathsf{T} = p_i \mathbf{d}^\mathsf{T} \bmod q$.

$\mathsf{Hyb}_1 \equiv \mathsf{Hyb}_2$. We claim that Hyb_1 and Hyb_2 are identical. It suffices to argue about the distribution of $\mathbf{B}$, since that of $\mathbf{d}, \mathbf{f}$ are unchanged, and $(\mathbf{l}_1, \ldots, \mathbf{l}_Q)$ is obtained by post-processing $(\mathbf{d}, \mathbf{B}, \mathbf{f})$.

First we notice that for any $\mathbf{d}^\mathsf{T} = (d_1, \ldots, d_t)$ that is primitive, i.e. $\mathbf{d}^\mathsf{T}\mathcal{R}_q^t = \mathcal{R}_q$, there must exist a matrix $\mathbf{W} \in \mathcal{R}_q^{t \times t}$ invertible over $\mathcal{R}_q$ such that $\mathbf{d}^\mathsf{T}\mathbf{W} = d \cdot \boldsymbol{\iota}_t^\mathsf{T} \bmod q$ for some $d \in \mathcal{R}_q^\times$.[28] In Hyb_2, after sampling a uniform primitive $\mathbf{d}$, we additionally sample a uniform $\mathbf{W}$ subject to these constraints.

Next, since $\mathbf{W}$ is invertible over $\mathcal{R}_q$, picking $\mathbf{B}$ uniformly subject to $\mathbf{d}^\mathsf{T}\mathbf{B} = \mathbf{f}^\mathsf{T} \bmod q$ as in Hyb_1 is equivalent to picking $\bar{\mathbf{B}}$ uniformly subject to $\mathbf{d}^\mathsf{T}\mathbf{W}\bar{\mathbf{B}} = \mathbf{f}^\mathsf{T} \bmod q$ and then setting $\mathbf{B} = \mathbf{W}\bar{\mathbf{B}} \bmod q$. Write such $\bar{\mathbf{B}} = \begin{pmatrix} \overline{\mathbf{A}} \\ \underline{\mathbf{a}}^\mathsf{T} \end{pmatrix}$ for some $\overline{\mathbf{A}}, \underline{\mathbf{a}}^\mathsf{T}$, so that $\mathbf{d}^\mathsf{T}\mathbf{W} \begin{pmatrix} \overline{\mathbf{A}} \\ \underline{\mathbf{a}}^\mathsf{T} \end{pmatrix} = \mathbf{f}^\mathsf{T} \bmod q$. Recall $\mathbf{d}^\mathsf{T}\mathbf{W} = d \cdot \boldsymbol{\iota}_t^\mathsf{T} \bmod q$, therefore we have $d \cdot \boldsymbol{\iota}_t^\mathsf{T} \begin{pmatrix} \overline{\mathbf{A}} \\ \underline{\mathbf{a}}^\mathsf{T} \end{pmatrix} = d\underline{\mathbf{a}}^\mathsf{T} = \mathbf{f}^\mathsf{T} \bmod q$, equivalently $\underline{\mathbf{a}}^\mathsf{T} = d^{-1} \cdot \mathbf{f}^\mathsf{T} \bmod q$. On the other hand, there are no constraints on $\overline{\mathbf{A}}$ and hence it can be arbitrary. The above equivalent sampling procedure is precisely Hyb_2.

[28] For example, we can pick the first $t-1$ columns of $\mathbf{W}$ to form a basis of the right-kernel of $\mathbf{d}^\mathsf{T}$ modulo every prime ideal dividing q. This is possible since $\mathbf{d}^\mathsf{T}$ is primitive, meaning that the right-kernel of $\mathbf{d}^\mathsf{T}$ modulo every prime ideal dividing q has rank $t-1$. For the last column, since $\langle d_1, \ldots, d_t \rangle_{\mathcal{R}_q} = \mathcal{R}_q$, there must exist a vector $\mathbf{w}_t$ such that $\mathbf{d}^\mathsf{T}\mathbf{w}_t = d \in \mathcal{R}_q^\times$ for some d.

$\mathsf{Hyb}_2 \equiv \mathsf{Hyb}_3$. We claim that Hyb_2 and Hyb_3 are identical. The only change is the sampling of $(\mathbf{W}, \mathbf{d}, d)$. It is clear that the following procedures result in identically distributed $(\mathbf{W}, \mathbf{d}, d)$:

– Sample uniform $\mathbf{d}$ subject to being primitive, then sample uniform $\mathbf{W}$ subject to invertible over $\mathcal{R}_q$ and $\mathbf{d}^\mathsf{T}\mathbf{W} = d \cdot \boldsymbol{\iota}_t^\mathsf{T} \bmod q$ for some $d \in \mathcal{R}_q^\times$,
– Sample uniform $d \leftarrow\$ \mathcal{R}_q^\times$ and $\mathbf{W} \leftarrow\$ \mathcal{R}_q^{t\times t}$ subject to being invertible modulo q, then compute $\mathbf{d}^\mathsf{T} := d \cdot \boldsymbol{\iota}_t^\mathsf{T}\mathbf{W}^{-1} \bmod q$.

$\mathsf{Hyb}_3 \approx_s \mathsf{Hyb}_4$. We claim that Hyb_3 and Hyb_4 are statistically close. The only difference between Hyb_3 and Hyb_4 is how $\mathbf{l}_i$ is computed. In the former, it is computed as $\mathbf{l}_i^\mathsf{T} := x_i \cdot p_i \cdot \mathbf{d}^\mathsf{T} \bmod q$ where $x_i \leftarrow\$ \mathcal{R}_q$, $p_i \cdot \mathbf{f}^\mathsf{T} \leftarrow\$ \mathcal{D}_{\mathcal{L}(\mathbf{f}^\mathsf{T}),s,\mathbf{e}_i}$, and $d \leftarrow\$ \mathcal{R}_q^\times$. We have

$$\begin{aligned}\mathbf{l}_i^\mathsf{T} := x_i \cdot p_i \cdot \mathbf{d}^\mathsf{T} &= x_i \cdot p_i \cdot (0, \dots, 0, d) \cdot \mathbf{W}^{-1} \bmod q \\ &= (0, \dots, 0, x_i \cdot p_i \cdot d) \cdot \mathbf{W}^{-1} \bmod q \\ &\approx_s (0, \dots, 0, x_i) \cdot \mathbf{W}^{-1} \bmod q,\end{aligned}$$

where the $\approx_s$ follows from Lemma 19 (which is applicable since $s \geq 2\chi_f k\varphi \geq 2\eta_\epsilon(\mathcal{L}(\mathbf{f}^\mathsf{T}))$ with overwhelming probability), which says that $p_i \in \mathcal{R}_q^\times$ with overwhelming probability, in which case $x_i \cdot p_i \cdot d$ and x_i are identically distributed – uniform over $\mathcal{R}_q$. The last line is precisely how $\mathbf{l}_i$ is computed in Hyb_4.

$\mathsf{Hyb}_4 \approx_c \mathsf{Hyb}_5$. This hop follows immediately from the $\mathsf{dNTRU}_{\mathcal{R},q,k,\chi_f}$ assumption (Definition 4).

$\mathsf{Hyb}_5 \equiv \mathsf{Hyb}_6$. We claim that Hyb_5 and Hyb_6 are identical. To see this, we observe that in Hyb_5, $\overline{\mathbf{A}}$, $\underline{\mathbf{a}}$ and $\mathbf{W}$ are all uniformly random subject to $\mathbf{W}$ being invertible modulo q. Therefore, $\mathbf{B}, \mathbf{W}$ could be equivalently sampled by independently sampling $\mathbf{B} \leftarrow\$ \mathcal{R}_q^{t\times k}$ and $\mathbf{W} \leftarrow\$ \mathcal{R}_q^{t\times t}$ subject to being invertible modulo q. Next, note that $(0, \dots, 0, x_i) \cdot \mathbf{W}^{-1}$ is a random $\mathcal{R}_q$ multiple of the last row of $\mathbf{W}^{-1}$, which is distributed uniformly randomly over $\mathcal{R}_q^t$ conditioned on being primitive. Denoting this vector $\hat{\mathbf{r}}$, we recover Hyb_6. ☐

Remark 7 (On Pseudorandom with Leakage from LWE). We briefly discuss the difficulty of constructing a primal trapdoor satisfying the Pseudorandom with Leakage property solely based on the LWE assumption (but not NTRU). A natural strategy of constructing a primal trapdoor, generalising the method presented above, is to sample matrices $\mathbf{B}, \mathbf{D}, \mathbf{F}$ such that $\mathbf{F}$ is Gaussian, and $\mathbf{D}$ and $\mathbf{B}$ are uniformly random subject to $\mathbf{D} \cdot \mathbf{B} = \mathbf{F} \bmod q$. For example, we could consider an "LWE-like" construction $\mathbf{B} = \mathbf{W} \cdot \begin{pmatrix}\mathbf{A} \\ \mathbf{SA} + \mathbf{F}\end{pmatrix} \bmod q$ and $\mathbf{D} = (-\mathbf{S}, \mathbf{I})\mathbf{W}^{-1} \bmod q$ where $\mathbf{A}, \mathbf{S}, \mathbf{W}$ are uniformly random subject to $\mathbf{W}$ being square and invertible modulo q. The tuples $(\mathbf{S}, \mathbf{F})$ and $(\mathbf{W}, \mathbf{F})$ could be regarded as LWE and NTRU secrets respectively. It is not difficult to show that, under either the LWE or the (matrix) NTRU assumption, i.e. even if one of $\mathbf{S}$ or $\mathbf{W}$ is public, the distribution of $\mathbf{B}$ is indistinguishable from uniformly random, without leakage. However, suppose that leakages of the form $\mathbf{L} = \mathbf{X}(-\mathbf{S}, \mathbf{I})\mathbf{W}^{-1}$ are known. On the one hand, it is clear that if the NTRU secret $\mathbf{W}$ is public then $\mathbf{B}$ is no longer pseudorandom, because one could recover $\mathbf{S}$ from $\mathbf{LW} = (-\mathbf{XS}, \mathbf{X}) \bmod q$. On the other hand, even if the LWE secret $\mathbf{S}$ is set to $\mathbf{0}$, then as shown we are still able to prove the pseudorandomness of $\mathbf{B}$ in the case $\mathbf{F}$ has only one row (over $\mathcal{R}$).

Remark 8. We are unable to extend Lemma 19 and Theorem 9 in the above to the setting with a hidden dense sublattice $\mathcal{L}(\mathbf{F})$ of module rank $\mathsf{nrows}_\mathcal{R}(\mathbf{F}) > 1$. The difficulty, roughly, is that for the former we could not impose reasonable conditions to rule out the possibility that a linear combination of two long vectors results in a short vector after reduction modulo q, while for the latter we no longer know how to decouple the correlation between the leakages and the matrix $\mathbf{B}$. However, for applications where $\mathbf{B}$ is only required to be pseudorandom without leakages, the above can be readily generalised to the setting $\mathsf{nrows}_\mathcal{R}(\mathbf{F}) > 1$, e.g. when setting $\mathbf{D} \cdot \mathbf{B} = \mathbf{F} \bmod q$ for some uniformly random matrix $\mathbf{D}$ invertible modulo q and a low-norm matrix $\mathbf{F}$.

B.4 Proof of Theorem 2

Proof. Follows directly from Theorems 8 and 9. $\square$

B.5 Proof of Theorem 3

Proof. First, we inspect the distribution $\mathcal{D}_{\mathcal{L}(\mathbf{f}^{\mathsf{T}}),s,-c\mathbf{f}}$ for any fixed $\mathbf{f} \in \mathcal{R}^k$ and $c \in \mathcal{K}$. For any $\mathbf{x} \in \mathcal{L}(\mathbf{f}^{\mathsf{T}})$, writing $\mathbf{x} = -p\mathbf{f}$ for some $p \in \mathcal{R}$, we have

$$\rho_{s,-c\mathbf{f}}(\mathbf{x}) = \exp(-\pi\|\mathbf{x} + c\mathbf{f}\|^2/s^2) = \exp(-\pi\|-p\mathbf{f} + c\mathbf{f}\|^2/s^2) = \exp(-\pi\|p - c\|_{\mathbf{s}}^2) = \rho_{\mathbf{s},c}(p),$$

where $\mathbf{s} = s^2/\sigma(\bar{\mathbf{f}}^{\mathsf{T}}\mathbf{f})$ with division computed component-wise, and the second-last equality in the chain follows from

$$\begin{aligned}\|-\mathbf{f}p + \mathbf{f}c\|^2/s^2 = \|\mathbf{f}p - \mathbf{f}c\|^2/s^2 &= \overline{\sigma((p-c)\mathbf{f})}^{\mathsf{T}}\sigma((p-c)\mathbf{f})/s^2 \\ &= \overline{\sigma(p-c)}^{\mathsf{T}}(\sum_{i=1}^{k}\overline{\mathsf{diag}(\sigma(f_i))}\mathsf{diag}(\sigma(f_i)))\sigma(p-c)/s^2 \\ &= \overline{\sigma(p-c)}^{\mathsf{T}}\mathsf{diag}(\sigma(\bar{\mathbf{f}}^{\mathsf{T}}\mathbf{f}))\sigma(p-c)/s^2 = \|p-c\|_{\mathbf{s}}^2.\end{aligned}$$

The above implies $\mathcal{D}_{\mathcal{L}(\mathbf{f}^{\mathsf{T}}),s,-c\mathbf{f}}(-p\mathbf{f}) = \mathcal{D}_{\mathcal{R},\mathbf{s},c}(p)$. In other words, for p sampled from $\mathcal{D}_{\mathcal{R},\mathbf{s},c}$, the induced distribution of $-p\mathbf{f}$ is precisely $\mathcal{D}_{\mathcal{L}(\mathbf{f}^{\mathsf{T}}),s,-c\mathbf{f}}$. Moreover, if c is such that $-c \cdot \mathbf{f} = \mathsf{proj}_{\mathcal{L}(\mathbf{f}^{\mathsf{T}})}(-\mathbf{e})$ for some $\mathbf{e} \in \mathcal{R}^k$, then we have $-p\mathbf{f} \sim \mathcal{D}_{\mathcal{L}(\mathbf{f}^{\mathsf{T}}),s,\mathsf{proj}_{\mathcal{L}(\mathbf{f}^{\mathsf{T}})}(-\mathbf{e})}$, which is identical to $\mathcal{D}_{\mathcal{L}(\mathbf{f}^{\mathsf{T}}),s,-\mathbf{e}}$ by Lemma 18.

Next, fix any $(\mathbf{B}, \mathsf{td} = (\mathbf{B}, \mathbf{f}, \mathbf{d})) \in \mathsf{pTrapGen}(1^t, 1^k, q)$, and any $\mathbf{r} \in \mathcal{R}_q^t, \mathbf{c} \in \mathcal{R}^k$ satisfying $\|\mathbf{c}^{\mathsf{T}} - \mathbf{r}^{\mathsf{T}}\mathbf{B} \bmod q\| \leq s$. Denote by $\mathbf{e} \coloneqq \mathbf{c}^{\mathsf{T}} - \mathbf{r}^{\mathsf{T}}\mathbf{B} \bmod q$, so that $\|\mathbf{e}\| \leq s$. Recall that $\mathsf{td} = (\mathbf{B}, \mathbf{f}, \mathbf{d})$ satisfies $\mathbf{d}^{\mathsf{T}}\mathbf{B} = \mathbf{f}^{\mathsf{T}} \bmod q$ by construction. The output of $\mathsf{Equivocate}$ is $\tilde{\mathbf{r}} = \mathbf{r} + p\mathbf{d} \bmod q$, for p sampled from $\mathcal{D}_{\mathcal{R},\mathbf{s},c}$, and with c satisfying $-c \cdot \mathbf{f} = \mathsf{proj}_{\mathcal{L}(\mathbf{f}^{\mathsf{T}})}(-\mathbf{e})$. Thus

$$\mathbf{c}^{\mathsf{T}} - \tilde{\mathbf{r}}^{\mathsf{T}}\mathbf{B} \bmod q = (\mathbf{r}^{\mathsf{T}}\mathbf{B} + \mathbf{e}^{\mathsf{T}}) - (\mathbf{r}^{\mathsf{T}} + p\mathbf{d}^{\mathsf{T}})\mathbf{B} \bmod q = \mathbf{e}^{\mathsf{T}} - p\mathbf{f}^{\mathsf{T}} \bmod q.$$

From the derivation in the previous paragraph, it holds that $-p\mathbf{f}$ follows $\mathcal{D}_{\mathcal{L}(\mathbf{f}^{\mathsf{T}}),s,-\mathbf{e}}$, equivalently $\mathbf{e} - p\mathbf{f}$ follows $\mathcal{D}_{\mathcal{L}(\mathbf{f}^{\mathsf{T}})+\mathbf{e},s}$. Hence $\|\mathbf{e} - p\mathbf{f}\| \leq \|\mathbf{e}\| + \|-p\mathbf{f}\| \leq s + s\sqrt{k\varphi} < q/2$, where the second inequality is due to $\|-p\mathbf{f}\| \leq s\sqrt{k\varphi}$ with overwhelming probability in $k\varphi$ by Corollary 3 and [SS11, Lemma 4.2, 4.4 (eprint version)], and the last inequality is implied by $q > 2\beta_{\Lambda}$. Therefore,

$$\mathbf{c}^{\mathsf{T}} - \mathbf{r}^{\mathsf{T}}\mathbf{B} \bmod q = \mathbf{e} - p\mathbf{f}$$

(the RHS without modding), and we conclude that $\mathbf{c}^{\mathsf{T}} - \mathbf{r}^{\mathsf{T}}\mathbf{B} \bmod q$ follows $\mathcal{D}_{\mathcal{L}(\mathbf{f}^{\mathsf{T}})+\mathbf{e},s}$.

Finally, by Lemma 16, with probability $1 - \mathsf{negl}(\lambda)$ over the randomness of $(\mathbf{B}, \mathsf{td}) \leftarrow \mathsf{pTrapGen}(1^t, 1^k, q)$, we have $\mathcal{D}_{\mathcal{L}(\mathbf{f}^{\mathsf{T}})+\mathbf{e},s} \approx_s \mathcal{D}_{\Lambda_q(\mathbf{B})+\mathbf{c},s}$. Now note that $\tilde{\mathbf{r}}^{\mathsf{T}}\mathbf{B} \bmod q = \mathbf{c}^{\mathsf{T}} - (\mathbf{c}^{\mathsf{T}} - \tilde{\mathbf{r}}^{\mathsf{T}}\mathbf{B}) \bmod q$, thus

$$\tilde{\mathbf{r}}^{\mathsf{T}}\mathbf{B} \bmod q \sim \mathbf{c} - \mathcal{D}_{\mathcal{L}(\mathbf{f}^{\mathsf{T}})+\mathbf{e},s} \bmod q \approx_s \mathbf{c} - \mathcal{D}_{\Lambda_q(\mathbf{B})+\mathbf{c},s} \bmod q = \mathcal{D}_{\Lambda_q(\mathbf{B}),s,\mathbf{c}} \bmod q. \quad \square$$

C Extended Cryptanalysis on the Equivocal LWE Assumption

We discuss a broad class of attacks commonly referred to as "zero-ising attacks" [CHL+15, MSZ16, CVW18, HJL21] against lattice-based iO and related assumptions. Roughly, these attacks attempt to manipulate the given problem instance with the goal of finding short vectors which are not intended to be revealed.

Recall that $(\tilde{\mathbf{R}}_b, \mathbf{L}_b, \mathbf{B}, \mathbf{C})$, where $\mathbf{L}_b$ is derived from $(\mathbf{P}, \mathsf{ctxt}_b)$, satisfies the relation

$$\begin{pmatrix}\mathbf{C} \\ \tilde{\mathbf{R}}_b\mathbf{B}\end{pmatrix} = \begin{pmatrix}\mathbf{I}_p \\ \mathbf{L}_b\end{pmatrix}\mathbf{R}\mathbf{B} + \begin{pmatrix}\mathbf{E} \\ \tilde{\mathbf{E}}_b\end{pmatrix} + \begin{pmatrix}\mathsf{Encode}(\mathsf{sk}) \\ \mathbf{0}\end{pmatrix} \bmod q \tag{13}$$

where $\mathbf{E} \leftarrow\!\$\ \mathcal{D}_{\mathcal{R},\chi}^{p\times k}$ and $\tilde{\mathbf{E}}_b \leftarrow\!\$\ \mathcal{D}_{\Lambda_q(\mathbf{B})^h,s,\mathbf{L}_b\cdot(\mathbf{E}+\mathsf{Encode}(\mathsf{sk}))+\mathbf{M}_b'}$, where the centre $\mathbf{L}_b \cdot (\mathbf{E} + \mathsf{Encode}(\mathsf{sk})) + \mathbf{M}_b'$ is the shortest representative (which is short relative to q) of $\mathbf{L}_b\cdot(\mathbf{E}+\mathsf{Encode}(\mathsf{sk}))+\mathbf{M}_b$ modulo q (Eq. (8)), and we abuse notation slightly to mean that the i-th row of $\tilde{\mathbf{E}}_b$ is a discrete Gaussian over $\Lambda_q(\mathbf{B})$ with parameter s with centre being the i-th row of $\mathbf{L}_b \cdot (\mathbf{E} + \mathsf{Encode}(\mathsf{sk})) + \mathbf{M}_b$.

In our context, we interpret zero-ising attacks as corresponding to finding (not necessarily short) vectors $(\mathbf{x}, \mathbf{t})$ or $\mathbf{u}$ such that

$$\left(\mathbf{t}^{\mathsf{T}} \mid -\mathbf{x}^{\mathsf{T}}\right)\begin{pmatrix}\mathbf{I}_p\\ \mathbf{L}_b\end{pmatrix} = \mathbf{0}^{\mathsf{T}} \bmod q \qquad\qquad \text{or} \qquad\qquad \mathbf{B}\mathbf{u} = \mathbf{0} \bmod q$$

with the goal of left-multiplying $\left(\mathbf{t}^{\mathsf{T}} \mid -\mathbf{x}^{\mathsf{T}}\right)$ or right-multiplying $\mathbf{u}$ to Eq. (13) to obtain non-trivial information about various secrets, e.g. the hidden dense sublattice of $\Lambda_q(\mathbf{B})$, the LWE error $\mathbf{E}$, the SKE secret key sk, etc. We refer to the above as "zero-ising attacks from the left" and "from the right" respectively. Below, we will examine each type of zero-ising attacks and argue that they are either infeasible or do not seem to be useful.

Zero-ising Attacks from the Right. We first examine the case of zero-ising attacks from the right which is clearer. Intuitively, since $\mathbf{B}$ is drawn from an equivocal distribution which is pseudorandom (with leakage) by definition, finding a (not necessarily short) preimage $\mathbf{u}$ such that $\mathbf{B}\mathbf{u} = \mathbf{0} \bmod q$ and such that it is useful for distinguishing Eq. (13) should be infeasible. Indeed, the ability to find such $\mathbf{u}$ would likely be sufficient for distinguishing the (standard) $\mathsf{dLWE}_{\mathcal{R},q,t,k,\chi}$ problem w.r.t. a uniformly random $\mathbf{B}$, which is assumed to be hard. Heuristically, the knowledge of hint_b does not seem to help solving SIS or LWE w.r.t. $\mathbf{B}$ other than (trivially) distinguishing the LWE samples that are equivocated. We elaborate on the heuristic argument below in more detail.

Case R1: $\mathbf{u}$ *is short.* Suppose the distinguisher finds a short vector $\mathbf{u}$ such that $\mathbf{B}\mathbf{u} = \mathbf{0} \bmod q$, then right-multiplying $\mathbf{u}$ to Eq. (13) yields

$$\mathbf{E} \cdot \mathbf{u} + \mathsf{Encode}(\mathsf{sk}) \cdot \mathbf{u} \approx \mathsf{Encode}(\mathsf{sk}) \cdot \mathbf{u} \tag{14}$$

which potentially reveals unintended information about sk. This is also known as the dual attack on the (standard) LWE problem, which amounts to solving the SIS problem w.r.t. $\mathbf{B}$. By pseudorandomness of $\mathbf{B}$ in our setting, and the heuristic that hint_b does not seem to help solving SIS, this is believed to be hard.

Case R2: $\mathbf{u}$ *is long.* By linear algebra, the distinguisher could easily find a long vector $\mathbf{u}$ such that $\mathbf{B}\mathbf{u} = \mathbf{0} \bmod q$. Heuristically, $\mathbf{u}$ being long should destroy the utility of Eq. (14) (e.g. the approximation no longer holds), since $\mathbf{E}$ is a Gaussian matrix supposedly hidden from the distinguisher and $\mathbf{E}\mathbf{u} \bmod q$ is heuristically uniformly random.

Zero-ising Attacks from the Left. We next analyse the case of zero-ising attacks from the left which is trickier and should be subject to further cryptanalysis.

Case L1: $\mathbf{x}$ *is short and* $\mathbf{t} = \mathbf{0}$. Suppose the distinguisher succeeds in finding a short vector $\mathbf{x}$ such that $\mathbf{L}_b^{\mathsf{T}}\mathbf{x} = \mathbf{0} \bmod q$, then left-multiplying $\left(\mathbf{0}^{\mathsf{T}} \mid -\mathbf{x}^{\mathsf{T}}\right)$ to Eq. (13) would yield

$$-\mathbf{x}^{\mathsf{T}}\tilde{\mathbf{R}}_b\mathbf{B} = \mathbf{x}^{\mathsf{T}}\tilde{\mathbf{E}}_b \approx \mathbf{0}^{\mathsf{T}} \bmod q.$$

In other words, the distinguisher would obtain a short vector in $\Lambda_q(\mathbf{B})$ and hence learn non-trivial information about the hidden dense sublattice, which might be disastrous. However, find such $\mathbf{u}$ amounts to solving the SIS problem (i.e. ISIS with target $\mathbf{t} = \mathbf{0}$) w.r.t. $\mathbf{L}_b$. If we assume the heuristic the hint hint_b does not help solving ISIS w.r.t. $\mathbf{L}_b$, which seems realistic since ISIS is a search problem and the hint does not seem to narrow down the search space in any way, then we can show that finding a solution of norm α for the latter against any fixed target is hard assuming the hardness of $\mathsf{SIS}_{\mathcal{R},q,p/d,h,\alpha}$.

More formally, we show that if there exists an efficient algorithm which, on input $(\mathbf{B}, \mathbf{C}, \mathbf{P}, \mathsf{ctxt}_b)$ as generated in the Equivocal LWE assumption, finds a short non-zero vector $\mathbf{x}$ of norm at most α satisfying $\mathbf{L}_b^\mathsf{T}\mathbf{x} = \mathbf{0} \bmod q$, where $\mathbf{L}_b^\mathsf{T} = \mathbf{G}^{-1}(\hat{\mathbf{H}}_b + \mathbf{1}_d \otimes \mathbf{P}) \bmod q$, then we can construct an efficient against $\mathsf{SIS}_{\mathcal{R},q,p/d,h,\alpha}$. We prove by a hybrid argument, gradually changing the distribution of the input $(\mathbf{B}, \mathbf{C}, \mathbf{P}, \mathsf{ctxt}_b)$. First, we replace $\mathbf{B}$ by a uniformly random matrix. This is computationally unnoticeable due to the pseudorandomness of $\mathbf{B}$. Then, we replace $\mathbf{C}$ by a uniformly random matrix. This is computationally unnoticeable due to the $\mathsf{dLWE}_{\mathcal{R},q,t,k,\chi}$ assumption. At this point, the only information about sk available to the adversary are $\mathbf{P} \leftarrow \mathsf{SKE.Enc}(\mathsf{sk}, \mathsf{tag}_{\mathsf{pad}}, \mathsf{msg}_{\mathsf{pad}})$ and $\mathsf{ctxt}_b \leftarrow \mathsf{SKE.Enc}(\mathsf{sk}, \mathsf{tag}^*, \mathsf{msg}_b)$. By the IND-\$-CPA security of SKE, we can replace $\mathbf{P}$ by a uniformly random matrix.

Now, we show a reduction from $\mathsf{SIS}_{\mathcal{R},q,p/d,h,\alpha}$ to solving SIS against $\mathbf{L}_b^\mathsf{T} = \mathbf{G}^{-1}(\hat{\mathbf{H}}_b + \mathbf{1}_d \otimes \mathbf{P}) \bmod q$ generated as in the last hybrid. On input a $\mathsf{SIS}_{\mathcal{R},q,p/d,h,\alpha}$ instance $\hat{\mathbf{P}}$, pick any index $j^* \in [d]$, generate $\hat{\mathbf{H}}_b$ and all other components except $\mathbf{P}$ as in the hybrid (which does not depend on $\mathbf{P}$), and set $\mathbf{P} \coloneqq \hat{\mathbf{P}} - \hat{\mathbf{H}}_{b,j^*} \bmod q$, where $\hat{\mathbf{H}}_{b,j^*}$ denotes the j^*-th block row of $\hat{\mathbf{H}}_b$. Pass $(\mathbf{B}, \mathbf{C}, \mathbf{P}, \mathsf{ctxt}_b)$ to the adversary and output the SIS solution $\mathbf{x}$ returned by the adversary. We observe that our reduction is successful if the adversary is, since

$$
\begin{aligned}
\mathbf{L}_b^\mathsf{T}\mathbf{x} &= \mathbf{0} \bmod q, \\
\mathbf{G}^{-1}(\hat{\mathbf{H}}_b + \mathbf{1}_d \otimes \mathbf{P})\mathbf{x} &= \mathbf{0} \bmod q, \\
(\hat{\mathbf{H}}_b + \mathbf{1}_d \otimes \mathbf{P})\mathbf{x} &= \mathbf{0} \bmod q, \\
(\hat{\mathbf{H}}_{b,j^*} + \mathbf{P})\mathbf{x} &= \mathbf{0} \bmod q, \\
\hat{\mathbf{P}}\mathbf{x} &= \mathbf{0} \bmod q.
\end{aligned}
$$

A similar analysis still holds if the SIS problem is replaced by an ISIS problem w.r.t. $\mathbf{L}_b^\mathsf{T}$ with any fixed (worst-case) target vector $\mathbf{t}$, in that we can obtain a reduction from $\mathsf{ISIS}_{\mathcal{R},q,p/d,h,\alpha}$ with a random target $\hat{\mathbf{t}}$, which in turn is implied by $\mathsf{SIS}_{\mathcal{R},q,p/d,h,2\alpha}$. To see this, let $\mathbf{t}_{j^*}$ denote the j^*-th block of $\mathbf{t}$. We modify the above reduction so that, on input an $\mathsf{ISIS}_{\mathcal{R},q,p/d,h,\alpha}$ instance $(\hat{\mathbf{P}}, \hat{\mathbf{t}})$, generate $\hat{\mathbf{H}}_{b,j^*}$ and all other components except $\mathbf{P}$ as in the above reduction, pick an invertible matrix $\mathbf{W}$ such that $\mathbf{W}\hat{\mathbf{t}} = \mathbf{t}_{j^*}$, and set $\mathbf{P} \coloneqq \mathbf{W}\hat{\mathbf{P}} - \hat{\mathbf{H}}_{b,j^*} \bmod q$. Clearly, $\mathbf{P}$ is still well-distributed as $\hat{\mathbf{P}}$ is uniform and $\mathbf{W}$ is invertible, and a solution $\mathbf{x}$ to $\mathbf{L}_b^\mathsf{T}\mathbf{x} = \mathbf{0} \bmod q$ also satisfies $\hat{\mathbf{P}}\mathbf{x} = \hat{\mathbf{t}} \bmod q$.

The parameter α governs the hardness of the SIS problem and the "usefullness" of the solution to the attack: For a solution $\mathbf{x}$ of norm α, the norm of the short vector $\mathbf{x}^\mathsf{T}\tilde{\mathbf{E}}_b \in \Lambda_q(\mathbf{B})$ is of norm at most $\alpha s\sqrt{hk}$, which is potentially useful when $\alpha s\sqrt{hk} < q/2$. We are therefore interested in the hardness of SIS with $2\alpha \leq q/(s\sqrt{hk})$.

Case L2: Both $\mathbf{x}$ and $\mathbf{t}$ are short and $\mathbf{t} \neq \mathbf{0}$. Suppose the distinguisher's strategy is to compute short $(\mathbf{x}, \mathbf{t})$ such that $\left(\mathbf{t}^\mathsf{T} \mid -\mathbf{x}^\mathsf{T}\right)\begin{pmatrix}\mathbf{I}_p \\ \mathbf{L}_b\end{pmatrix} = \mathbf{0}^\mathsf{T} \bmod q$, or in other words $\mathbf{L}_b^\mathsf{T}\mathbf{x} = \mathbf{t} \bmod q$. Since $\mathbf{L}_b$, $\mathbf{x}$ and $\mathbf{t}$ are all short, reduction modulo q has no effect, and we in fact have $\mathbf{L}_b^\mathsf{T}\mathbf{x} = \mathbf{t}$ without modular reduction. We can therefore rewrite $\left(\mathbf{t}^\mathsf{T} \mid -\mathbf{x}^\mathsf{T}\right)$ as

$$
\left(\mathbf{t}^\mathsf{T} \mid -\mathbf{x}^\mathsf{T}\right) = \left(\mathbf{x}^\mathsf{T}\mathbf{L}_b \mid -\mathbf{x}^\mathsf{T}\right) = \mathbf{x}^\mathsf{T}\left(\mathbf{L}_b \mid -\mathbf{I}_h\right)
$$

which is always in the $\mathcal{R}$-span of the rows of $\left(\mathbf{L}_b \mid -\mathbf{I}_h\right)$. Consequently, whatever information that can be learnt by left-multiplying $\left(\mathbf{t}^\mathsf{T} \mid -\mathbf{x}^\mathsf{T}\right)$ to Eq. (13) could also be learnt by left-multiplying $\left(\mathbf{L}_b \mid -\mathbf{I}_h\right)$ to Eq. (13) instead. It therefore suffices to analyse the latter case.

Left-multiplying $\left(\mathbf{L}_b \mid -\mathbf{I}_h\right)$ to Eq. (13) yields

$$
-\tilde{\mathbf{E}}_b + \mathbf{L}_b \cdot (\mathbf{E} + \mathsf{Encode}(\mathsf{sk})) \bmod q.
$$

Since $\mathbf{M}_b$ could be derived by the distinguisher via $(\mathbf{L}_b, \mathbf{M}_b) = \Phi(\mathsf{ctxt}_b)$, offsetting the above with $\mathbf{M}_b$ does not affect the knowledge of the distinguisher and is more convenient to analyse. Since $\mathbf{L}_b \cdot (\mathbf{E} + \mathsf{Encode}(\mathsf{sk})) + \mathbf{M}'_b$ is a short representative of $\mathbf{L}_b \cdot (\mathbf{E} + \mathsf{Encode}(\mathsf{sk})) + \mathbf{M}_b$ modulo q, we obtain the leakage matrix

$$\mathbf{Z}_b := -\tilde{\mathbf{E}}_b + \mathbf{L}_b \cdot (\mathbf{E} + \mathsf{Encode}(\mathsf{sk})) + \mathbf{M}'_b \tag{15}$$

where the equality holds without modulo q.

We are not able to characterise what can be learnt from the leakage $\mathbf{Z}_b$ in Eq. (15). Instead, we discuss heuristic evidence of such leakage not being useful, and invite further cryptanalysis.

Recall that $\tilde{\mathbf{E}}_b$ follows the distribution $\mathcal{D}_{\Lambda_q(\mathbf{B})^h, s, \mathbf{L}_b \cdot (\mathbf{E}+\mathsf{Encode}(\mathsf{sk}))+\mathbf{M}'_b}$. This means $\mathbf{Z}_b$ in Eq. (15) follows the distribution $\mathcal{D}_{\Lambda_q(\mathbf{B})^h + \mathbf{L}_b \cdot (\mathbf{E}+\mathsf{Encode}(\mathsf{sk}))+\mathbf{M}'_b, s}$. Note that $\mathbf{L}_b \cdot (\mathbf{E} + \mathsf{Encode}(\mathsf{sk})) + \mathbf{M}'_b = \mathbf{L}_b\mathbf{C} + \mathbf{M}_b = \mathbf{W}_b \bmod \Lambda_q(\mathbf{B})$ with congruence taken for individual rows. Therefore, the distribution of $\mathbf{Z}_b$ can also be expressed as

$$\mathbf{Z}_b \sim \mathcal{D}_{\Lambda_q(\mathbf{B})^h + \mathbf{W}_b, s},$$

i.e. spherical Gaussian distributions over cosets of $\Lambda_q(\mathbf{B})$ given by the rows of $\mathbf{W}_b$. We note that the description of $\mathcal{D}_{\Lambda_q(\mathbf{B})^h + \mathbf{W}_b, s}$ consists of only public information (although it is not publicly sampleable), and this view of $\mathbf{Z}_b$ does not seem helpful from a distinguisher's perspective. Rather, it serves as evidence for the leakage being "minimal" or "as random as possible subject to closeness constraint".

Alternatively, we may view each row of the expression $-\tilde{\mathbf{E}}_b + \mathbf{L}_b \cdot (\mathbf{E} + \mathsf{Encode}(\mathsf{sk}))$ in Eq. (15) as an "integer LWE" (ILWE) sample [BDE+18] (generalised to the $\mathcal{R}$ setting). To recall, ILWE is similar to LWE except that there is no reduction modulo q. In our case, $\mathbf{L}_b$ plays the role of the public matrix, $\mathbf{E}+\mathsf{Encode}(\mathsf{sk})$ the LWE secret and $\tilde{\mathbf{E}}_b$ the error. If the adversary somehow obtains the value $-\tilde{\mathbf{E}}_b + \mathbf{L}_b \cdot (\mathbf{E} + \mathsf{Encode}(\mathsf{sk}))$ (e.g. when $\mathbf{M}'_b = \mathbf{0}$), they can attempt to solve the ILWE problem to recover $\mathbf{E} + \mathsf{Encode}(\mathsf{sk})$, from which they can recover sk.

It is straightforward to verify that the least-square attack presented in [BDE+18, Section 4.1] can be used to recover $\mathbf{E} + \mathsf{Encode}(\mathsf{sk})$ from $-\tilde{\mathbf{E}}_b + \mathbf{L}_b \cdot (\mathbf{E} + \mathsf{Encode}(\mathsf{sk}))$ given enough number of samples, i.e. when the matrix $\mathbf{L}_b$ is tall enough. Recall that $\mathbf{L}_b \in \mathcal{R}^{h \times p\lceil \log q \rceil}$. The attack in [BDE+18, Section 4.1] applies when φh is (a logarithmic factor) larger than $\|\tilde{\mathbf{E}}_b\|^2 / \|\mathbf{L}_b\|^2$. Setting the Gaussian parameter s to be large enough (so that $\|\tilde{\mathbf{E}}_b\|$ is large enough) defeats this attack. We refer to Section 5.2 for concrete parameter suggestions.

Furthermore, it was shown in [BDE+18, Section 3] that the least-square attack is almost optimal in the case where the ILWE error consists of independent Gaussian vectors sampled from the entire space. Although this analysis does not directly apply to our setting, since the error $\tilde{\mathbf{E}}_b$ has dependencies on the secret $\mathbf{E}$ and its rows are restricted to be drawn from the lattice $\Lambda_q(\mathbf{B})$, it suggests that any attack that recovers $\mathbf{E} + \mathsf{Encode}(\mathsf{sk})$ must somehow exploit the structure and correlation of the error term $\tilde{\mathbf{E}}_b$. We note, however, that this heuristic analysis concerns about the hardness of recovering $\mathbf{E} + \mathsf{Encode}(\mathsf{sk})$ entirely, but not, say, whether $\mathbf{Z}_0$ is indistinguishable from $\mathbf{Z}_1$.

Finally, we caution that if the Gaussian parameter s is too large, an adversary might be able to learn information about the "shape" of the hidden sublattice of $\Lambda_q(\mathbf{B})$, since the rows of $\tilde{\mathbf{E}}_b$ are short vectors in $\Lambda_q(\mathbf{B})$. However, we are unable to find concrete evidence for or against this intuition.

Case L3: $(\mathbf{x}, \mathbf{t})$ *is long.* The distinguisher can easily find long vectors $(\mathbf{x}, \mathbf{t})$ satisfying $\left(\mathbf{t}^\mathsf{T} \mid -\mathbf{x}^\mathsf{T}\right) \begin{pmatrix} \mathbf{I}_p \\ \mathbf{L}_b \end{pmatrix} = \mathbf{0}^\mathsf{T} \bmod q$ by linear algebra. Heuristically, left-multiplying a long $\left(\mathbf{t}^\mathsf{T} \mid -\mathbf{x}^\mathsf{T}\right)$ to Eq. (13) would yield

$$\mathbf{t}^\mathsf{T}\mathbf{E} - \mathbf{x}^\mathsf{T}\tilde{\mathbf{E}}_b + \mathbf{t}^\mathsf{T}\mathsf{Encode}(\mathsf{sk}) \bmod q$$

where the term $\mathbf{t}^\mathsf{T}\mathbf{E} - \mathbf{x}^\mathsf{T}\tilde{\mathbf{E}}_b \bmod q$ should destroy all utility (analogous to the heuristics of Case R2). Without control over the distribution of $(\mathbf{x}, \mathbf{t})$, however, it is difficult to formally reason about the inefficacy of this attack strategy. Below, we instead inspect possible strategies for a distinguisher to find such $(\mathbf{x}, \mathbf{t})$.

As we have shown in Case L1, for any fixed target $\mathbf{t}$, it can be proven assuming standard SIS that solving the ISIS instance $(\mathbf{L}_b^\mathsf{T}, \mathbf{t})$ is hard without using the hint. Furthermore, even if the hint is taken into account, it is unclear how to use it to narrow down the search space. Under these constraints, some natural strategies of finding $(\mathbf{x}, \mathbf{t})$ satisfying $\left(\mathbf{t}^\mathsf{T} \mid -\mathbf{x}^\mathsf{T}\right)\begin{pmatrix}\mathbf{I}_p\\ \mathbf{L}_b\end{pmatrix} = \mathbf{0}^\mathsf{T} \bmod q$ are (mixtures of):

(i) first fix $\mathbf{x}$, then compute $\mathbf{t} \coloneqq \mathbf{L}_b^\mathsf{T}\mathbf{x} \bmod q$, or
(ii) first fix $\mathbf{t}$, then solve the equation $\mathbf{t} = \mathbf{L}_b^\mathsf{T}\mathbf{x} \bmod q$ for (high-norm) $\mathbf{x}$.

For the first case, we may heuristically treat $\mathbf{t}$ as being uniformly random, in which case $\mathbf{t}^\mathsf{T}\mathbf{E} \bmod q$ would be statistically or computationally uniformly random by appealing to either the leftover hash lemma or an LWE assumption with appropriate parameters.

For the second case, we may heuristically treat $\mathbf{x}$ as being uniformly random. Letting $\mathbf{R}'_b = \tilde{\mathbf{R}}_b - \mathbf{L}_b\mathbf{R} \bmod q$, we can express

$$\tilde{\mathbf{R}}_b\mathbf{B} = (\mathbf{L}_b\mathbf{R} + \mathbf{R}'_b)\mathbf{B} = \mathbf{L}_b\mathbf{R}\mathbf{B} + \tilde{\mathbf{E}}_b \bmod q$$
$$\mathbf{R}'_b\mathbf{B} \bmod q = \tilde{\mathbf{E}} \sim \mathcal{D}_{\Lambda_q(\mathbf{B})^h, s, \mathbf{L}_b\cdot(\mathbf{E}+\mathsf{Encode}(\mathsf{sk}))+\mathbf{M}'_b},$$

where we recall the centre $\mathbf{L}_b \cdot (\mathbf{E}+\mathsf{Encode}(\mathsf{sk}))+\mathbf{M}'_b$ is the shortest representative of $\mathbf{L}_b \cdot (\mathbf{E}+\mathsf{Encode}(\mathsf{sk}))+\mathbf{M}'_b \bmod q$. Since the admissibility check guarantees that $\|\mathbf{L}_b \cdot (\mathbf{E}+\mathsf{Encode}(\mathsf{sk}))+\mathbf{M}'_b \bmod q\| \le s$ (Eqs. (7) and (8)), the Pseudorandomn with Leakage property (Definition 8, Fig. 2) of $\mathcal{E}$ then guarantees that $(\mathbf{B}, \mathbf{x}^\mathsf{T}\mathbf{R}'_b)$ where $\mathbf{B} \leftarrow\!\!\$\, \mathcal{E}$ is indistinguishable to $(\mathbf{B}, \hat{\mathbf{x}}^\mathsf{T}\hat{\mathbf{R}})$, where $\mathbf{B}$, $\hat{\mathbf{x}}$ and $\hat{\mathbf{R}}$ are uniformly random with appropriate dimensions. Now, consider left-multiplying $\left(\mathbf{t}^\mathsf{T} \mid -\mathbf{x}^\mathsf{T}\right)$ to Eq. (13) to obtain

$$\begin{aligned}
\mathbf{t}^\mathsf{T}\mathbf{C} - \mathbf{x}^\mathsf{T}\tilde{\mathbf{R}}_b\mathbf{B} &= \mathbf{t}^\mathsf{T}\mathbf{C} - \mathbf{x}^\mathsf{T}(\mathbf{L}_b\mathbf{R} + \mathbf{R}'_b)\mathbf{B} \bmod q\\
&= \mathbf{t}^\mathsf{T}(\mathbf{C} - \mathbf{R}\mathbf{B}) - \mathbf{x}^\mathsf{T}\mathbf{R}'_b\mathbf{B} \bmod q\\
&= \mathbf{t}^\mathsf{T}(\mathbf{E} + \mathsf{Encode}(\mathsf{sk})) - \mathbf{x}^\mathsf{T}\mathbf{R}'_b\mathbf{B} \approx_c \mathbf{t}^\mathsf{T}(\mathbf{E} + \mathsf{Encode}(\mathsf{sk})) - \hat{\mathbf{x}}^\mathsf{T}\hat{\mathbf{R}}\mathbf{B} \bmod q.
\end{aligned}$$

Note that $\hat{\mathbf{x}}^\mathsf{T}\hat{\mathbf{R}}\mathbf{B} \bmod q$ is uniformly random over $\Lambda_q(\mathbf{B}) \bmod q$, which does not completely mask $\mathbf{t}^\mathsf{T}(\mathbf{E} + \mathsf{Encode}(\mathsf{sk}))$. Nevertheless, it is unclear what useful information can be obtained from $\mathbf{t}^\mathsf{T}(\mathbf{E} + \mathsf{Encode}(\mathsf{sk})) - \hat{\mathbf{x}}^\mathsf{T}\hat{\mathbf{R}}\mathbf{B} \bmod q$.

D Extended Discussions and Proofs for XiO Construction

D.1 Verbal Description of XiO Construction

Obfuscator. To obfuscate a circuit $\Gamma : [h] \times [k] \to \{0,1\}$, the obfuscator does the following:

- Sample key $\mathbf{s}_d \leftarrow\!\!\$\, \mathcal{R}_q^{n-1}$ and encrypt the description $\boldsymbol{\gamma} \in \{0,1\}^\ell$ of the circuit Γ and the vector $\mathbf{0}_h$, under tag tag^* and $\mathsf{tag}_{\mathsf{pad}}$ respectively, using the tagged symmetric-key encryption scheme SKE described in Fig. 5. Parse the obtained ciphertexts $\mathsf{ctxt}, \mathbf{P}$ as
$$\mathsf{ctxt} = (\mathbf{V}_0\|\mathbf{V}_1\|\ldots\|\mathbf{V}_d) \in \mathcal{R}_q^{n\times \ell m} \times (\mathcal{R}_q^{n\times m^2})^d$$
$$\mathbf{P} = (\mathbf{p}_1\|\ldots\|\mathbf{p}_h) \in (\mathcal{R}_q^{n})^h$$
- Sample an equivocal matrix $\mathbf{B} \in \mathcal{R}_q^{t\times k}$ together with its trapdoor td by running $\mathsf{pTrapGen}(1^t, 1^k, q)$, then encrypt ($k$-copies of) the GSW secret key $\mathbf{s}_d$ via the (packed) dual-Regev encryption [GPV08] with respect to matrix $\mathbf{B}$, i.e., sample LWE secret $\mathbf{R} \leftarrow\!\!\$\, \mathcal{R}_q^{km\times t}$ and short error $\mathbf{E} \leftarrow\!\!\$\, \mathcal{D}_{\mathcal{R},\chi}^{km\times k}$, and compute
$$\mathbf{C} \coloneqq \mathbf{R}\mathbf{B} + \mathbf{E} + \mathbf{I}_k \otimes ((-\mathbf{s}_d^\mathsf{T} \parallel 1)\mathbf{G})^\mathsf{T} \bmod q \in \mathcal{R}_q^{km\times k}.$$
- For each $(i,j) \in [h]\times[k]$, run the algorithm $\mathsf{Evaluate}$, defined in Theorem 7, on input ctxt and $U(\cdot,(i,j))$ to obtain $\mathbf{H}_{i,j} \in \mathcal{R}_q^{n\times m}$. Let $\mathbf{h}_{i,j} \coloneqq \mathbf{H}_{i,j}\mathbf{G}^{-1}(\boldsymbol{\tau}) \in \mathcal{R}_q^n$, where $\boldsymbol{\tau} \coloneqq \left(0 \ldots 0 \ \lfloor q/2\rceil\right)^\mathsf{T} \in \mathcal{R}_q^n$. Arrange these vectors in matrix form
$$\hat{\mathbf{H}}^\mathsf{T} \coloneqq \begin{pmatrix}\mathbf{h}_{1,1}^\mathsf{T} & \ldots & \mathbf{h}_{1,k}^\mathsf{T}\\ \vdots & \ddots & \vdots\\ \mathbf{h}_{h,1}^\mathsf{T} & \ldots & \mathbf{h}_{h,k}^\mathsf{T}\end{pmatrix} \in \mathcal{R}^{h\times kn}. \tag{16}$$

– Compute

$$\mathbf{L}^{\mathsf{T}} := (\mathbf{I}_k \otimes \mathbf{G})^{-1}(\hat{\mathbf{H}} + \mathbf{1}_k \otimes \mathbf{P}) \in \mathcal{R}^{km\times h}, \tag{17}$$

by running the algorithm Linear defined in Fig. 6.
– For $i \in [h]$, sample $\tilde{\mathbf{r}}_i \leftarrow \mathsf{Equivocate}(\mathsf{td}, \mathbf{l}_i^{\mathsf{T}}\mathbf{R}, \mathbf{l}_i^{\mathsf{T}}\mathbf{C} + \mathbf{m}_i^{\mathsf{T}}, s)$, where $\mathbf{m}_i^{\mathsf{T}}$ denotes the i-th row of $\boldsymbol{\Gamma}$ multiplied by $-\lfloor q/2 \rfloor$, using the trapdoor td of $\mathbf{B}$. Let $\mathsf{hint} := \tilde{\mathbf{R}} \in \mathcal{R}_q^{h\times t}$ to have the i-th row given by $\tilde{\mathbf{r}}_i^{\mathsf{T}}$.
– The obfuscated circuit consists of $\tilde{\Gamma} := (\mathbf{B}, \mathbf{C}, \mathbf{P}, \mathsf{ctxt}, \mathsf{hint})$.

To make sense of the obfuscator, we make the following observations. By the properties of $\mathsf{Evaluate}$ algorithm (Theorem 7), for $(i,j) \in [h] \times [k]$, it holds that

$$\mathbf{H}_{i,j} = \mathbf{K}_{i,j} + \gamma_{i,j}\mathbf{G}$$

for some $\mathbf{K}_{i,j} \in \mathcal{R}_q^{n\times m}$ such that $\|(-\mathbf{s}_d^{\mathsf{T}} \parallel 1)\mathbf{Z}_{i,j}\| \le \sqrt{\varphi m}\eta\chi$. Thus

$$\hat{\mathbf{H}} = \mathbf{K} + \boldsymbol{\Gamma}^{\mathsf{T}} \otimes \boldsymbol{\tau} \bmod q$$

for $\mathbf{K} \in \mathcal{R}^{kn\times h}$, where the (i,j)-th block of $\mathbf{K}$ is given by $\mathbf{K}_{i,j}\mathbf{G}^{-1}(\boldsymbol{\tau})$, and $\boldsymbol{\Gamma}$ is the truth table of Γ arranged in matrix form, i.e., $\gamma_{i,j} = \Gamma(i,j)$ for all $(i,j) \in [h] \times [k]$. That is, $\hat{\mathbf{H}}$ is a GSW ciphertext under key $\mathbf{s}_d$ encrypting $\boldsymbol{\Gamma}^{\mathsf{T}}$. Further

$$\mathbf{1}_k \otimes \mathbf{P} = \mathbf{1}_k \otimes (\mathbf{p}_1 \| \ldots \| \mathbf{p}_h)$$

is also a GSW ciphertex under key $\mathbf{s}_d$ encrypting 0. Since $\mathbf{L}^{\mathsf{T}} := (\mathbf{I}_k \otimes \mathbf{G})^{-1}(\hat{\mathbf{H}} + \mathbf{1}_k \otimes \mathbf{P})$ and $\mathbf{C}$ is a dual-Regev ciphertext encrypting the secret key $\mathbf{s}_d$ of the ciphertext $\hat{\mathbf{H}} + \mathbf{1}_k \otimes \mathbf{P}$, left-multiplying $\mathbf{L}$ to $\mathbf{C}$ results in a dual-Regev ciphertext of $\boldsymbol{\Gamma}$. Upon examination, we can see that

$$\mathbf{LC} \approx \mathbf{LRB} + \lfloor q/2 \rfloor \boldsymbol{\Gamma} \bmod q \in \mathcal{R}_q^{h\times k}.$$

Therefore revealing $\mathsf{hint} = \tilde{\mathbf{R}}$, with $\tilde{\mathbf{R}}\mathbf{B} \approx \mathbf{LRB} \bmod q$, allows to recover the truth table $\boldsymbol{\Gamma}$ from $\mathbf{LC}$.

Evaluator. An evaluator, to evaluate an obfuscated circuit $\tilde{\Gamma} := (\mathbf{B}, \mathbf{C}, \mathbf{P}, \mathsf{ctxt}, \mathsf{hint})$ at $(i,j) \in [h] \times [k]$, exploits the aforementioned property that

$$\mathbf{LC} \approx \tilde{\mathbf{R}}\mathbf{B} + \lfloor q/2 \rfloor \boldsymbol{\Gamma} \bmod q.$$

Specifically, the evaluator performs the following steps:

– For each $j' \in [k]$, compute $\mathbf{H}_{i,j'} \leftarrow \mathsf{Evaluate}(\mathsf{ctxt}, U(\cdot, (i,j')))$ and let $\mathbf{h}_{i,j} = \mathbf{H}_{i,j'}\mathbf{G}^{-1}(\boldsymbol{\tau})$. Pack $(\mathbf{h}_{i,j'})_{j'\in[k]}$ into a column vector $\hat{\mathbf{h}}_i \in \mathcal{R}^{kn}$.
– Compute $\mathbf{l}_i = \mathbf{G}^{-1}(\hat{\mathbf{h}}_i + \mathbf{1}_k \otimes \mathbf{p}_i) \in \mathcal{R}^{km}$ via the subroutine Linear.
– Parse

$$\mathbf{B} = \begin{pmatrix}\mathbf{b}_1 \ldots \mathbf{b}_k\end{pmatrix} \in \mathcal{R}_q^{t\times k}, \qquad \mathbf{C} = \begin{pmatrix}\mathbf{c}_1 \ldots \mathbf{c}_k\end{pmatrix} \in \mathcal{R}_q^{km\times k}, \qquad \text{and} \qquad \tilde{\mathbf{R}} = \begin{pmatrix}\tilde{\mathbf{r}}_1^{\mathsf{T}} \\ \vdots \\ \tilde{\mathbf{r}}_h^{\mathsf{T}}\end{pmatrix} \in \mathcal{R}_q^{h\times t}.$$

– Compute $\tilde{\gamma}_{i,j} = \mathbf{l}_i^{\mathsf{T}}\mathbf{c}_j - \tilde{\mathbf{r}}_i^{\mathsf{T}}\mathbf{b}_j$ and return $\lfloor (2/q)\tilde{\gamma}_{i,j} \rceil \bmod 2$.

D.2 XiO Correctness Proof (Theorem 4)

Proof. Fix arbitrary $j' \in [k]$. Let

$$\mathbf{H}_{i,j'} \leftarrow \mathsf{Evaluate}((\mathbf{V}_i)_{i\in[0,d]}, U(\cdot,(i,j'))),$$
$$\mathbf{h}_{i,j'} = \mathbf{H}_{i,j'}\mathbf{G}^{-1}(\boldsymbol{\tau}).$$

By the properties of $\mathsf{Evaluate}$ (Theorem 7), it holds that

$$\mathbf{h}_{i,j} = \mathbf{z}_{i,j} + \gamma_{i,j}\boldsymbol{\tau} \bmod q,$$

where $\mathbf{z}_{i,j}$ is such that $\bar{e}_{i,j} := \langle(-\mathbf{s}_d^{\intercal}, 1), \mathbf{z}_{i,j}\rangle$ with $\|\bar{e}_{i,j}\| \leq \varphi^{3/2} m\eta\chi$. Define

$$\hat{\mathbf{h}}_i^{\intercal} := \left(\mathbf{h}_{i,1}^{\intercal} \dots \mathbf{h}_{i,k}^{\intercal}\right),$$

and parse

$$\mathbf{B} = \left(\mathbf{b}_1 \dots \mathbf{b}_k\right) \in (\mathcal{R}_q^t)^k, \quad \mathbf{C} = \left(\mathbf{c}_1 \dots \mathbf{c}_k\right) \in (\mathcal{R}_q^{km})^k, \quad \mathbf{E} = \left(\hat{\mathbf{e}}_1 \dots \hat{\mathbf{e}}_k\right) \in (\mathcal{R}^{km})^k.$$

Let $\boldsymbol{\iota}_j \in \mathcal{R}^k$ denote the j-th unit vector in $\mathcal{R}^k$. By definition of $\mathbf{C}$ and $\mathbf{l}_i$, we have that

$$\mathbf{c}_j = \mathbf{R}\mathbf{b}_j + \hat{\mathbf{e}}_j + \boldsymbol{\iota}_j \otimes \mathbf{G}^{\intercal}\begin{pmatrix}-\mathbf{s}_d\\1\end{pmatrix} \bmod q,$$
$$\mathbf{l}_i = \mathbf{G}^{-1}(\hat{\mathbf{h}}_i + \mathbf{1}_k \otimes \mathbf{p}_i) \in \mathcal{R}^{km}.$$

It follows that

$$\begin{aligned}
\mathbf{l}_i^{\intercal}\mathbf{c}_j &= \mathbf{l}_i^{\intercal}\left(\mathbf{R}\mathbf{b}_j + \hat{\mathbf{e}}_j + \boldsymbol{\iota}_j \otimes \mathbf{G}^{\intercal}\begin{pmatrix}-\mathbf{s}_d\\1\end{pmatrix}\right) \bmod q\\
&= \mathbf{l}_i^{\intercal}\mathbf{R}\mathbf{b}_j + \mathbf{l}_i^{\intercal}\hat{\mathbf{e}}_j + \mathbf{l}_i^{\intercal}\left(\boldsymbol{\iota}_j \otimes \mathbf{G}^{\intercal}\begin{pmatrix}-\mathbf{s}_d\\1\end{pmatrix}\right) \bmod q\\
&= \mathbf{l}_i^{\intercal}\mathbf{R}\mathbf{b}_j + \mathbf{l}_i^{\intercal}\hat{\mathbf{e}}_j + \left(-\mathbf{s}_d^{\intercal}, 1\right)\mathbf{G}\mathbf{l}_{i,j} \bmod q\\
&= \mathbf{l}_i^{\intercal}\mathbf{R}\mathbf{b}_j + \mathbf{l}_i^{\intercal}\hat{\mathbf{e}}_j + \left(-\mathbf{s}_d^{\intercal}, 1\right)(\mathbf{h}_{i,j} + \mathbf{p}_i) \bmod q\\
&= \mathbf{l}_i^{\intercal}\mathbf{R}\mathbf{b}_j + \mathbf{l}_i^{\intercal}\hat{\mathbf{e}}_j + (\bar{e}_{i,j} + \left(-\mathbf{s}_d^{\intercal}, 1\right)(\mathbf{V}_{d+i}\mathbf{G}^{-1}(\boldsymbol{\tau}) + \gamma_{i,j}\boldsymbol{\tau})) \bmod q\\
&= \mathbf{l}_i^{\intercal}\mathbf{R}\mathbf{b}_j + \mathbf{l}_i^{\intercal}\hat{\mathbf{e}}_j + (\bar{e}_{i,j} + \mathbf{e}_{d+i}^{\intercal}\mathbf{G}^{-1}(\boldsymbol{\tau}) + \gamma_{i,j}\lfloor q/2\rfloor) \bmod q.
\end{aligned}$$

In other words,

$$\mathbf{l}_i^{\intercal}\mathbf{c}_j - \gamma_{i,j}\lfloor q/2\rfloor = \underbrace{\mathbf{l}_i^{\intercal}\mathbf{R}}_{\text{LWE secret}}\ \mathbf{b}_j + \underbrace{\mathbf{l}_i^{\intercal}\hat{\mathbf{e}}_j + \bar{e}_{i,j} + \mathbf{e}_{d+i}^{\intercal}\mathbf{G}^{-1}(\boldsymbol{\tau})}_{\text{noise}} \bmod q \tag{18}$$

is an LWE sample with respect to (columns of) $\mathbf{B}$, with

$$\left\|\mathbf{l}_i^{\intercal}\hat{\mathbf{e}}_j\right\| \leq \varphi^{3/2}km\chi, \qquad \|\bar{e}_{i,j}\| \leq \varphi^{3/2}m\eta\chi, \qquad \text{and} \qquad \left\|\mathbf{e}_{d+i}^{\intercal}\mathbf{G}^{-1}(\boldsymbol{\tau})\right\| \leq \varphi^{3/2}m\chi,$$

with overwhelming probability. By the properties of $\mathsf{Equivocate}$, we have

$$\tilde{\mathbf{R}}\mathbf{B} = \mathbf{L}\mathbf{R}\cdot\mathbf{B} + \tilde{\mathbf{E}}$$

where the norm of each entry of $\tilde{\mathbf{E}}$ is bounded by $\sqrt{\varphi}s$. Thus, parsing $\tilde{\mathbf{R}} = \begin{pmatrix}\tilde{\mathbf{r}}_1^{\intercal}\\ \vdots \\ \tilde{\mathbf{r}}_h^{\intercal}\end{pmatrix}$ one has

$$\mathbf{l}_i^{\intercal}\mathbf{c}_j - \tilde{\mathbf{r}}_i^{\intercal}\mathbf{b}_j = \mathbf{l}_i^{\intercal}\mathbf{c}_j - (\mathbf{l}_i^{\intercal}\mathbf{R}\mathbf{b}_j + \tilde{e}_{i,j})$$

$$
\begin{aligned}
&= \mathbf{l}_i^{\mathsf{T}}\mathbf{c}_j - \mathbf{l}_i^{\mathsf{T}}\mathbf{R}\mathbf{b}_j - \tilde{e}_{i,j} \\
&= \gamma_{i,j}\lfloor q/2 \rfloor + \underbrace{\mathbf{l}_i^{\mathsf{T}}\hat{\mathbf{e}}_j + \bar{e}_{i,j} + \mathbf{e}_{d+i}^{\mathsf{T}}\mathbf{G}^{-1}(\boldsymbol{\tau}) - \tilde{e}_{i,j}}_{\text{error}} \\
&\approx \gamma_{i,j}\lfloor q/2 \rfloor \bmod q.
\end{aligned}
$$

Therefore, as long as

$$
q \geq 4\left(\varphi^{3/2}km\chi + \varphi^{3/2}m\eta\chi + \varphi^{3/2}m\chi + \sqrt{\varphi}s\right), \tag{19}
$$

correctness of the evaluation is guaranteed. ☐

D.3 XiO Security Proof (Theorem 5)

Proof. To prove the claim, we need to show that for any pair of circuits $(\Gamma_0, \Gamma_1) \in \mathcal{C}_\lambda$ such that $|\Gamma_0| = |\Gamma_1|$ and $\Gamma_0(i,j) = \Gamma_1(i,j)$ on all inputs $(i,j) \in [h] \times [k]$, the distributions

$$
D_0 \coloneqq \mathsf{Obf}(\Gamma_0) \text{ and } D_1 \coloneqq \mathsf{Obf}(\Gamma_1)
$$

are computationally indistinguishable.

In the following, let $\boldsymbol{\gamma}^{(0)}$ and $\boldsymbol{\gamma}^{(1)}$ denote the descriptions of Γ_0 and Γ_1 respectively, and let $\boldsymbol{\Gamma}$ denote the common truth table of both circuits Γ_0 and Γ_1.

Recall that, for $b \in \{0,1\}$, by construction a sample of D_b takes the form $\mathsf{Obf}(\Gamma_b) = (\mathbf{B}, \mathbf{C}, \mathbf{P}, \mathsf{ctxt}_b, \mathsf{hint}_b)$.

First, we consider a pair of modified distributions D_b' which are identical to D_b except that $\tilde{\mathbf{R}}_b$ is sampled using the inefficient procedure

$$
\mathsf{hint}_b = \tilde{\mathbf{R}}_b \leftarrow \mathsf{HintGen}(\mathbf{W}_b, \mathbf{B}),
$$

where $\mathbf{W}_b \coloneqq \mathbf{L}_b\mathbf{C} - \lfloor q/2 \rfloor \boldsymbol{\Gamma} \bmod q \in \mathcal{R}_q^{h\times k}$ and $\mathsf{HintGen}$ is defined as in Fig. 4. By Theorem 3 (induced distribution of $\mathsf{Equivocate}$), we conclude that D_b is statistically close to D_b' for $b \in \{0,1\}$. Therefore, in order to show that $D_0 \approx_c D_1$, it suffices to argue that D_0' and D_1' are computationally indistinguishable.

To show that D_0' and D_1' are computationally indistinguishable, we invoke the $\mathsf{EqLWE}_{\mathsf{params}}$ assumption (Definition 9). Notice that for the choice of params we have that

- The tagged symmetric-key encryption scheme SKE is defined in Fig. 5. Its encryption algorithm on input key $\mathbf{s}_d$, tag $\mathsf{tag}^* = 0$ and message $\boldsymbol{\gamma}$ outputs

  $$
  \mathsf{ctxt} = \left\{ \begin{array}{cl} \mathbf{V}_0 & \coloneqq \mathsf{GSW}(\mathbf{s}_0, \boldsymbol{\gamma}) \\ (\mathbf{V}_i)_{i\in[d]} & \coloneqq (\mathsf{GSW}(\mathbf{s}_1, \tilde{\mathbf{s}}_0) \| \ldots \| \mathsf{GSW}(\mathbf{s}_d, \tilde{\mathbf{s}}_{d-1}) \end{array} \right\},
  $$

  where $\mathbf{s}_i \leftarrow\!\!\$\ \mathcal{R}_q^{n-1}$ and $\tilde{\mathbf{s}}_i = \mathbf{G}^{-1}\left(\begin{bmatrix} -\mathbf{s}_i \\ 1 \end{bmatrix}\right) \in \mathcal{R}^m$ for $i \in [0, d-1]$ and on input key $\mathbf{s}_d$, tag $\mathsf{tag}_{\mathsf{pad}} = 1$ and message $\mathbf{0}_h$ outputs

  $$
  \mathbf{P} = \left\{ (\mathbf{p}_i)_{i\in[h]} \coloneqq (\mathsf{GSW}(\mathbf{s}_d, 0) \| \ldots \| \mathsf{GSW}(\mathbf{s}_d, 0)) \cdot (\mathbf{I}_h \otimes \mathbf{G}^{-1}(\boldsymbol{\tau}), \right\}
  $$

  with GSW being the subroutine defined in Fig. 5.
- The encoding $\mathsf{Encode} : \mathcal{R}_q^{n-1} \to \mathcal{R}_q^{km\times k}$ is defined as $\mathsf{Encode}(\mathbf{s}_d) \coloneqq \mathbf{I}_k \otimes ((-\mathbf{s}_d^{\mathsf{T}} \parallel 1)\mathbf{G})^{\mathsf{T}} \bmod q$.
- The distribution $\mathcal{E} \sim \mathcal{R}_q^{t\times k}$ is defined as $\mathcal{E} \coloneqq \left\{\mathbf{B} \;\middle|\; (\mathbf{B}, \mathsf{td}) \leftarrow \mathsf{pTrapGen}(1^t, 1^k, q)\right\}$.

Notice that,

- from Eq. (18) and its derived norm bounds in the proof of Theorem 4, with overwhelming probability in λ it holds that

  $$
  \|\mathbf{W}_b - \mathbf{L}_b\mathbf{R}\mathbf{B}\| \leq \sqrt{k} \cdot (\varphi^{3/2}km\chi + \varphi^{3/2}m\eta\chi + \varphi^{3/2}m\chi) \leq s.
  $$

  Hence $\mathsf{assert}(\|\mathbf{W}_b - \mathbf{L}_b\mathbf{R}\mathbf{B} \bmod q\| \leq s) = 1$ for $b \in \{0,1\}$ holds with overwhelming probability;

– for adversary's choices of the message msg_b as $\boldsymbol{\gamma}_b$, for $b \in \{0,1\}$, the circuit Φ, on input $\mathsf{ctxt}_{\mathsf{msg}_b}$ outputs $(\hat{\mathbf{H}}_b, \mathbf{M}_b) \in \mathcal{R}_q^{kn \times h} \times \mathcal{R}_q^{h \times k}$, where $\hat{\mathbf{H}}_b$ is defined as in Eq. (16) and

$$\mathbf{M}_b \coloneqq \begin{pmatrix} \mathbf{m}_1^\mathsf{T} \\ \vdots \\ \mathbf{m}_h^\mathsf{T} \end{pmatrix},$$

where $\mathbf{m}_i^\mathsf{T}$ denotes the i-th row of $\boldsymbol{\Gamma}$ multiplied by $\lfloor q/2 \rfloor$. Note that $\mathbf{M}_0 = \mathbf{M}_1 \bmod q$ holds by construction.

It follows that

$$D'_b = \mathsf{inst}_b, \text{ where } \mathsf{inst}_b \leftarrow \mathsf{Exp}^b(\mathsf{msg}_0, \mathsf{msg}_1, \Phi)$$

for Exp^b defined as in Fig. 4. Invoking the $\mathsf{EqLWE}_{\mathsf{params}}$ assumption, we deduce the computational indistinguishability of $D'_0 = \mathsf{inst}_0$ and $D'_1 = \mathsf{inst}_1$, which concludes the proof. □

D.4 XiO Corollary Proof (Corollary 1)

Proof. Recall from Section 5.1 that the necessary conditions on the choice of params for $\mathsf{EqLWE}_{\mathsf{params}}$ to be plausible are that

– SKE is IND-\$-CPA-secure (Definition 15),
– $\mathcal{E} \sim \mathcal{R}_q^{t \times k}$ is an s-equivocal distribution (Definition 7).

By Theorems 2 and 3, under the $\mathsf{dNTRU}_{\mathcal{R},q,k,\chi_f}$ assumption, $\mathcal{E} \coloneqq \{\mathbf{B} \mid (\mathbf{B}, \mathsf{td}) \leftarrow \mathsf{pTrapGen}(1^t, 1^k, q)\}$ follows a s-equivocal distribution.

Therefore, it remains to show that SKE can be proven IND-\$-CPA secure. We do this in the following lemma.

Lemma 20. *Under the* $\mathsf{dLWE}_{\mathcal{R},q,n-1,\chi}$ *assumption, the tagged symmetric-key encryption scheme* SKE *described in Fig. 5 is IND-\$-CPA secure.*

Proof. We will prove that statement for $\mathsf{tag} = \mathsf{tag}^*$ only. The proof for $\mathsf{tag} = \mathsf{tag}_{\mathsf{pad}}$ is identical and therefore omitted. In order to prove IND-\$-CPA security of SKE it suffices to prove that

$$H_0 \coloneqq \left\{ (\mathbf{V}_i)_{i \in [0,d]} \;\middle|\; \begin{array}{c} \mathbf{s}_i \leftarrow\!\!\$\; \mathcal{R}_q^{n-1} \text{ for } i \in [0,d] \\ \mathbf{V}_0 = \mathsf{GSW}(\mathbf{s}_0, \mathsf{msg}) \\ \mathbf{V}_i = \mathsf{GSW}(\mathbf{s}_i, \tilde{\mathbf{s}}_{i-1}) \text{ for } i \in [d] \end{array} \right\}$$

and

$$H_d \coloneqq \left\{ (\mathbf{V}_i)_{i \in [0,d]} \;\middle|\; \begin{array}{c} \mathbf{V}_0 \leftarrow\!\!\$\; \mathcal{R}_q^{n \times \ell m} \\ \mathbf{V}_i \leftarrow\!\!\$\; \mathcal{R}_q^{n \times m^2} \text{ for } i \in [d] \end{array} \right\}$$

are computationally indistinguishable, where $\tilde{\mathbf{s}}_i \coloneqq \mathbf{G}^{-1}\left(\begin{bmatrix} -\mathbf{s}_i \\ 1 \end{bmatrix}\right)$. This follows from a simple hybrid argument. For $j \in [d]$ define the following distributions

$$H_j \coloneqq \left\{ (\mathbf{V}_i)_{i \in [0,d]} \;\middle|\; \begin{array}{c} \mathbf{s}_i \leftarrow\!\!\$\; \mathcal{R}_q^{n-1} \text{ for } i \in [0,d-j] \\ \mathbf{V}_0 = \mathsf{GSW}(\mathbf{s}_0, \mathsf{msg}) \\ \mathbf{V}_i = \mathsf{GSW}(\mathbf{s}_i, \tilde{\mathbf{s}}_{i-1}) \text{ for } i \in [1, d-j], \mathbf{V}_i \leftarrow\!\!\$\; \mathcal{R}_q^{n \times m^2} \text{ for } i \in [d-j+1, d] \end{array} \right\}.$$

We will show that $H_{j-1} \approx_c H_j$ for all $j \in [d+1]$, from which the lemma follows. We will have that

– $H_{j-1} \approx_c H_j$ for $j \in [d]$: it follows under $\mathsf{dLWE}_{\mathcal{R},q,n-1,m^2,\chi}$ with respect to LWE secret $\mathbf{s}_i$.
– $H_{d-1} \approx_c H_d$: it follows under $\mathsf{dLWE}_{\mathcal{R},q,n-1,\ell m,\chi}$ with respect to LWE secret $\mathbf{s}_0$.

We are going to formally show that $H_0 \approx_c H_1$ assuming $\mathsf{dLWE}_{\mathcal{R},q,n-1,m^2,\chi}$. The other reductions are almost identical, except a change in the assumption parameter (specifically the number of LWE samples), and are therefore omitted.

The reduction works as follows

- it parses $\tilde{\mathbf{A}} = \mathbf{A}_0 \in \mathcal{R}_q^{(n-1)\times m^2}$ and $\tilde{\mathbf{c}} = \mathbf{c}_0 \in \mathcal{R}_q^{m^2}$ obtained from the $\mathsf{dLWE}_{\mathcal{R},q,n-1,m^2,\chi}$ instance,
- it samples $\mathbf{s}_i \leftarrow\$ \mathcal{R}_q^{n-1}$ for $i \in [0, d-1]$,
- it computes $\mathbf{V}_0 = \mathsf{GSW}(\mathbf{s}_0, \boldsymbol{\gamma})$, and $\mathbf{V}_i = \mathsf{GSW}(\mathbf{s}_i, \tilde{\mathbf{s}}_{i-1})$, where $\tilde{\mathbf{s}}_{i-1} \coloneqq \mathbf{G}^{-1}\left(\begin{bmatrix} -\mathbf{s}_{i-1} \\ 1 \end{bmatrix}\right)$ for $i \in [d-1]$,
- it sets $\mathbf{V}_d = \begin{pmatrix} \mathbf{A}_0 \\ \mathbf{c}_0^{\mathtt{T}} \end{pmatrix} + \left(\mathbf{G}^{-1}\begin{pmatrix} -\mathbf{s}_{d-1} \\ 1 \end{pmatrix}\right)^{\mathtt{T}} \otimes \mathbf{G}$,
- it outputs $(\mathbf{V}_i)_{i\in[0,d]}$.

Observe that

- if $(\tilde{\mathbf{A}}, \tilde{\mathbf{c}})$ is a structured $\mathsf{dLWE}_{\mathcal{R},q,n-1,m^2,\chi}$ instance, then the output of the reduction is identical to H_0,
- if $(\tilde{\mathbf{A}}, \tilde{\mathbf{c}})$ is a uniform random instance, then then the output of the reduction is identical to H_1.

We conclude that $H_0 \approx_c H_1$ as claimed.

Putting everything together yields $H_0 \approx_c H_d$, which implies the IND-\$-CPA security of SKE as claimed. □

Given Lemma 20, Corollary 1 follows as required. □

φ	$\in \mathsf{poly}(\lambda)$	$\geq \lambda$	Degree of the cyclotomic ring $\mathcal{R}$
q	$\in o(N^{(9/4)g}) \cdot \mathsf{poly}(\lambda)$	prime unramified in $\mathcal{R}$, $q\mathcal{R}$ splits into $g \geq 1$ prime ideals, $\geq \max \left\{ \begin{array}{c} 4\left(\varphi^{3/2}km\chi+\varphi^{3/2}m\eta\chi+\varphi^{3/2}m\chi+\sqrt{\varphi}s\right), \\ (4\beta_{\Lambda}\chi_f\sqrt{(k-1)\varphi}(\varphi+1)^{1+\frac{2}{\varphi}}/\varphi^{\frac{1}{4}})^g, \\ c\cdot s\sqrt{hk} \end{array} \right\}$ $\leq 2^{o(\lambda)}\chi$	Modulus: lower-bounded to satisfy constraints from correctness (Theorem 4), desired properties of Π_{PTD} (Theorems 2 and 3) and hardness requirements of SIS ($c > 2$, cf. Section 5.2), and upper-bounded to satisfy hardness requirements of dLWE
β_{Λ}	$\in o(N^2) \cdot \mathsf{poly}(\lambda)$	$\geq s(\sqrt{k\varphi}+1)$	Sparseness of $\Lambda_q(\mathbf{B})/\mathcal{L}(\mathbf{f}^{\mathrm{T}})$ (Theorems 2 and 3)
s	$\in o(N^{3/2}) \cdot \mathsf{poly}(\lambda)$	$\geq \max \left\{ \begin{array}{c} 2\chi_f k\varphi, \\ \varphi^{3/2}h\sqrt{kn\lceil\log q\rceil}, \\ \sqrt{k}\cdot(\varphi^{3/2}km\chi+\varphi^{3/2}m\eta\chi+\varphi^{3/2}m\chi) \end{array} \right\}$	Gaussian parameter for Π_{PTD}: Lower-bounded to satisfy constraints from desired properties of Π_{PTD} (Theorems 2 and 3), suggested parameters of EqLWE (Eq. (9)), and distance between $\mathbf{LC} - \lfloor q/2 \rfloor \boldsymbol{\Gamma}$ and $\Lambda_q(\mathbf{B})$
χ	$\in O(\lambda)$		Gaussian parameter for the noise of GSW and Packed-Dual-Regev encryption
χ_f	$\in O(\lambda)$		Gaussian parameter used in $\Pi_{\mathsf{PTD}}.\mathsf{pTrapGen}$
h	$\in o(N)$		Height of truth table
k	$\in o(\sqrt{N})$	$\geq 2(t-1)\log q + 1$	Width of truth table and equivocal matrix $\mathbf{B}$
t	$\in \mathsf{poly}(\lambda)$		Height of equivocal matrix $\mathbf{B}$
n	$\in O(1)$		Height of GSW ciphertexts $\mathbf{V}_i$ for $i \in [0, d+h]$
m	$= n\lceil\log q\rceil$		Width of gadget matrix $\mathbf{G}$
d	$\in O(\ell \log \ell)$		Depth of universal circuit U
η	$\in \tilde{O}(\lambda^2)$		Norm growth of error term in $\mathsf{Evaluate}$'s output (Theorem 7)

Table 1. Parameters for obfuscator construction from Fig. 6, when instantiating ($\mathsf{pTrapGen}, \mathsf{Equivocate}$) with construction Π_{PTD} from Section 4.2.